

# Single-Stage PFC LLC Converter Design

with the STMicroelectronics L6790A

## Worked design

90 to 264 Vac, 47 to 63 Hz

25 V / 26.3 A = 657.5 W

# Single-Stage PFC LLC Converter Design

## Before you start

The L6790A behaviour described here comes from a **draft** datasheet dated 30 April 2026. That draft has items marked TBD and at least one place where it contradicts itself. Check every controller constant against the released datasheet before the design goes to production.

The worked design has not been built. Only the numbers marked as measured are measured; the rest are calculated.

Some of the LLC theory figures are taken from the application notes in the reference list. Each one names its source in the caption.

## Revision history

| Version | Date           | Change                                                                                                                                                                                                                                                                                     |
|---------|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| V1.0    | August 2026    | First issue. Worked design: 25 V / 26.3 A, 657.5 W.                                                                                                                                                                                                                                        |
| V1.1    | September 2026 | Background section on LLC and power factor correction added, with circuit figures. Transformer design, the closed form for $f_{sw}(\theta)$ , the compensator design method, the gain-margin check and the input capacitor added. Worked design moved to the $N_p:N_s = 15:2$ transformer. |
| V1.2    | September 2026 | Renamed single-stage PFC LLC throughout. Verification margins defined. Worked values moved out of the design procedure and into the design example, where each one names the equation it comes from and shows the numbers put into it.                                                     |

## Conventions

- Frequencies are switching frequencies unless written  $f_l$ , which is the line frequency.
- $f_n = f_{sw}/f_r$  is the switching frequency normalised to the series resonance. Gain curves are always drawn against it, so that one curve serves every tank. Figures reproduced from other documents may write  $f_s$  for  $f_{sw}$ ; that is the same quantity.
- $M$  is the voltage gain of the resonant network: the output voltage referred to the primary, divided by the fundamental of the square wave driving it.  **$M = 1$  at the series resonance**, whatever the load.
- $Z_0 = \sqrt{L_r/C_r}$  is the characteristic impedance of the tank, and  $Q = Z_0/R_{ac}$  is its quality factor — how heavily the tank is loaded.  $Q = 0$  is no load; larger  $Q$  is more power and a flatter, lower gain curve.
- $V_{o,eff} = V_{out} + N_{rect} V_f$  is the output voltage the primary actually sees, with the rectifier drop added.  $N_{rect}$  is the number of rectifier drops in the conduction path: 1 for a centre tap, 2 for a full bridge.
- $n$  is the equivalent-model turns ratio used in the gain equation;  $n_T$  is the physical turns ratio of a transformer whose leakage is integrated into  $L_r$ .
- $\lambda = L_r/L_m$ , and  $m = 1 + 1/\lambda$  is the ratio of total primary to resonant inductance used in older literature.
- $\theta$  is the line phase angle, zero at the mains zero crossing.
- “Equivalent input” is what the resonant tank sees after topology morphing, which is not the mains voltage.

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# Single-Stage PFC LLC Converter Design

## 1 Introduction

A conventional universal-input LLC power supply has two stages: a boost power-factor corrector that holds a 400 V bus, and an LLC converter that steps that bus down. The capacitor between them does two jobs. It stores the energy that a unity-power-factor input cannot deliver evenly, and it gives the LLC an almost constant input, so the resonant tank only has to cover a narrow range.

A **single-stage PFC LLC** removes both the boost stage and that capacitor. The rectified mains feeds the resonant tank directly. The result is a converter whose input is a 100/120 Hz half sine, whose gain is no longer something the controller can set, and whose output capacitor must absorb everything the bus capacitor used to. It is a different converter, not an LLC with one stage removed, and most of the design habits from the two-stage case do not apply.

This note explains how such a converter works and how to design one with the STMicroelectronics **L6790A** controller. One worked design runs through it, from specification to component values: **90 to 264 Vac in, 25 V / 26.3 A = 657.5 W out**, a low-voltage, high-current dc rail.

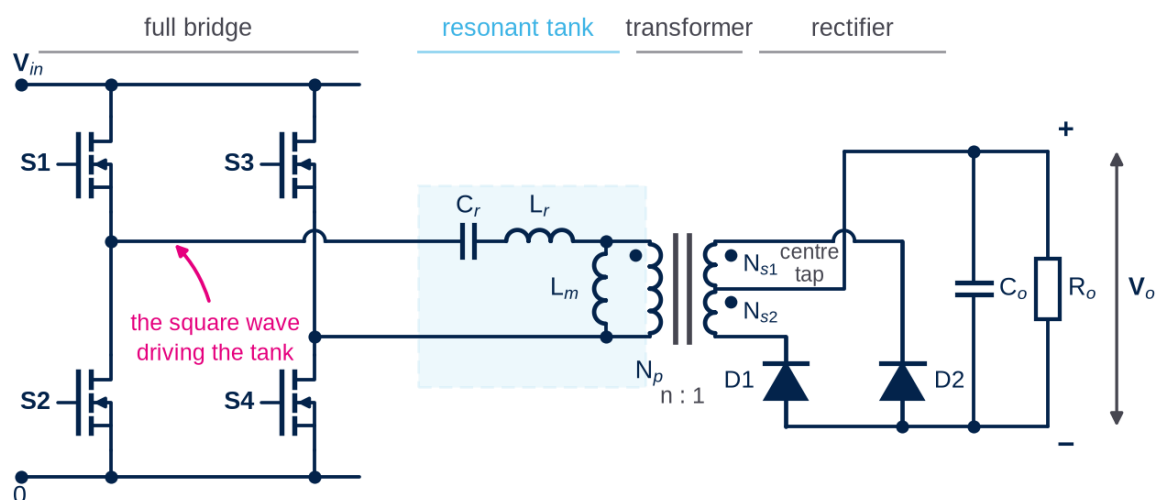
**Scope.** Covered: the power stage, the resonant tank, the output bank, the controller network and the voltage loop.  
Not covered: EMI filtering, magnetics construction, layout and safety approval.

## 2 The two converters this one is made of

What comes later is easier to follow if the two halves are understood separately first. This section covers an ordinary LLC and ordinary power factor correction. A reader who has designed both can skip to Section 3.

### 2.1 What an LLC converter is

An LLC converter is a square-wave generator that drives a resonant network; the network feeds a transformer and a rectifier. The switches only chop the input into a square wave whose frequency can be varied. There is no duty cycle to set and no inductor current to program.



**Figure 1: An LLC stage: a square-wave generator, the resonant network, a transformer and a rectifier. The switches only set the frequency — the tank decides how much power flows and the transformer sets the voltage.**

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The three tank elements are named in the order they appear in “LLC”: the series inductance  $L_r$ , the magnetising inductance  $L_m$  of the transformer, and the series capacitance  $C_r$ . The middle one is not a separate part: it is the transformer’s own magnetising inductance. So only  $C_r$  and  $L_r$  are added components, and  $L_r$  is often the leakage inductance of the transformer itself.

- **The square wave sets the frequency, and nothing else.** The output voltage is controlled by moving  $f_{sw}$ , because the impedance of the tank, and so its voltage division, depends on frequency.
- **The tank current is almost a sine wave.** That is the purpose: the switches turn on and off while the current is small, or while it flows through the body diode, so most of the switching loss disappears.
- **The transformer steps the voltage and provides isolation,** and its leakage inductance can be used as  $L_r$  instead of being a problem. That is the usual construction at these power levels.

## 2.2 Why an LLC, and not something simpler

An LLC is more work than a forward or a flyback converter, and it does not regulate by duty cycle, which is what most designers are used to. At a few hundred watts and above, it is worth the extra work for these reasons:

- **Soft switching over the whole load range, on both sides.** The primary switches turn on at zero voltage, and below resonance the secondary rectifiers turn off at zero current. A phase-shifted full bridge loses ZVS at light load; an LLC does not, because the magnetising current that does the work is present at any load.
- **The frequency can be raised.** In a hard-switched converter, switching loss limits the frequency, and with it the size of the magnetics and the filter. Remove that loss and hundreds of kHz become practical.
- **The transformer leakage is used, not fought.** In a forward converter the leakage inductance is a problem that needs a snubber. Here it can be  $L_r$ , so the resonant inductor costs nothing and the snubber is not needed.
- **No output inductor.** The rectifier feeds the output capacitor directly. That removes a wound component, and its loss, from the high-current side, where it hurts most.
- **Low noise.** The currents are sine waves rather than trapezoids, and the switching edges are slow and damped, so the conducted and radiated emissions are much lower than in a hard-switched converter of the same power.

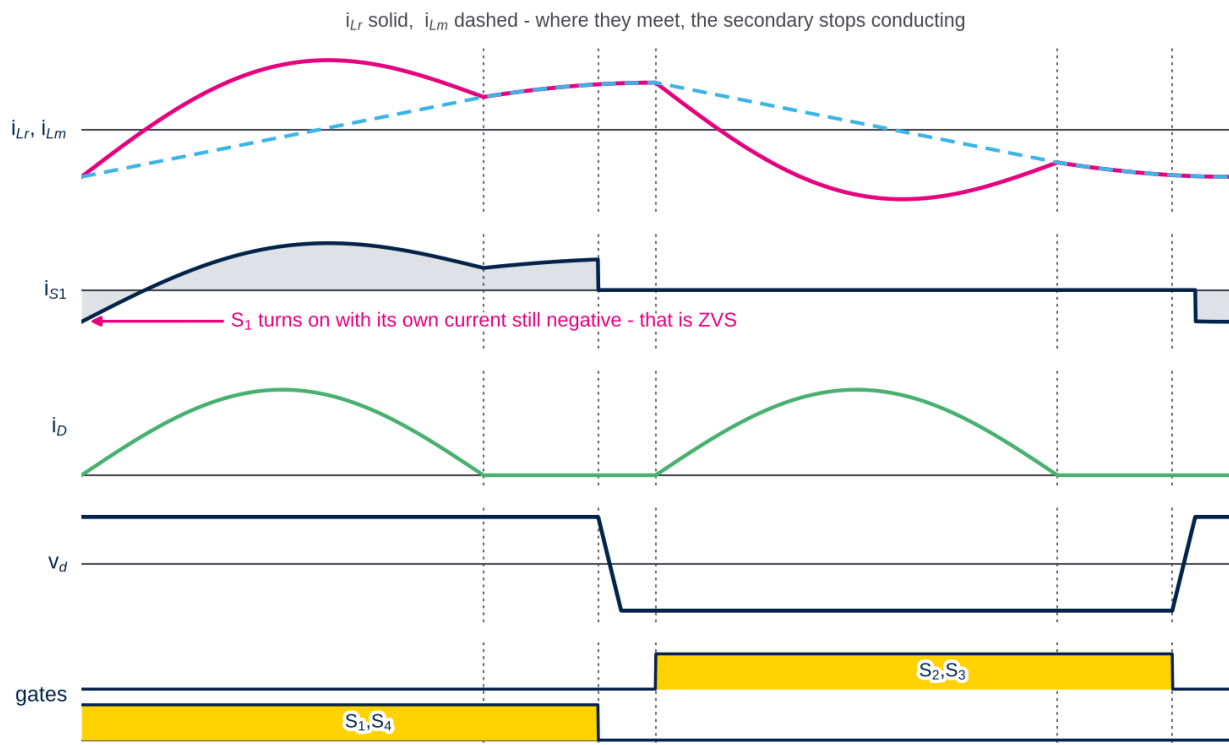
The costs are real too. Regulation is by frequency, so the operating frequency moves with line and load, and the magnetics must work over that whole range. The magnetising current circulates whether or not the load takes power, so light-load efficiency is not a strong point. And the design does not come from a duty-cycle expression: it comes from the gain curve, which is what the rest of this section is about.

## 2.3 Why resonance buys anything

In a hard-switched converter a switch turns on with the full input voltage across it, so the energy stored in its output capacitance is lost in the channel every cycle. That loss grows with frequency. It is what limits the frequency, and with it the size of the magnetics.

An LLC that runs **in its inductive region** presents an inductive load to the bridge. The tank current then lags the drive voltage. In the dead time between the two switches of a leg, that lagging current discharges the mid-point capacitance by itself, and the next switch turns on at zero volts. **That is zero-voltage switching, and it is the whole reason for the topology.** Section 2.10 shows the mechanism and what it requires of the design.

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**Figure 2: One switching period below resonance.**  $i_{Lr}$  is the tank current and  $i_{Lm}$  the magnetising current; the difference between them is what crosses to the secondary, and it is gone once the two meet. The switch turns on while its own current is still negative, and that is what discharges the mid-point.

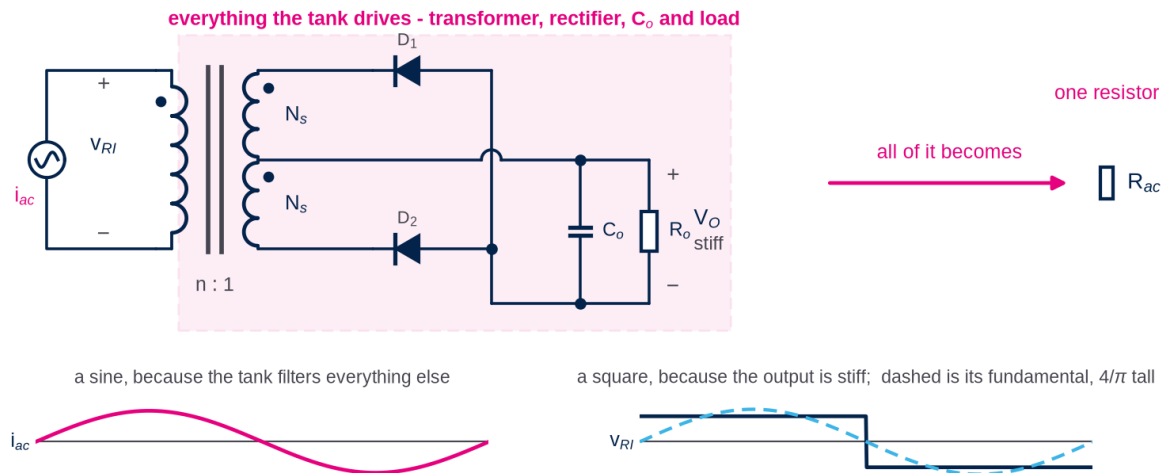
The two frequencies that bound this are the subject of Section 2.5, and the operating regions they divide are in Section 2.6.

The condition is *inductive operation*, not a particular frequency. Go low enough in frequency and the tank becomes capacitive: the current leads, the dead time works against the transition instead of for it, and the bridge hard-switches into a low impedance. It is the one failure mode that destroys parts rather than just heating them. **Where that happens is not a fixed frequency:** the boundary moves with load, and Section 2.9 is about exactly that.

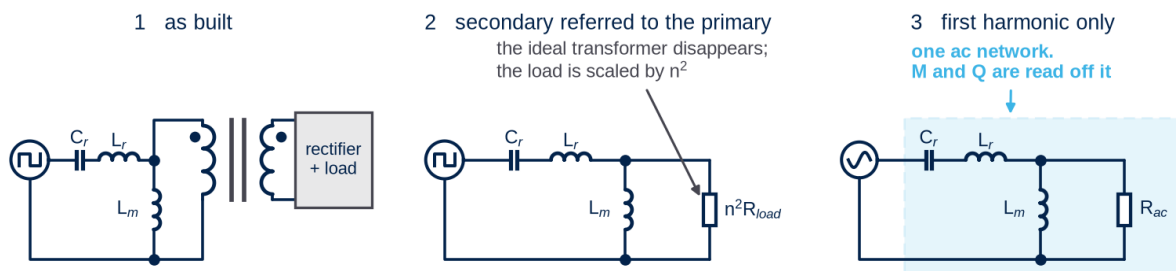
## 2.4 How the circuit becomes $M(f_n, Q)$

Every gain curve in every LLC note comes from three simplifications, applied in order. They are worth doing once, because each one throws something away, and it is useful to know what.

# Single-Stage PFC LLC Converter Design



**Figure 3: Why the rectifier and the load can be replaced by one resistance.** The output is stiff, so the voltage the tank sees is a square wave; the tank filters everything except the fundamental, so the current is a sine wave. A sine of current in phase with a square of voltage delivers the same fundamental power as a resistor would take.



**Figure 4: The reduction, in three steps: as built; the secondary referred to the primary; the fundamental only.** What is left is one ac network whose voltage transfer is the gain  $M$  and whose damping is  $Q$ .

- **Refer the secondary to the primary.** The ideal transformer disappears and the load resistance is multiplied by  $n^2$ .
- **Replace the rectifier and the output filter by one resistor.** The rectifier draws a square current wave and the output is stiff, so at the fundamental frequency the whole of it looks like a resistor.
- **Keep only the fundamental of the drive.** The tank is a filter centred near the switching frequency, so the harmonics of the square wave contribute little. This is the *first harmonic approximation*, and it is why every LLC design ends with a simulation or a measurement rather than with the equations.

What is left is one ac network: a voltage divider whose ratio changes with frequency. Two numbers describe it, and both are used on every page from here on:

- **The gain  $M$ :** the voltage across  $R_{ac}$  divided by the fundamental that drives it. Because  $C_r$  and  $L_r$  cancel at the series resonance  $f_r$ ,  **$M = 1$  there at any load**. Below  $f_r$  the network can step up ( $M > 1$ ); above it, it steps down.
- **The quality factor  $Q = Z_0/R_{ac}$ ,** where  $Z_0 = \sqrt{L_r/C_r}$  is the characteristic impedance of the tank.  $Q$  says how heavily the tank is loaded:  **$Q$  rises with output power**. At  $Q = 0$  (no load) the gain curve is tall and peaked; loading flattens the curve and pulls the peak down.

Frequency is always written normalised,  $f_n = f_{sw}/f_r$ , so that one family of curves serves every tank.  $M(f_n, Q)$  always means the gain of the right-hand circuit above.

## 2.5 The two resonances

# Single-Stage PFC LLC Converter Design

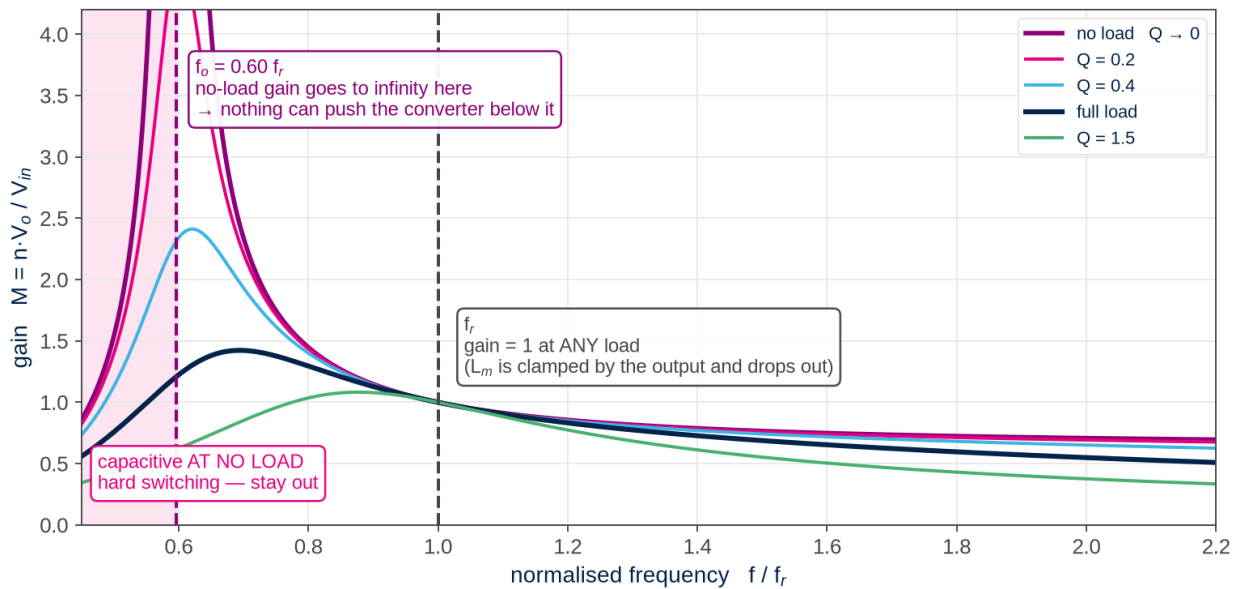
Every LLC has two. While the secondary conducts, the reflected output voltage clamps  $L_m$  and it drops out of the circuit, leaving the series resonance

$$f_r = \frac{1}{2\pi\sqrt{L_r C_r}} \quad (1)$$

At no load nothing clamps  $L_m$ , and the tank resonates with both inductances:

$$f_o = \frac{1}{2\pi\sqrt{(L_r + L_m) C_r}} \quad (2)$$

At  $f_r$  the gain is exactly 1 for any load, because the load term disappears. At  $f_o$  the no-load gain is unlimited. Between the two the tank can boost; above  $f_r$  it cannot. An LLC fed from a fixed dc bus runs close to  $f_r$  nearly all the time, and drops into that band only when the bus sags.



**Figure 5: The two resonances, and the band between them in which the tank can boost. The shaded edge is the no-load position of the capacitive boundary; under load it moves up.**

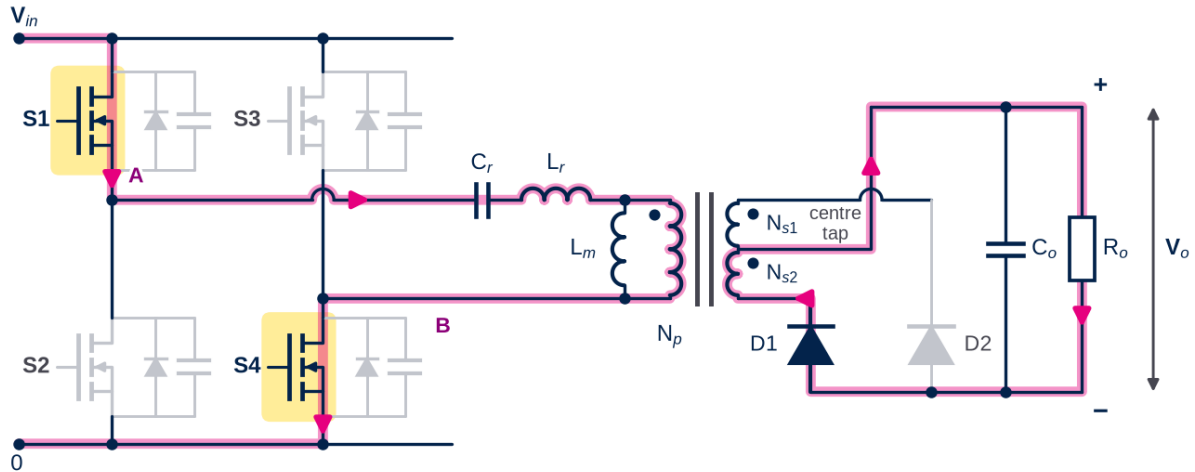
The two expressions are not much use without knowing what each one does to the current, because the current is where the difference shows.

First, the two things the circuit can be doing. Below resonance a half cycle contains both, in this order: power delivery first, then freewheeling, **with the same pair of switches on throughout**. At resonance the half cycle is power delivery only; above resonance the power delivery is cut short.

# Single-Stage PFC LLC Converter Design

## 1 POWER DELIVERY

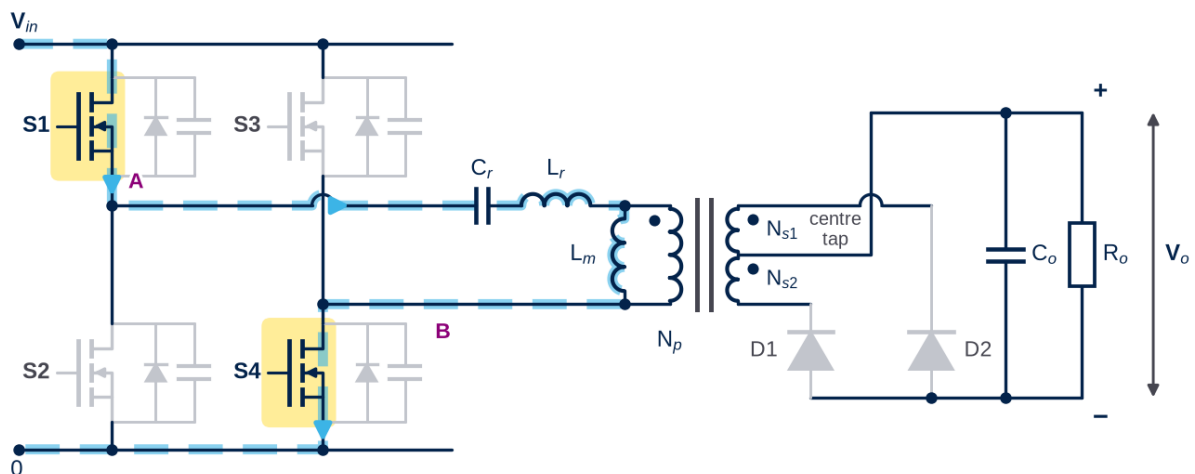
S1, S4 on · D1 conducts · lower half of  $N_s$



$L_m$  is clamped, so  $i_{Lm}$  is a straight ramp and  $i_{Lr} - i_{Lm}$  is the half sine that crosses.  
What is traced is the load component;  $i_{Lm}$  flows in  $L_m$  too, but it reverses inside this interval.

## 2 FREEWHEELING

S1, S4 STILL on · secondary off



$i_{Lr}$  has fallen to  $i_{Lm}$ . Every rectifier is off,  $L_m$  is free again, and the tank rings at  $f_o$ .

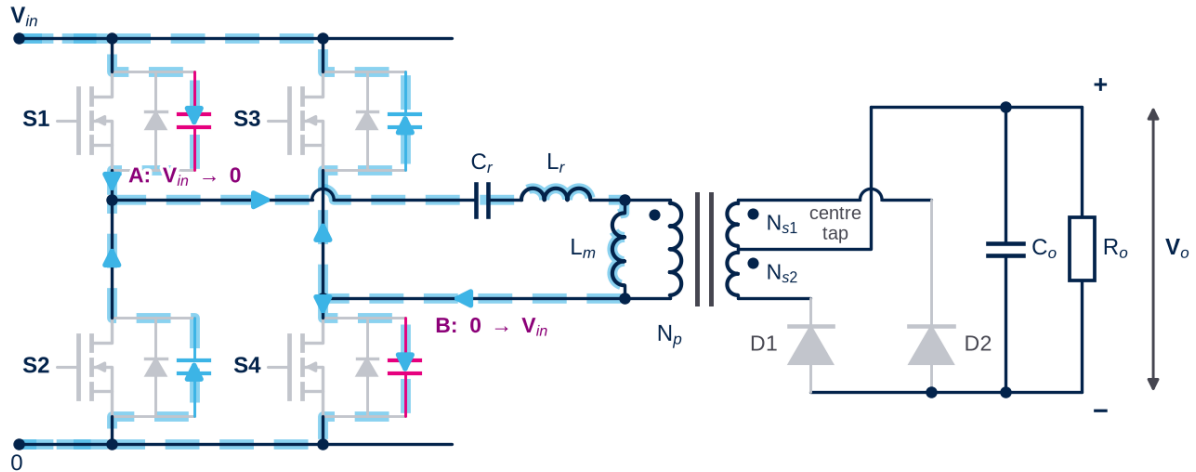
— load current     
 - - - magnetising current only     
  gated on  
 body diode conducting = the ZVS window     
  $C_{oss}$  charging,  $V_{ds}$  rising     
  $C_{oss}$  discharging

**Figure 6: One switching period, first half, steps 1 and 2.** 1 Power delivery —  $S_1, S_4$  are on, the resonant current exceeds the magnetising current and the difference passes through the transformer to  $D_1$  and the load.  $L_m$  is clamped to the reflected output, so this interval resonates at  $f_r$ . 2 Freewheeling — the resonant current has fallen to the magnetising current, nothing is left for the secondary, the rectifiers are off and  $L_m$  joins the resonance: this interval rings at  $f_o$ . The same pair of switches is still on. Each switch is drawn with its body diode and its  $C_{oss}$  beside it, greyed while they do nothing.

# Single-Stage PFC LLC Converter Design

## 3 DEAD TIME (a)

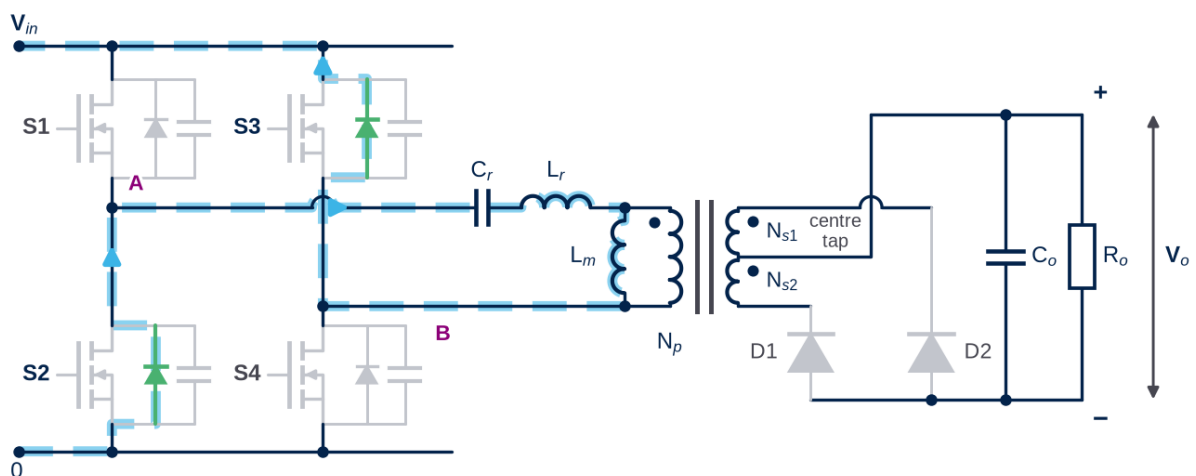
all four off ·  $C_{oss}$  swaps the midpoints over



All four channels are off, so the ONE tank current can reach the rails only THROUGH the four  $C_{oss}$ . It arrives at A from both, so A is pulled down; it leaves B into both, so B is pushed up. That swap IS the transition.

## 4 DEAD TIME (b) — ZVS

body diodes of S2 and S3 clamp



$V_{ds} = 0$  on S2 and S3 now, because their own body diodes are clamping them. Gate them on HERE.

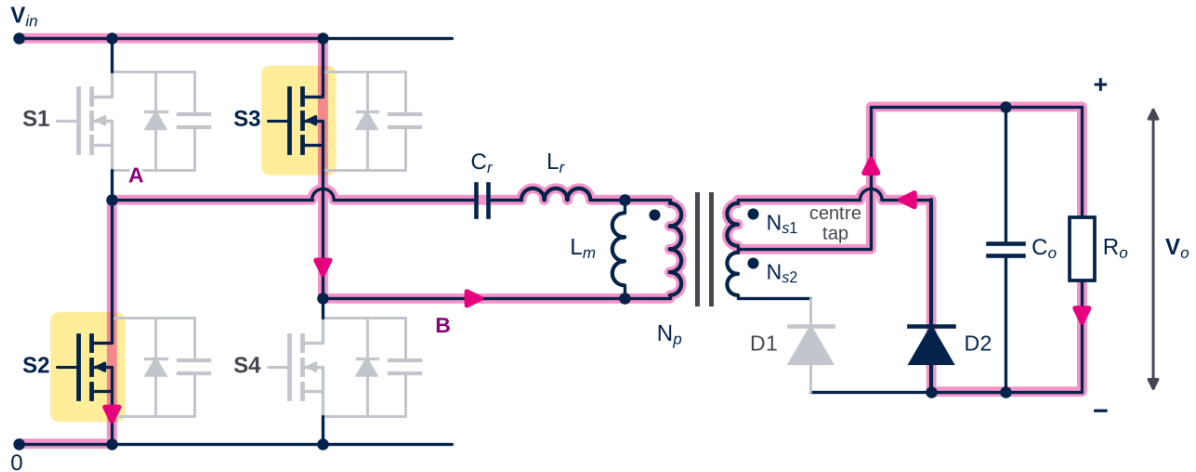
— load current    — magnetising current only     gated on  
 body diode conducting = the ZVS window    —  $C_{oss}$  charging,  $V_{ds}$  rising    —  $C_{oss}$  discharging

**Figure 7: Steps 3 and 4: the dead time. 3** All four switches are off, so the tank current — now the magnetising current alone — can only reach the rails through the four  $C_{oss}$ . It arrives at A from both, so A is pulled down, and leaves B into both, so B is pushed up: that swap is the transition. **4** The swing is complete and the body diodes of  $S_2, S_3$  clamp their  $V_{ds}$  at zero. This is the ZVS window: the gate that follows has to arrive inside it.

# Single-Stage PFC LLC Converter Design

## 5 POWER DELIVERY

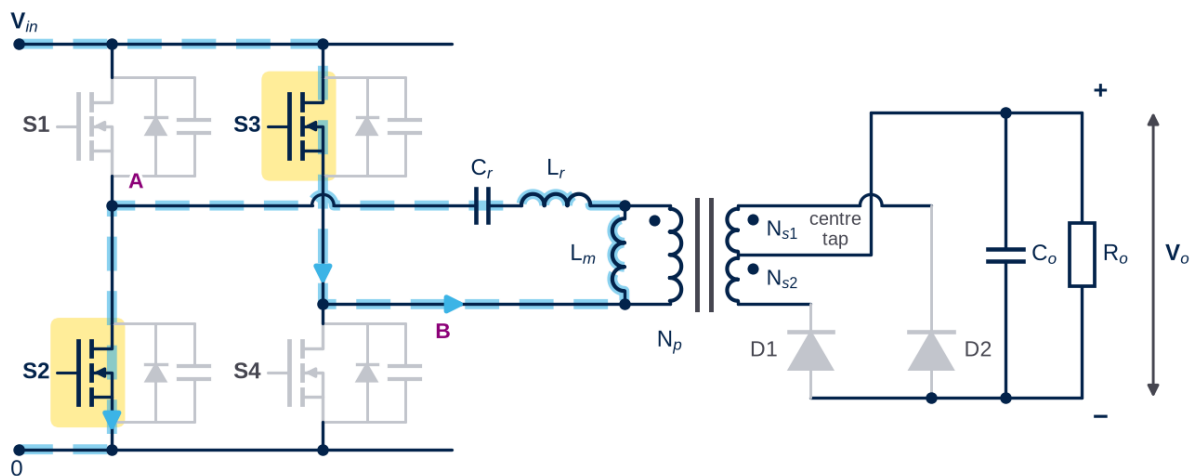
S2, S3 on · D2 conducts · upper half of  $N_s$



The mirror of 1. Just after turn-on the old current still runs back through the two channels. Again only the load component is traced -  $i_{Lm}$  reverses inside this interval as well.

## 6 FREEWHEELING

S2, S3 STILL on · secondary off



The mirror of 2, and the easy one to miss: it ends at the right-hand edge of a waveform plot.

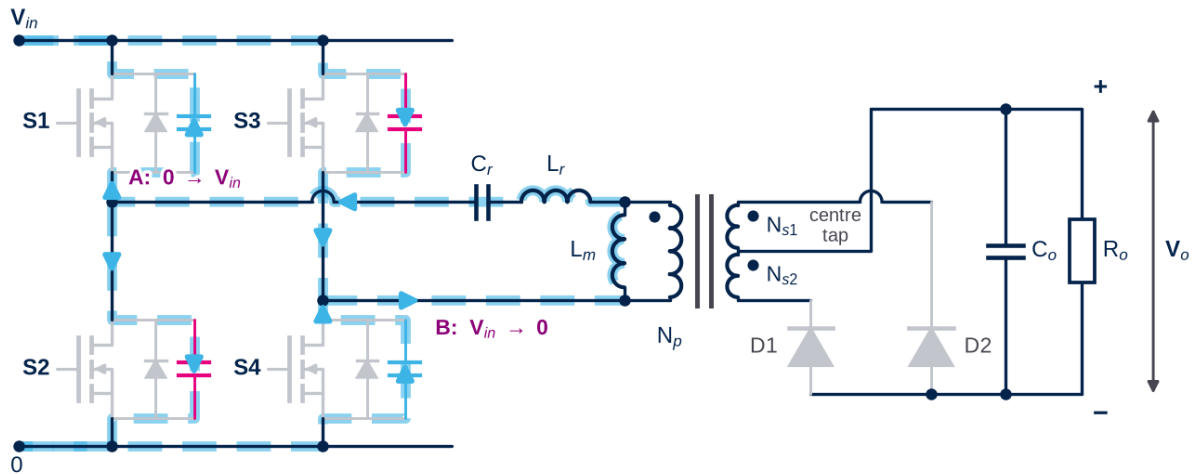
— load current      — magnetising current only       gated on  
 body diode conducting = the ZVS window        $C_{oss}$  charging,  $V_{ds}$  rising        $C_{oss}$  discharging

**Figure 8: Second half, steps 5 and 6, the mirror image of 1 and 2.  $S_2, S_3$  carry the power delivery (5, through  $D_2$ ) and then the freewheeling (6), again with the same pair on throughout.**

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## 7 DEAD TIME (a)

all four off ·  $C_{OSS}$  swaps the midpoints back

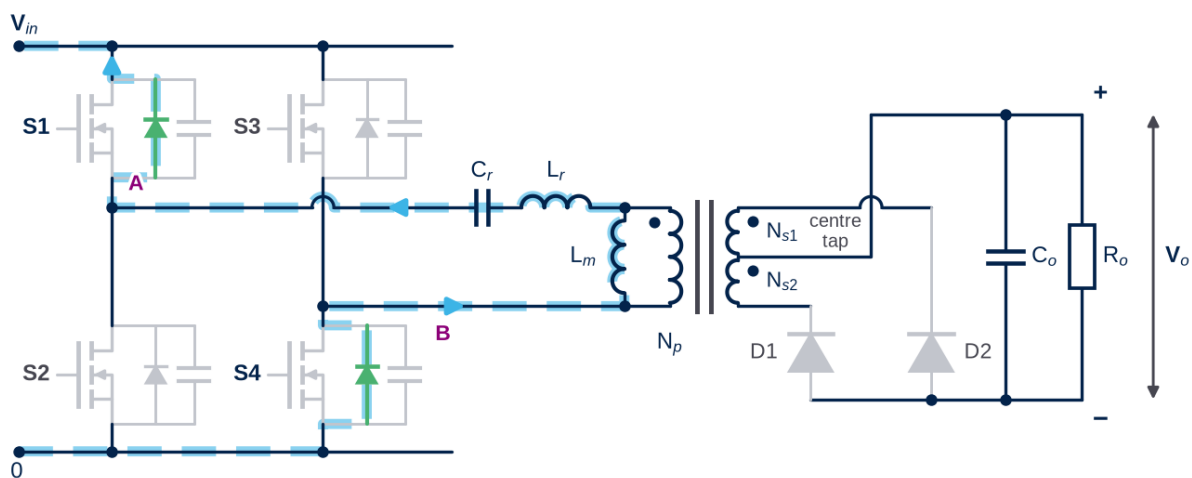


The mirror of 3: the current now arrives at B and leaves A, so B falls and A rises.

Through the capacitances again, never a channel. Magenta:  $C_{OSS}$  charging,  $V_{ds}$  rising; cyan: discharging.

## 8 DEAD TIME (b) — ZVS

body diodes of S1 and S4 clamp

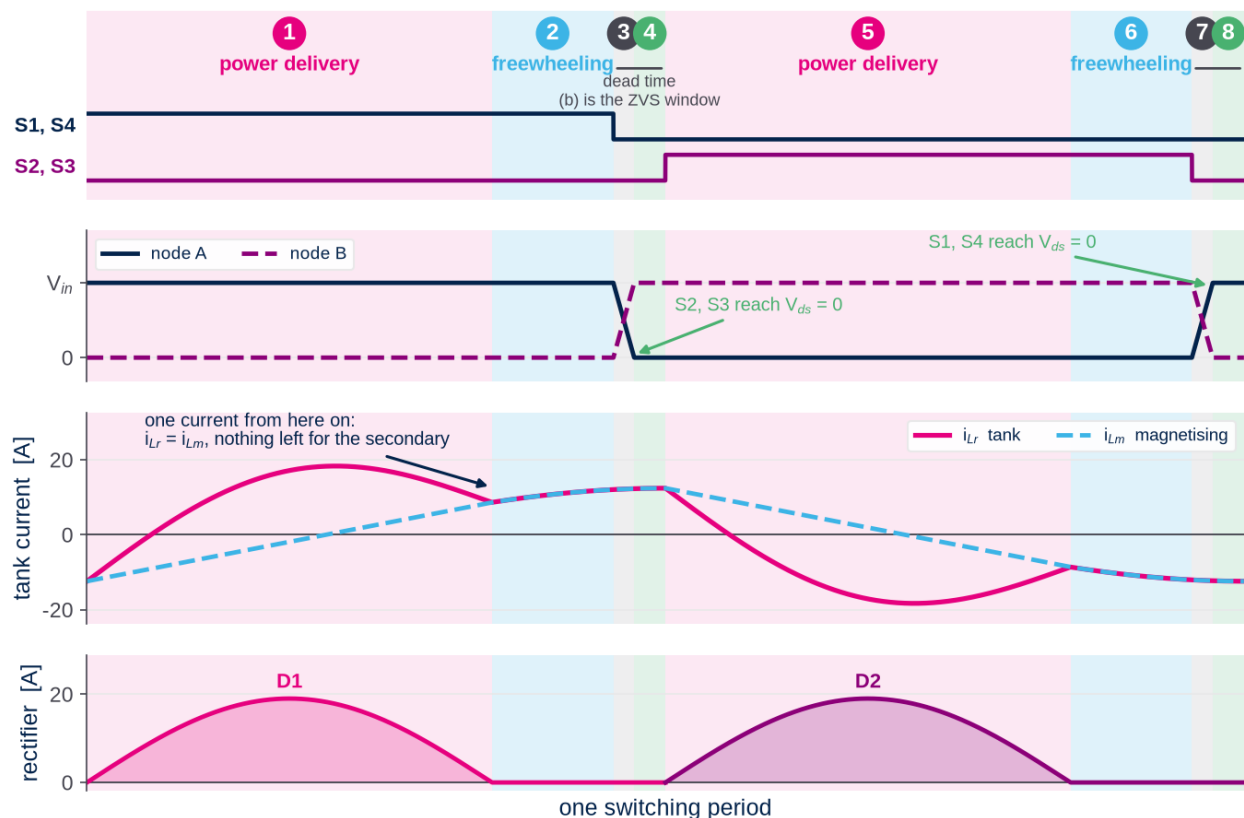


The same for S1 and S4, and the period closes. ZVS is in 4 and 8 - and in none of the other six.

█ load current      █ magnetising current only      █ gated on  
█ body diode conducting = the ZVS window      █  $C_{oss}$  charging,  $V_{ds}$  rising      █  $C_{oss}$  discharging

**Figure 9: Steps 7 and 8, the second dead time. The mid-points swing back (7) and the body diodes of  $S_1, S_4$  clamp (8), which is where  $S_1, S_4$  then close on zero volts and step 1 begins again. ZVS happens in steps 4 and 8, and in none of the other six.**

# Single-Stage PFC LLC Converter Design

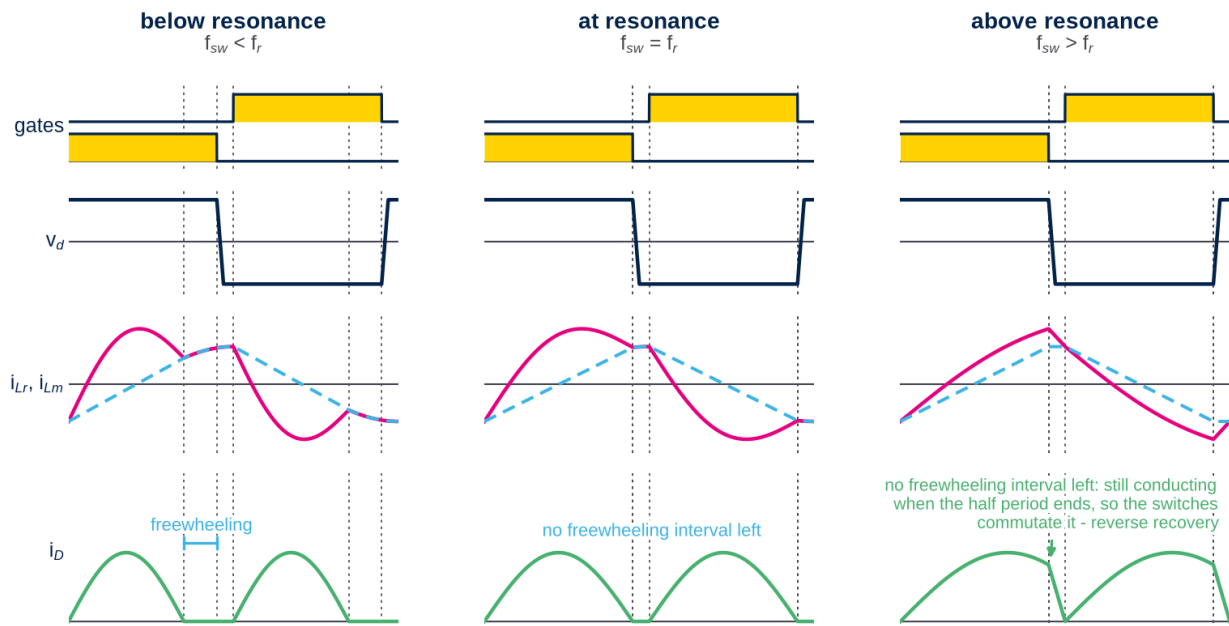


**Figure 10:** Where each step sits on the waveforms. The numbered bands are the eight panels above, in time order across one switching period: the gates, the two mid-point voltages, the tank and magnetising currents, and the two rectifier currents. Read a panel, then find its band here. Drawn below resonance at  $f_{sw} / f_r = 0.70$ , with the dead time wider than scale so that steps 3, 4, 7 and 8 have room to be seen; the current amplitudes are this design's own  $I_{Lr,pk}$  and  $I_{Lm,pk}$  at the low-line corner.

**Steps 1 and 2 are the two resonances**, and they are the two parts of *one* half cycle: the same pair of switches stays on through both. **ZVS happens in neither**. It happens in the dead time, steps 3–4 and 7–8, when both switches of a leg are off and the tank current carries the mid-point to the other rail — the step in  $v_d$  between the gate pulses in Figure 2.

**The word “freewheeling” is used for two different intervals**. Here it means steps 2 and 6: the secondary is off,  $L_m$  has rejoined the resonance, and the same pair of switches is still on. Other documents, among them Toshiba's *Resonant Circuits and Soft Switching* (2019), use the word for the **dead time**, when the current circulates in the body diodes. Both intervals carry current without delivering power, but the dead time exists at every frequency and steps 2 and 6 exist only below resonance. Check which one a source means before comparing numbers.

# Single-Stage PFC LLC Converter Design



**Figure 11: The three cases side by side. Gate signals, the bridge voltage, the resonant current  $i_{Lr}$  with the magnetising current  $i_{Lm}$  dashed over it, and the rectifier current. What changes across the three columns is how the resonant half period compares with the switching half period. Below resonance there is time left over, and it is spent freewheeling; at resonance the two coincide; above resonance the half period ends first and cuts off the rectifier current while it is still flowing.**

- **Below  $f_r$  (left).** The resonant half period is shorter than the switching half period.  $i_{Lr}$  finishes its half sine early and **lands on  $i_{Lm}$** . From that instant the two are one current and no power crosses to the secondary: that is the freewheeling interval.
- **At  $f_r$  (centre).** The two half periods are the same length.  $i_{Lr}$  meets  $i_{Lm}$  exactly when the bridge switches, and the rectifier current reaches zero at the same instant. This is the most efficient point an LLC has, and it is where a conventional design puts its nominal input.
- **Above  $f_r$  (right).** The switching half period ends first, so the resonant half sine is **cut off part way**:  $i_{Lr}$  is still well above  $i_{Lm}$  when the bridge switches. The primary switches turn off more current, and the rectifier is interrupted while it is still conducting: hard commutation on the secondary.

**That freewheeling interval is what  $f_o$  is.** While it lasts nothing clamps  $L_m$ , so the tank is  $L_r + L_m$  with  $C_r$ , ringing at  $f_o$ . It looks flat only because  $f_o$  is so much lower than  $f_r$ . Lower the switching frequency towards  $f_o$  and the interval grows until it fills the whole half period, and the converter delivers no power. So the two frequencies are the two ends of one range: at  $f_r$  the flat interval has just disappeared, at  $f_o$  it has taken over, and everything useful happens between them.

**Symbols are not standard across the literature.** Before using a manufacturer's application note alongside this one, check which frequency that document calls  $f_o$ . At least one widely used guide gives that name to the series resonance, which is the opposite of the meaning here.

A conventional LLC picks one side and stays there, usually just above  $f_r$  at nominal input, where the gain is close to 1 and the circulating current is smallest. Which side the converter designed here runs on is a different question. It is answered in Section 4.5, once the rest of the method is in place.

## 2.6 The gain function and the three regions

With the usual first-harmonic approximation the tank gain is

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$$M(f_n, Q, \lambda) = \frac{1}{\sqrt{\left(1 + \lambda - \frac{\lambda}{f_n^2}\right)^2 + Q^2 \left(f_n - \frac{1}{f_n}\right)^2}} \quad (3)$$

with  $f_n$ ,  $Q$  and  $\lambda$  as defined in the conventions. Two checks can be made on it by eye before trusting any curve drawn from it. At  $f_n = 1$  the second term under the root disappears and the first becomes 1, so  **$M = 1$  at any  $Q$** . That is the series resonance, and it is why every gain curve passes through the same point. **At no load** ( $Q = 0$ , so the second term is gone) raising  $f_n$  without limit only brings  $M$  down to  $1/(1+\lambda)$  and no further; that floor is the subject of Section 4.8. Under load the  $Q$  term grows with  $f_n$  and the gain keeps falling towards zero.

The curve has three regions. Below the capacitive/inductive boundary the tank is capacitive and the bridge hard-switches, which must never be allowed. Between that boundary and  $f_r$  the tank is inductive and boosts; above  $f_r$  it is inductive and bucks. In the figure the lower boundary is drawn for the full-load curve. At no load it would sit at  $f_o$ ; it moves up as load is added, and the next sections are about that.

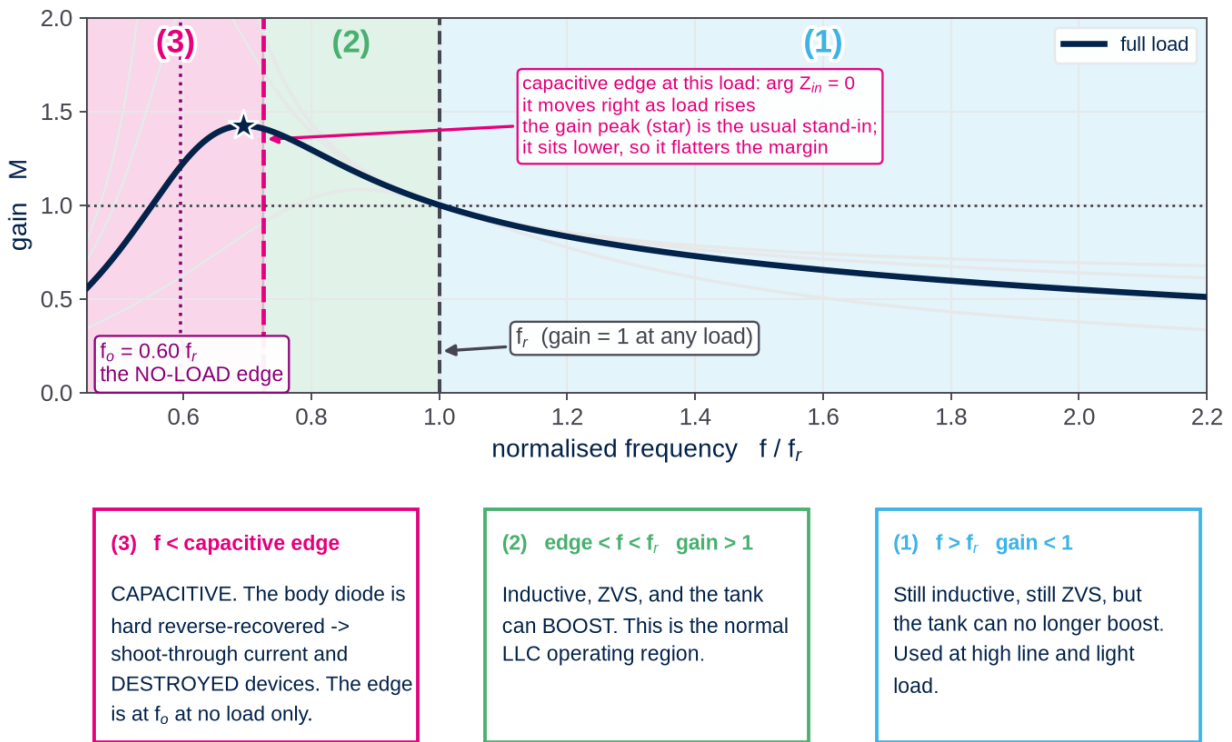


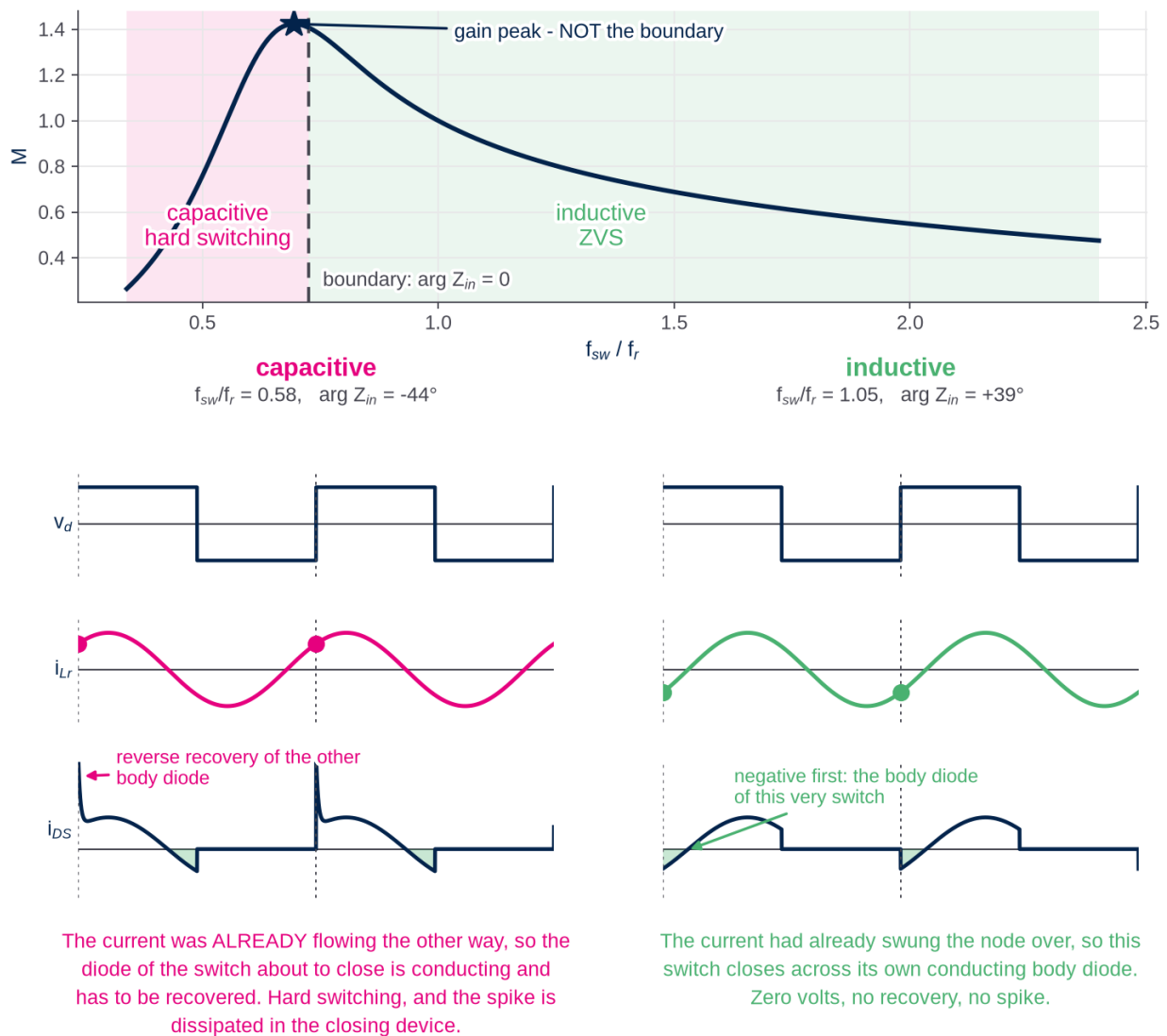
Figure 12: The three operating regions, shaded for the full-load curve drawn. Only the inductive ones are usable, and the boundary between them moves up as the converter is loaded. Region framing after ROHM TechWeb.

## 2.7 Capacitive and inductive, and why those words

The names have nothing to do with which component dominates. They describe **what the bridge sees**: the phase of the tank input current relative to the square wave that drives it. The tank is a series network. At low frequency  $C_r$  dominates and the current *leads*: the bridge is driving something that behaves like a capacitor. At high frequency the inductances dominate and the current *lags*: the bridge is driving something that behaves like an inductor. The word describes the load the bridge works into, not a part.

That phase is the whole of soft switching, which is why the distinction matters more than it sounds:

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**Figure 13: The same converter on either side of the boundary. Above: where the boundary sits on the gain curve. Note that it is not at the peak, but a little above it. Below, for each case: the bridge voltage  $v_d$ , the tank current at the phase the tank actually presents there, and the drain current of the switch that is closing. On the left the current leads and is positive at turn-on, and the drain current carries the recovery spike of the opposite body diode. On the right it lags and is still negative, so the switch closes on its own conducting body diode. The two drain currents are drawn to one scale.**

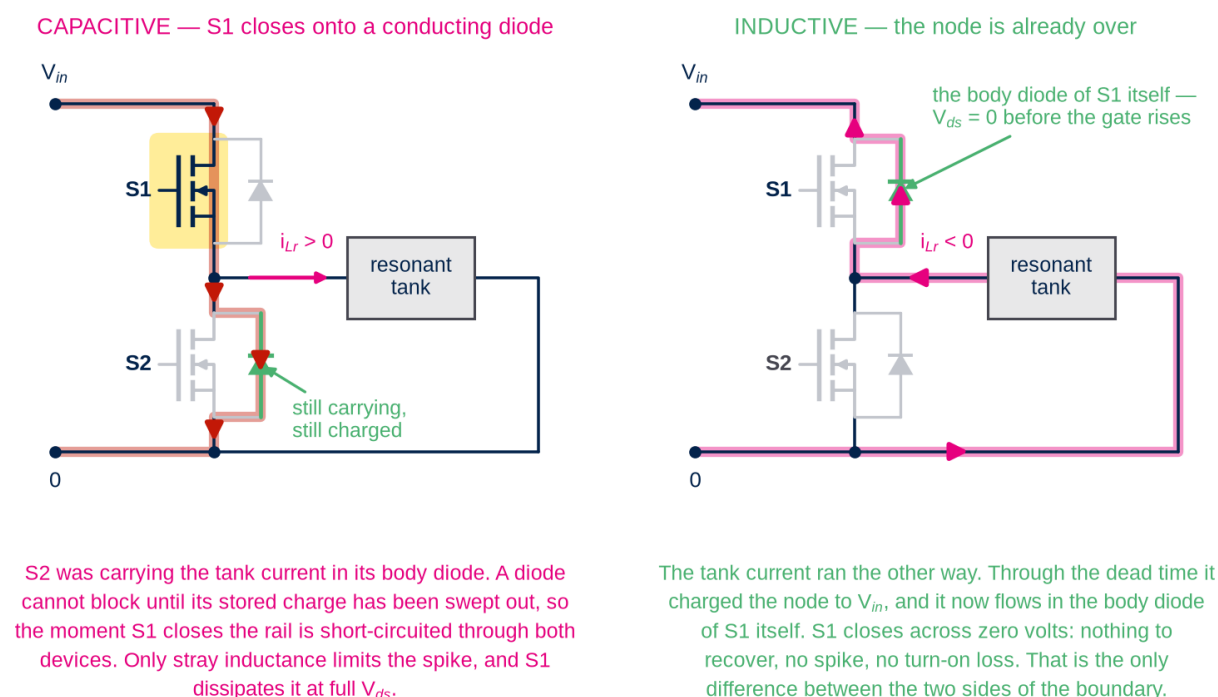
- **Inductive: the current lags.** When a switch turns off, the current is still flowing in the direction that pushes the bridge node towards the other rail. The dead time lets it do exactly that, and the next device turns on at zero volts. **ZVS.**
- **Capacitive: the current leads.** By the time the switch turns off, the current has already reversed, so it pushes the node *back* where it came from. The next device then turns on into the full rail voltage. Worse, the body diode of the device that was conducting is forced off with reverse recovery through a low impedance. **Hard switching, and the failure mode that destroys parts.**

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This is why the capacitive region is not a performance trade-off to be balanced against something else. It is a boundary the design stays on one side of, and the controller has a dedicated protection for the case where it does not.

## 2.8 Why a diode has to recover on one side and not on the other

The drain-current row above is the difference that reaches the hardware, so it is worth following one switching instant through in both cases. Take the leg alone: a high-side switch, a low-side switch, and the tank hanging off the mid-point. Whichever device has just been turned off, **the tank current does not stop** — it is the current in an inductor. The only question is what carries it during the dead time, and the answer decides everything.



**Figure 14: One switching instant in the same leg, either side of the boundary. Left: the tank current is already positive, so it is flowing in the body diode of the low-side device; when the high-side device is gated on, the rail is short-circuited through both until that diode recovers. Right: the tank current is negative, it has carried the mid-point up to the rail during the dead time, and the closing device finds its own body diode already conducting at zero volts. After ON Semiconductor AN-4151.**

- **Inductive.** At turn-off the current is flowing in the direction that carries the mid-point towards the *other* rail. It charges one  $C_{oss}$  and discharges the other, the node arrives, and the body diode of the incoming device picks the current up and holds the node there. That device is then gated on with  $V_{ds}$  already at zero. When its channel takes over, the current in its own body diode falls away with no reverse voltage applied to it, so **there is nothing to recover**.
- **Capacitive.** At turn-off the current has already reversed. It pushes the mid-point back onto the rail it came from, so the body diode of the device that just turned off picks it up and clamps the node there. **The node never moves.** The incoming device is then gated on across the full rail, straight onto a conducting diode.

A diode carrying forward current holds stored charge in its junction, and it cannot block reverse voltage until that charge has been swept out. For as long as that takes, both devices in the leg are conducting and the input rail is short-circuited through them. Three things follow, and all three are bad:

# Single-Stage PFC LLC Converter Design

- The current is limited only by the stray inductance of the loop, so it is a **spike, not a current** — many times the tank current, as the drain-current row shows.
- It flows in the incoming device while that device still stands at the full rail voltage, so it is dissipated there, in a few tens of nanoseconds, every cycle.
- When the diode finally does snap off, the  $di/dt$  in the loop inductance appears across it as overshoot, which is what actually breaks the part.

**This is the reason the AN asks for a fast-recovery body diode on the primary** (Section 6.24). A superjunction device stores a great deal of charge in a body diode that was never designed to be commutated, and in a full bridge that diode conducts before every turn-on even when the converter is behaving. It is also the reason the oscillator floor exists:  $f_{Min}$  above  $f_o$  is what keeps the converter out of the left-hand picture, and the anti-capacitive protection is what catches it if that fails.

## 2.9 The two boundaries are not the same boundary

This is the point that is most easily confused, because both pairs of words describe positions on the same frequency axis. They are different lines on it.

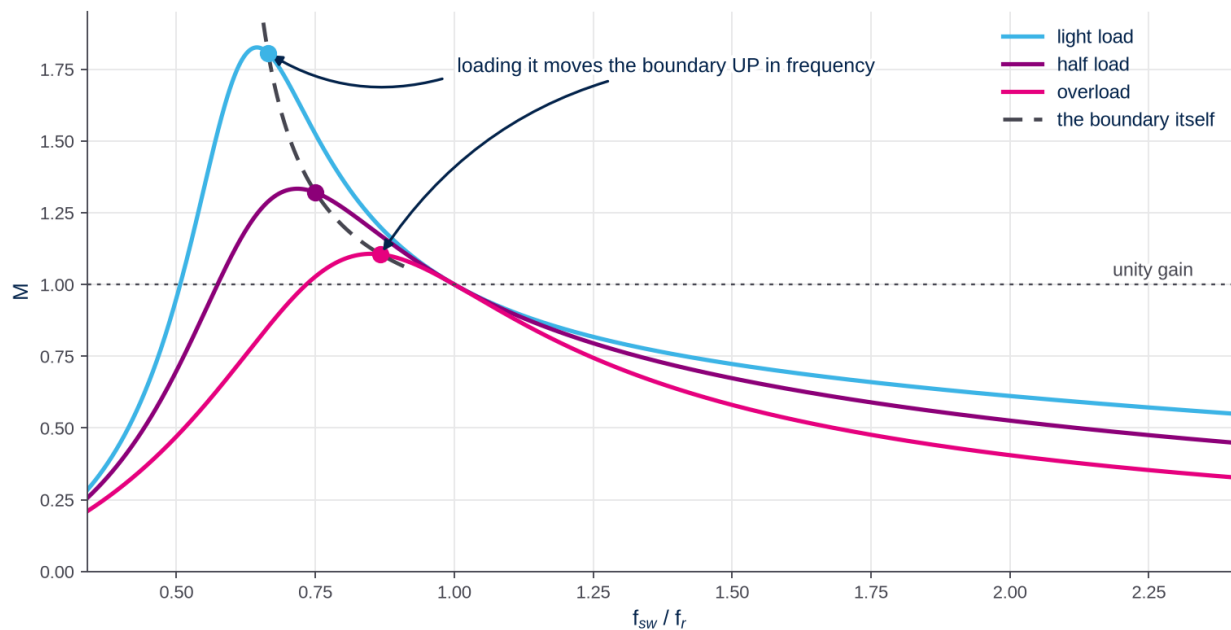
|                       | capacitive / inductive                       | below / above resonance                                |
|-----------------------|----------------------------------------------|--------------------------------------------------------|
| the boundary          | where $\arg Z_{in} = 0$                      | $f_r$ , the series resonance                           |
| does it move?         | <b>yes — with load</b>                       | no, it is fixed by $L_r$ and $C_r$                     |
| what it decides       | ZVS or hard switching, on the <b>primary</b> | boost or buck, and ZCS or not, on the <b>secondary</b> |
| which side is allowed | inductive only, always                       | both sides are legitimate                              |

**Table 1: Two classifications, two boundaries, two consequences.**

The peak-gain frequency sits between  $f_o$  and  $f_r$ , and it **rises as the converter is loaded**. At no load it coincides with  $f_o$ ; at full load it has moved some way up towards  $f_r$ . So the capacitive region is not a fixed band below  $f_o$ . It grows as load is added, and a frequency that is safely inductive at light load can be capacitive at overload.

**The edge is not quite the gain peak.** The tank is capacitive while the phase of its input impedance is negative, so the edge is where  $\arg Z_{in} = 0$ , and that sits a little *above* the peak of the gain curve. The peak is the usual stand-in because it needs only the gain curve, but it is optimistic in the dangerous direction: it calls a band inductive while the bridge is still hard-switching. Section 7.5 gives both frequencies for the worked design; the verdict there comes from Section 6.5, which measures the condition directly as the time the tank current takes to reach zero after the bridge changes state.

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**Figure 15:** The boundary is not fixed. Loading the converter pushes it to a higher frequency, so a frequency that is inductive at light load can be capacitive at overload. The dashed curve is the boundary itself, traced out as  $Q$  varies.

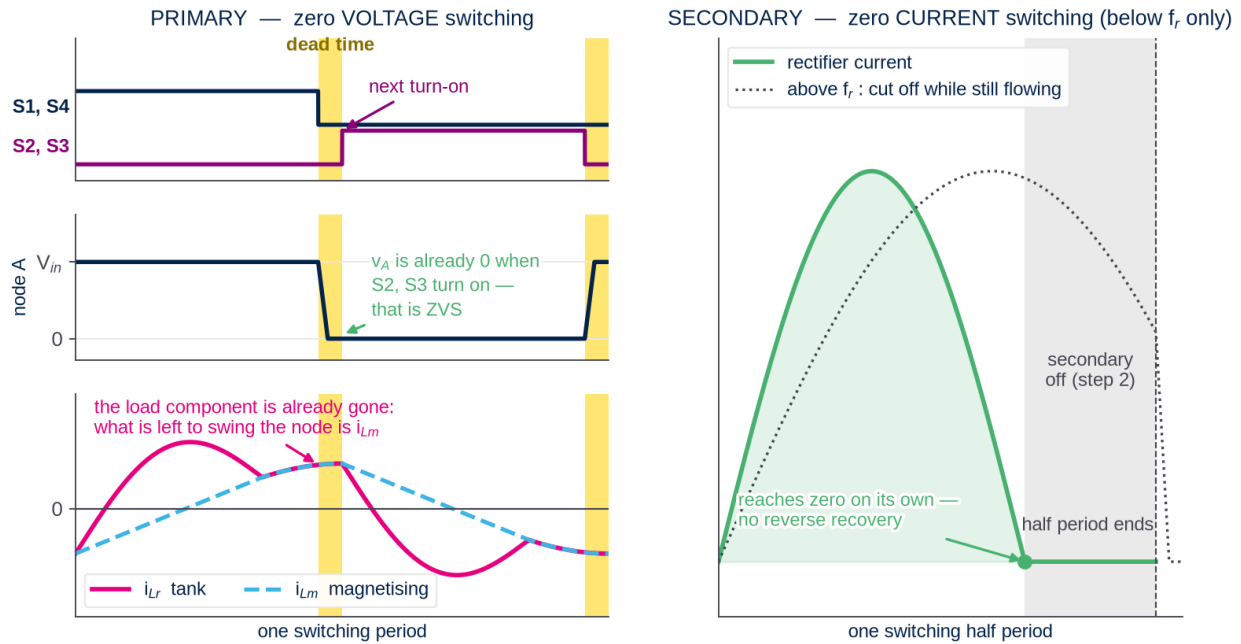
- A converter can be **below  $f_r$  and still inductive**. That is the ordinary boosting region, and it is where an LLC normally runs.
- It can be **below  $f_r$  and capacitive**, which is the fault. Going below  $f_r$  is not the danger; going below the boundary is.
- It cannot be **above  $f_r$  and capacitive**, because the boundary is always below  $f_r$ . Everything above  $f_r$  is inductive at every load.

**What this means for the design rule.** Keep the oscillator floor  $f_{Min}$  above  $f_o$ . That is the *no-load* position of the boundary, so it is the weakest form of the requirement: necessary, but not enough on its own. The real check is the ZVS sweep in Section 6.5, which is run under load and at every phase of the line cycle. Behind that, the controller has an anti-capacitive protection.

## 2.10 ZVS and ZCS are not the same thing

Both are ways of switching a device while one of its two quantities, voltage or current, is zero, so that their product, the loss, is zero. Here they apply to different devices, for different reasons, and only one of them is guaranteed.

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**Figure 16: Zero-voltage switching on the primary and zero-current switching on the secondary. Different devices, different mechanisms, different conditions.**

- **ZVS, zero-voltage switching, on the primary switches.** The device turns on with no voltage across it. It works because the tank current is still flowing when the previous device turns off, and that current discharges the bridge node during the dead time. It needs **inductive operation**, that is, anywhere above the capacitive/inductive boundary. So it is available on both sides of  $f_r$ , and  $f_o$  is that limit only at no load. It is required everywhere, and Section 6.5 says how much margin this design has.
- **ZCS, zero-current switching, on the secondary rectifiers.** The device turns off with no current through it, so a diode is not forced out of conduction and a synchronous rectifier has no reverse recovery. It happens because the resonant current reaches zero by itself, which only occurs **below  $f_r$** . Above  $f_r$  the rectifier is cut off while it is still conducting, and reverse recovery returns.

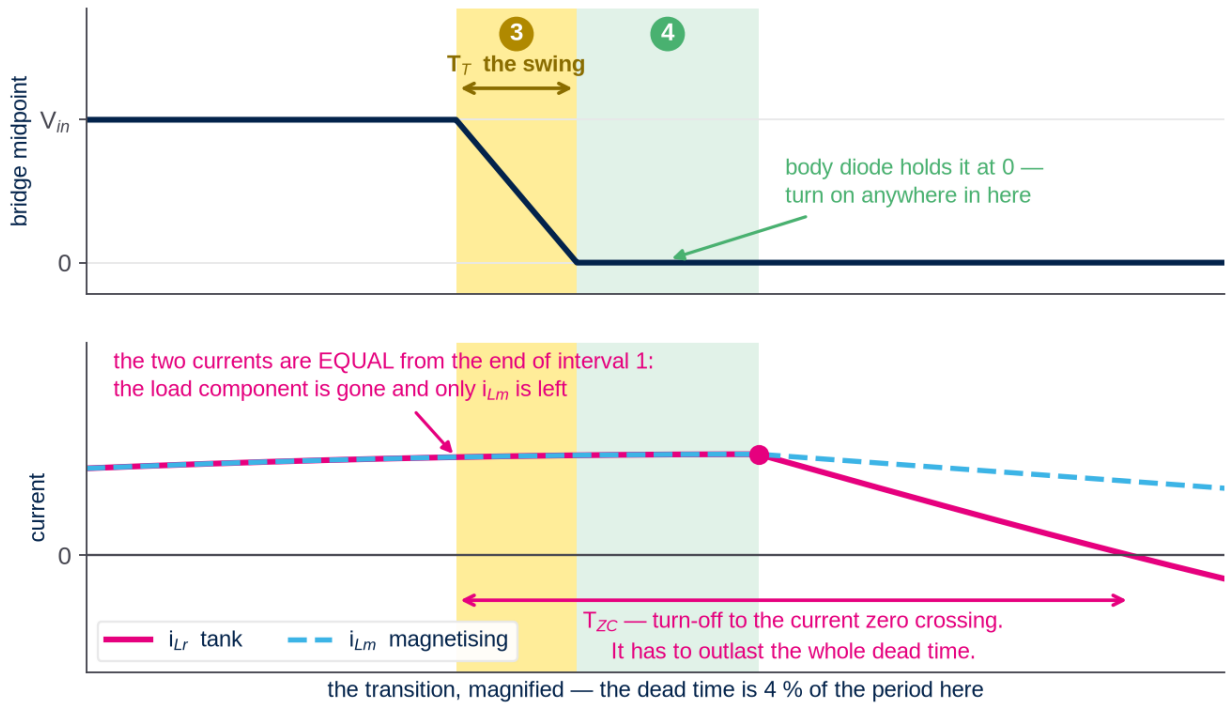
**So ZCS is not a property of the circuit but of the operating point.** Below  $f_r$  the secondary rectifiers turn off at zero current and reverse recovery never appears; above  $f_r$  they do not, and it does. A converter that stays on one side settles the question once. A converter whose input moves far enough to cross  $f_r$  has to be checked on both sides: its secondary body diode and synchronous-rectifier dead time matter whenever it runs above  $f_r$ .

## 2.11 Sizing the dead time so that ZVS actually happens

ZVS is what makes an LLC efficient, and it is a charge problem rather than a voltage problem. During the dead time the magnetising current must move enough charge to swing the bridge node across the rail before the opposite device turns on. The design quantity is  $T_{ZC}$ , the time the tank current takes to reach zero after the bridge transition, and the requirement is simply that it must be longer than the dead time:

$$T_{ZC} > t_D \quad (4)$$

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**Figure 17: How ZVS happens: the magnetising current, not the load current, discharges the bridge node during the dead time. Two conditions are needed: the current must still be flowing when the gates drop ( $T_{ZC} > t_D$ ), and the swing must finish inside the dead time.**

**Use the output charge, not the small-signal capacitance.** A MOSFET datasheet quotes  $C_{oss}$ ,  $C_{o(er)}$  and  $C_{o(tr)}$ , and they differ by three to five times. ZVS is a charge question, so  $C_{o(tr)}$  (equivalently  $Q_{oss}$ ) is the one to use. Sizing the dead time from the headline  $C_{oss}$  is optimistic by a large factor.

## 2.12 Not a flyback, and why that changes the core

Almost everybody meets a transformer first in a flyback, and almost every habit formed there is wrong here. Two of them cost money: **why a core area matters at all in a converter that stores none of the energy it delivers**, and **why the saturation test current is nothing like the peak winding current**. Both come from one difference, which is when the two windings conduct.

**In a flyback they never conduct together.** The secondary polarity dot is inverted, so the rectifier can only conduct while the switch is off. At every instant exactly one winding is carrying current, and therefore at every instant the whole winding current is magnetising current. Energy goes into the core during the on-time and comes out of it during the off-time, and every joule that reaches the output was first stored in the core. The flux follows the *current*, the core is gapped so that it can hold that energy without saturating, and  $A_e$  is sized from the peak current.

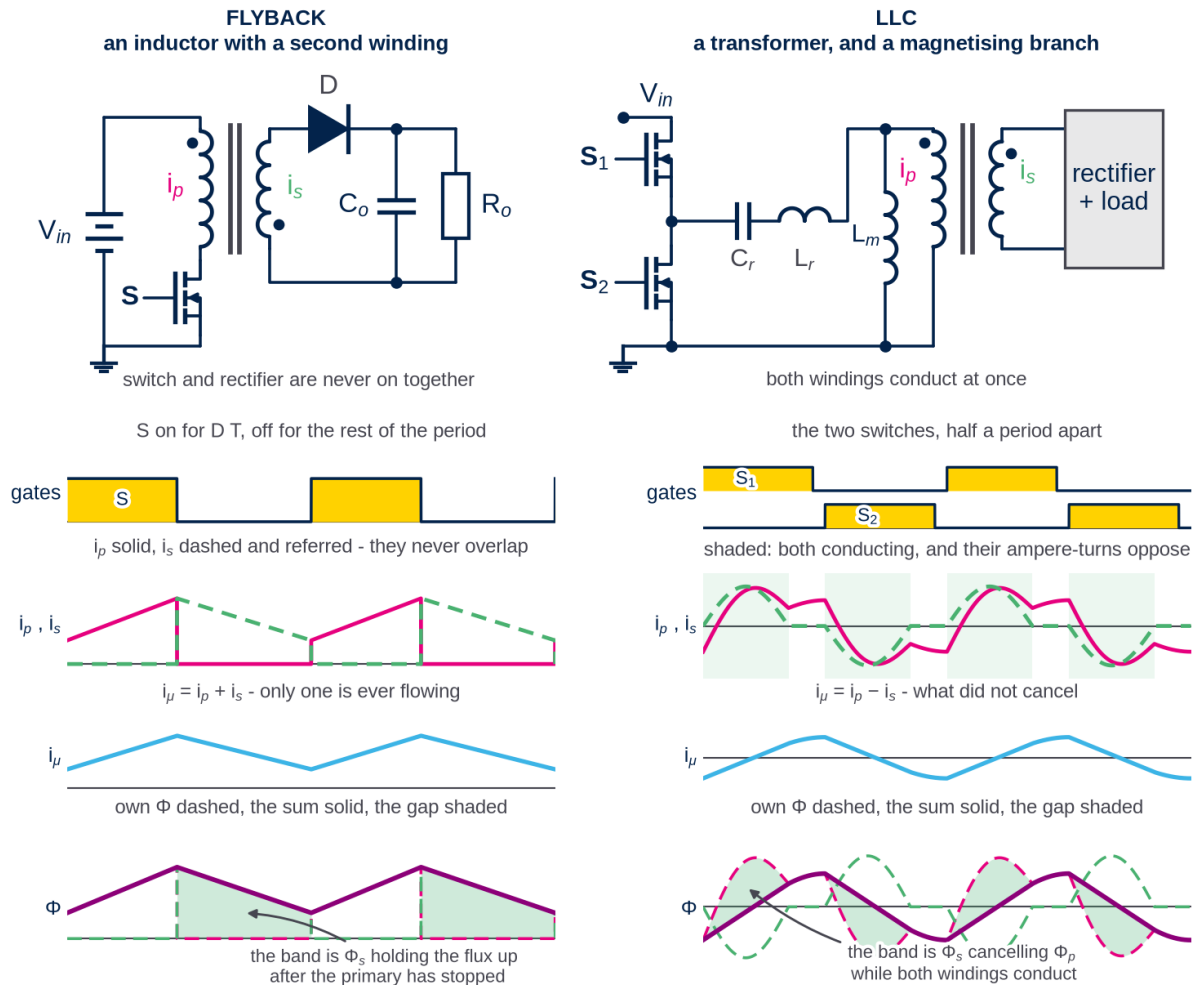
**In an LLC they conduct together.** While the secondary is rectifying, both windings are carrying current and their ampere-turns oppose. Only what is left after the subtraction magnetises the core:

$$N_p i_p - N_s i_s = N_p i_\mu \quad (5)$$

The load current passes *through*, by transformer action; it never enters the core as stored energy. What the core does hold is the magnetising energy  $\frac{1}{2} L_{\mu} i_\mu^2$ , and in this converter that is not a by-product but the mechanism:  $i_\mu$  is the current that charges and discharges the bridge node during the dead time, which is to say it is what makes ZVS possible (Section 2.11). An LLC transformer is gapped too, but the gap is there to hold  $L_{\mu}$  at the value the tank

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asked for, not to store what goes to the load.



**Figure 18: The two transformers from the same four rows. Left: the switch and the rectifier are never on together, so whichever winding conducts carries all of the magnetising current, and the flux is a one-sided ramp that follows it. Right: both windings conduct at once, their ampere-turns oppose, and only the difference  $i_\mu = i_p - i_s$  magnetises the core. The  $i_s$  traces are referred to the primary, with the sign each polarity dot gives. In the bottom row the dashed curves are what each winding would drive on its own, the solid one is their sum, and the shaded band is what the secondary did to the core: taken away at the same instant on the right, handed over afterwards on the left.**

**“An LLC transformer stores no energy” is a slogan, not a statement.** It stores the magnetising energy, twice every switching period, and the design leans on it. What it does not store is the energy delivered to the load.

So why does  $A_e$  come into it at all? Because Faraday's law does not ask whether anything is being stored:

$$B(t) = \frac{1}{NA_e} \int v dt \quad (6)$$

A winding with a voltage across it makes flux; flux in a finite area is a flux density; and ferrite saturates at a flux density. That is equally true of a flyback, an LLC, a mains transformer and a plain choke. **What differs is only the input to the calculation.** In a flyback the volt-seconds are the input voltage times the on-time, which is another way of writing the peak current. In an LLC the conducting secondary clamps the winding to the output for a resonant half period, so the volt-seconds — and with them the flux — are set by the **output voltage alone**.

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Section 6.11 turns that into an equation with neither the input voltage nor the load in it.

Three consequences follow, and they are worth having in mind long before a core is chosen:

- **The flux does not rise with load.** Twice the output current is twice  $i_s$ , and the primary has to carry that extra current as well — so both grow together and their difference,  $i_p$ , does not move at all. ( $i_p$  itself does not simply double: it is  $i_p$  plus the reflected load current, and only the second term grows.) A core that is adequate at full load is adequate in overload; what overload threatens is the copper and the semiconductors, not the core.
- **The flux does rise with output voltage.** Anything that lets  $V_{out}$  climb raises the flux in proportion, so the ceiling to design against is the over-voltage threshold and not the nominal output.
- **The flux does not fall with switching frequency,** below resonance. The rectifier clamps the winding for a resonant half period  $T_r/2$  whatever  $f_{sw}$  is doing, which is why  $f_r$  and not  $f_{sw}$  appears in the flux equation.

The same difference decides what the vendor is asked to test. A DC-overlap measurement is made with the secondary open, which removes the cancellation and puts the transformer back into the flyback condition — all of the test current is magnetising current. Section 6.19 takes that to a number.

## 2.13 How an LLC is normally designed

The sequence below is the standard one, and every step exists because of something already established above. It is worth having in mind before Section 6, which follows the same order and departs from it in exactly two places.

- **1 Turns ratio from the nominal point.** Choose  $n$  so that the converter runs at or near  $M = 1$  at nominal input, that is, at  $f_r$ . That is where the circulating current is lowest and where the converter should normally run.
- **2 The gain range the tank must cover.** From the input range and the output tolerance, find  $M_{min}$  and  $M_{max}$ .  $M_{max}$  is needed at minimum input and full load,  $M_{min}$  at maximum input and minimum load.
- **3 Load resistance referred to the primary,  $R_{ac}$ ,** which gives  $Q$  its meaning at full load.
- **4  $m$  (equivalently  $\lambda$ ) and  $Q$ , together, from the peak-gain chart.** They are not independent: for each  $m$  there is a curve of attainable peak gain against  $Q$ , and the design has to sit under that curve with margin. A larger  $m$  gives less circulating current and less peak gain; a smaller  $m$  gives the opposite.
- **5 The component values.**  $Q$  and  $R_{ac}$  give  $Z_0$ ;  $Z_0$  with the chosen  $f_r$  gives  $C_r$  and  $L_r$ ;  $m$  then gives  $L_m$ .
- **6 Verify.** Check that the gain at the worst corner is still under the peak with margin, that ZVS holds at the worst point rather than at the nominal one, and that the currents and the flux are ones real parts can carry.

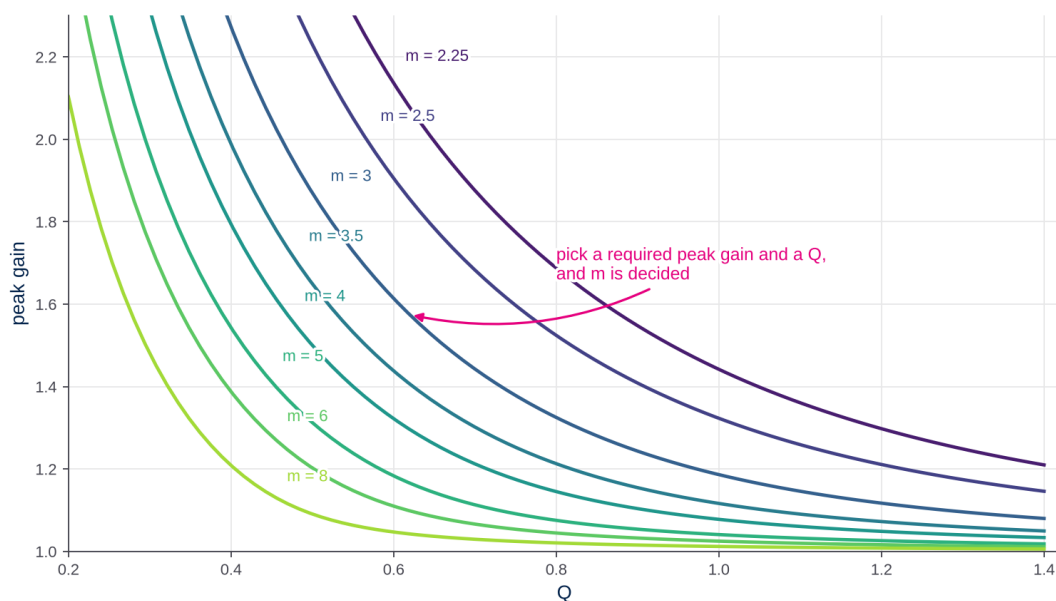


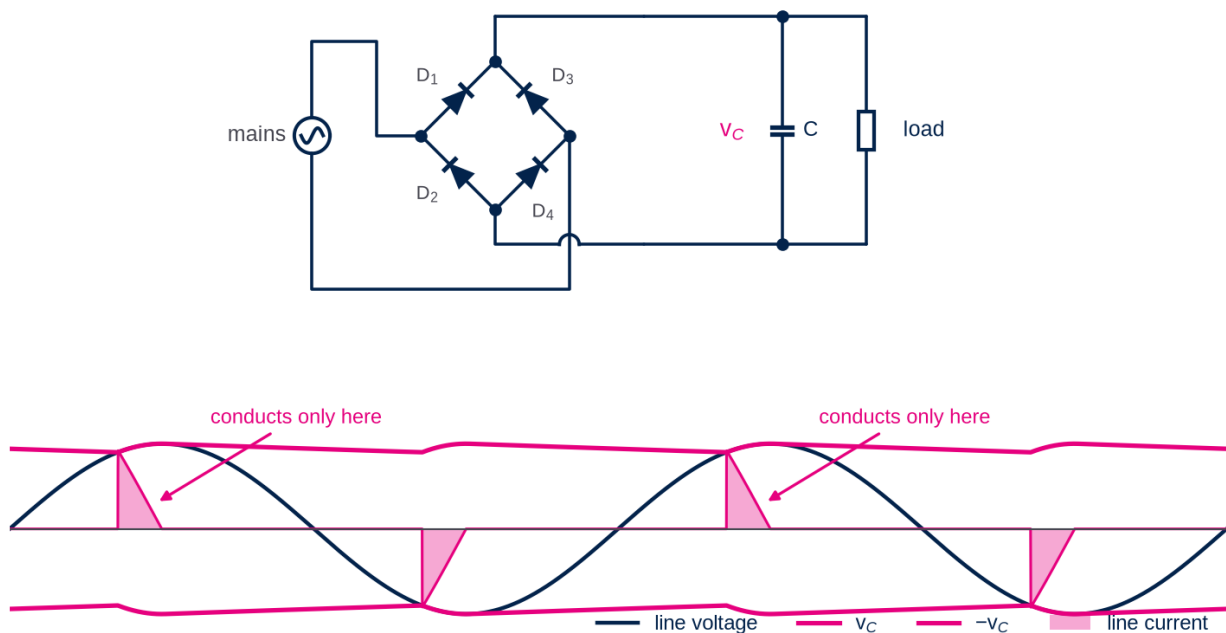
Figure 19: Step 4, and the reason  $m$  and  $Q$  cannot be chosen separately: the peak gain a tank can reach depends on both. Choosing a required gain and a  $Q$  leaves only a band of usable  $m$ .

# Single-Stage PFC LLC Converter Design

**That is the first half.** Everything above is the LLC on its own: a tank, a gain curve, and a frequency that moves the operating point along it. None of it involves the mains. The rest of this section is the other half: what a power factor corrector is for, and how the ordinary one works. It can be read without reference to any of the above. Section 3 is where the two halves are put together.

## 2.14 What power factor correction is

Start with what happens if nothing is done. A bridge rectifier feeding a capacitor holds the bus near the peak of the mains, so the diodes can only conduct during the short window in which the mains voltage is above the capacitor voltage. All the charge the load will use over the whole cycle has to arrive inside that window.



**Figure 20: A capacitor-input rectifier and what it draws.  $v_C$  stays near the crest, so the diodes conduct only in the narrow windows where the mains is above it, and the current inside them is correspondingly tall.**

All the consequences follow from that one fact. The peak current is several times what a resistor of the same average power would draw, so the wiring, the fuse, the bridge and the source all have to be sized for it. And a pulse train that narrow is mostly harmonics.

## 2.15 Power factor, and why it is not the same as distortion

**Power factor** is the ratio of real power to apparent power:

$$PF = \frac{P}{V_{rms} I_{rms}} = \cos \varphi_1 \times \frac{I_{1,rms}}{I_{rms}} \quad (7)$$

It has two independent factors, and they fail for different reasons. **Displacement** is the phase between the voltage and the fundamental of the current; it is what a motor gets wrong. **Distortion** is how much of the rms current is at the fundamental at all; it is what a rectifier gets wrong. The current in the figure above is roughly in phase, so its displacement factor is near 1, and it still scores only about 0.6, because the rest of the rms is harmonics.

The related figure of merit is **total harmonic distortion**, the harmonic content as a fraction of the fundamental:

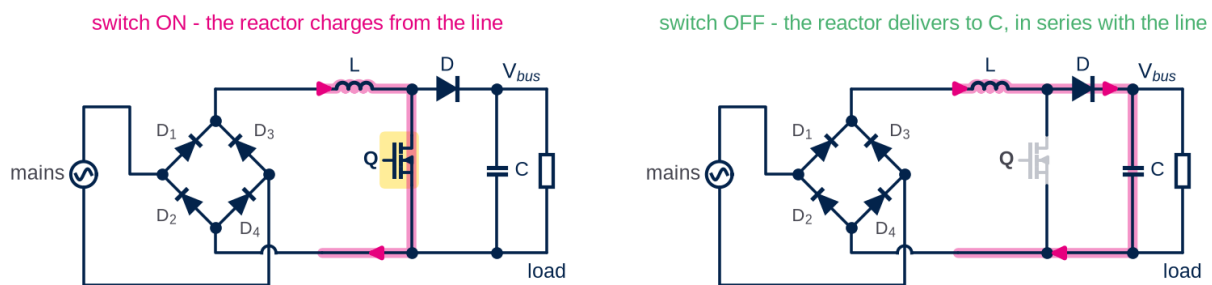
# Single-Stage PFC LLC Converter Design

$$THD = \frac{\sqrt{I_{rms}^2 - I_{1,rms}^2}}{I_{1,rms}}, \quad \frac{I_{1,rms}}{I_{rms}} = \frac{1}{\sqrt{1 + THD^2}} \quad (8)$$

**The two are not interchangeable.** A converter can hold PF above 0.99 and still fail a harmonic limit, because the standard limits *individual* harmonics in amperes rather than one ratio. Harmonics do not deliver power; they heat the neutral, saturate distribution transformers and disturb other equipment on the same supply, which is why they are regulated. IEC 61000-3-2 sets the limits, and the limit depends on the class. A mains rectifier of this kind is Class D: exempt below 75 W, and held to a milliamp-per-watt line above it. **But Class D only reaches 600 W**; a supply above that falls under the absolute Class A limits. Settle the class first.

## 2.16 How a corrector fixes it

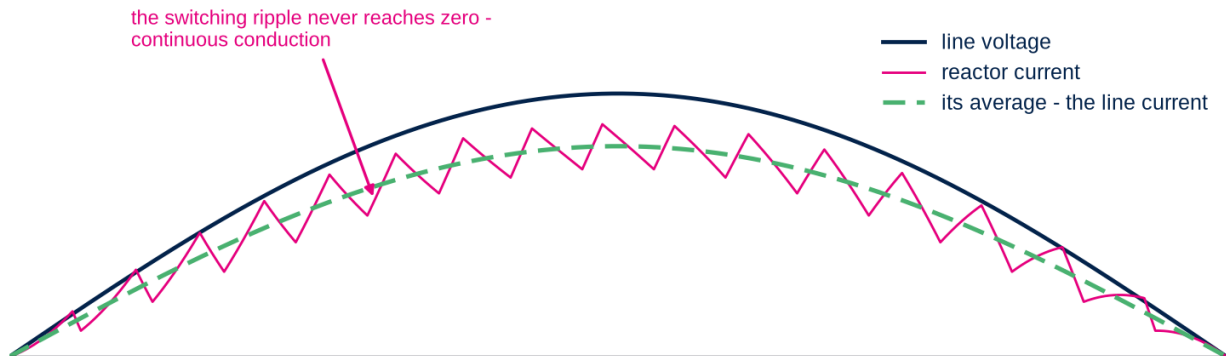
Force the input current to follow the input voltage, and the converter looks like a resistor to the mains: both factors go to one at once. The standard implementation is a **boost converter placed directly after the bridge**, switching fast enough that the mains looks constant within one switching cycle.



**Figure 21: A boost corrector, with the current path traced for each half of the switching period: the reactor charges from the line while the switch is on, and delivers to the bus in series with the line while it is off.**

- **Why a boost.** The input is a rectified sine that passes through zero, so the stage must be able to step up by an unlimited ratio near the zero crossing; only a boost can. Its input current is also continuous, which is what makes shaping it possible at all.
- **The inner loop** regulates the inductor current to a reference. It has to be fast compared with the switching frequency.
- **The outer loop** regulates the bus voltage, and it is deliberately made **slower than  $2f_i$** . If it were fast, it would fight the ripple on the bus, modulate the current level within the line cycle, and flatten the very shape the corrector exists to make. A slow voltage loop is not a compromise here; it is a requirement.
- **The multiplier** joins them: the shape comes from the mains, the level from the output. That is the whole control law.

# Single-Stage PFC LLC Converter Design

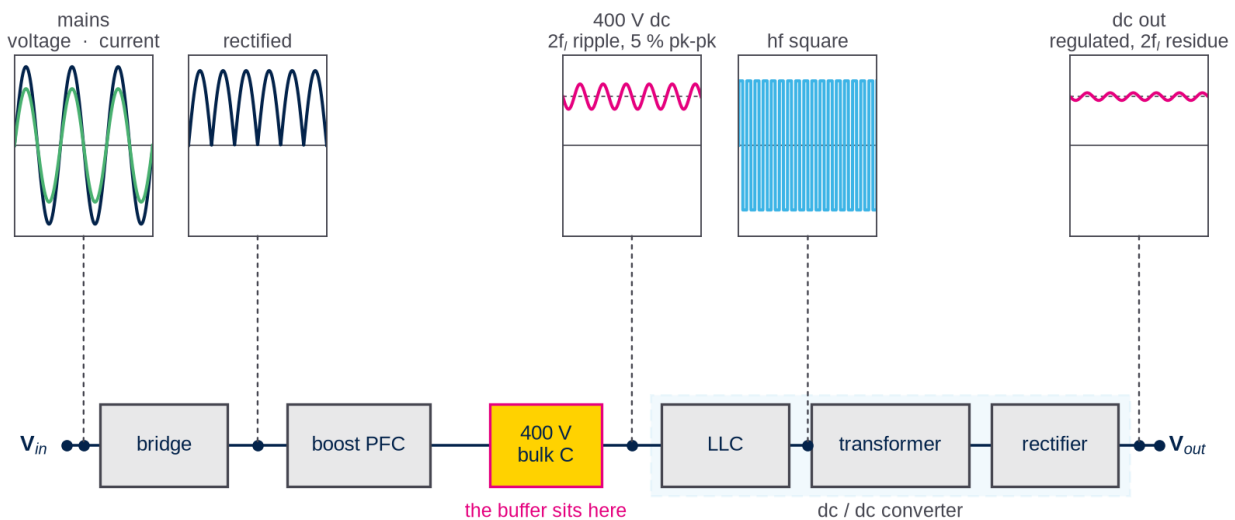


**Figure 22: The result, in continuous conduction mode. Switching ripple rides on the reactor current and never brings it to zero; its average follows the input voltage, which is what the two loops were arranged to produce.**

**This comes back later.** Take the boost stage away and the same conflict returns in a sharper form: the outer loop is then the *only* loop, its output is a power command, and any  $2f_l$  ripple that gets through it becomes input current distortion directly. That is Section 6.26.

The corrector delivers what was asked for, and creates a new problem in doing so. A current that follows the voltage means an input power that follows  $\sin^2\theta$ : zero twice per cycle, twice the average at the crests. The load still wants constant power. **Something has to store the difference**, and where that something sits is the subject of the next section.

## 2.17 Why they are normally two stages



**Figure 23: The usual two-stage arrangement, with the waveform at every node. The correction happens in the first block and the LLC works from the dc bus it produces — and the buffer that absorbs the twice-line-frequency pulsation sits between them, at 400 V. The ripple on the dc nodes is drawn larger than it is.**

The bus capacitor between a boost PFC and an LLC does two jobs at once, and they are often confused with each other:

- it **buffers** the difference between the pulsating input power and the constant output power;

# Single-Stage PFC LLC Converter Design

- it **isolates** the LLC from the mains waveform, so the tank sees a nearly constant input and has to cover only a narrow gain range.

Removing the boost stage removes both services at once. The buffering has to be done somewhere else, and the tank now has to work directly from a rectified sine that goes to zero a hundred times a second. What that costs, and why it works anyway, is the rest of this note.

## 3 Why single stage, and what it costs

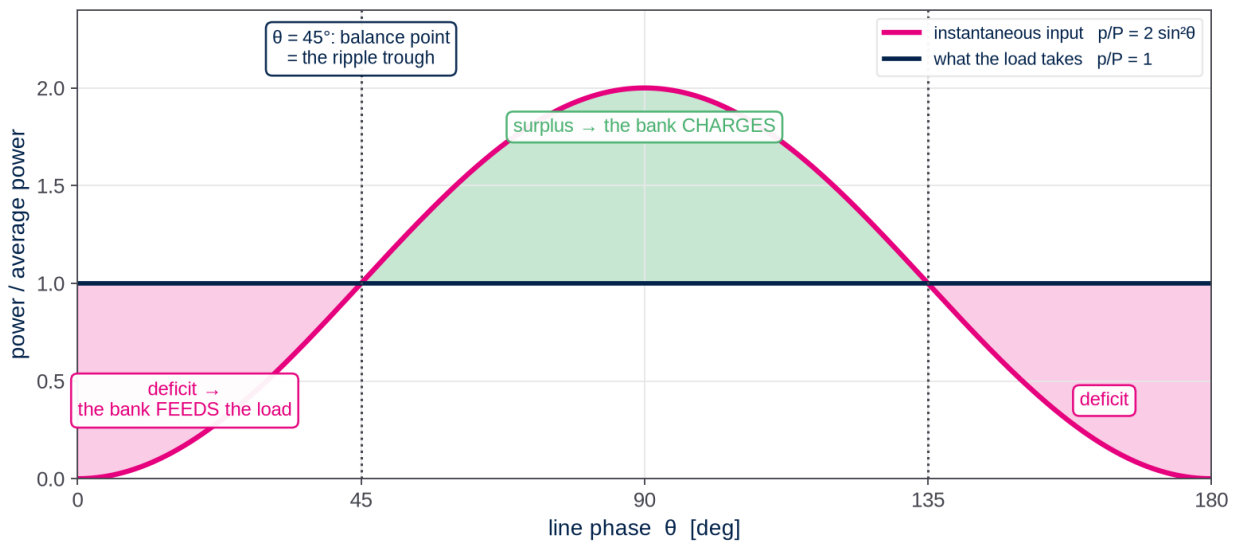
**Two steps of that recipe are the ones this converter breaks.** Step 1 has no nominal point to settle at, because the input sweeps a half sine; and step 2 asks for a gain range that no single tank can cover, which is why the bridge itself has to change. Sections 4.1 and 5 are those two departures.

### 3.1 The energy a unity power factor cannot deliver

Drawing a sinusoidal current in phase with a sinusoidal voltage makes the instantaneous input power a raised sine squared. It swings between zero and twice the average, at twice the line frequency:

$$p_{in}(t) = V_{ac} I_{ac} [1 - \cos(2\omega_l t)] \quad (9)$$

The load wants constant power. The difference has to be stored and returned a few milliseconds later. This requirement does not depend on topology; it follows from asking for unity power factor at all. Figure 24 shows the imbalance that must be buffered.



**Figure 24: Unity power factor forces a  $2f_l$  energy imbalance. The two shaded areas are equal, and they are what a capacitor somewhere must absorb and give back.**

### 3.2 Moving the buffer from 400 V to the output

In the two-stage converter that buffer is the bus capacitor, at 400 V. In the single-stage converter there is no bus, so the buffer moves to the output. **The joules do not change — the voltage they sit at does**, and a capacitor only ever returns the energy between its starting voltage and the lowest the load will accept:

$$E = \frac{1}{2} C (V^2 - V_{min}^2) \quad (10)$$

# Single-Stage PFC LLC Converter Design

Hold-up is the clearest way to see the price, because both architectures must ride out the same  $T_{\text{hold}}$  at full power, so they must store the same energy. On a 400 V bus that may fall to about 320 V the usable window is  $400^2 - 320^2$ ; on an output of a few tens of volts that may fall to  $V_{o,\text{min}}$  it is  $V_{\text{out}}^2 - V_{o,\text{min}}^2$ , smaller by a factor in the hundreds. The capacitance grows by the same factor. That single ratio is the price of the architecture, and it should be settled before anything else is designed: if the enclosure cannot house the bank, the output voltage is what has to change. Section 7.10 works it out for the worked design, where ripple, not hold-up, finally sets the bank.

**The trade in one line.** Removing the boost stage removes a switch, an inductor, a diode and a 400 V electrolytic, and their losses with them. It adds a very large low-voltage capacitor bank, a 2f<sub>r</sub> ripple on the output that the load must tolerate, and a resonant tank that has to work over a much wider range. Whether that is a good trade depends almost entirely on how much output ripple the load will accept.

## 3.3 What the load must tolerate

Because the buffer is on the output, the output carries the 2f<sub>r</sub> ripple directly; there is no second stage to reject it. The ripple is set by the bank and the load, not by the control loop, and it cannot be regulated away — attempting that puts distortion on the input current instead (Section 6.26). **How tight the figure has to be is a property of the load, not of the converter**, and it has to come from the specification rather than from a design rule; the worked design allows a few per cent peak-to-peak.

## 4 Operating principle

### 4.1 Gain is a boundary condition, not a control variable

In a two-stage LLC the controller commands gain: it moves the switching frequency until the output is right. In a single-stage PFC LLC it cannot, because **both ports are voltage sources**. The output is a millifarad-class capacitor whose voltage cannot move within a switching cycle, and the input is the rectified mains, which the converter does not control either. The instantaneous gain is therefore forced:

$$M(\theta) = \frac{n V_{o,\text{eff}}}{V_{\text{drive}}(\theta)} = \frac{n V_{o,\text{eff}}}{\sqrt{2} V_{ac,\text{eq}} \sin \theta} \quad (11)$$

with  $\theta$  the line phase angle. In a two-stage converter the output voltage is free to move — the loop moves it — so commanding a gain is the same as commanding an output. Here both voltages are pinned from outside, so their ratio is fixed moment by moment and the controller cannot change it.

So what does moving the switching frequency do? It moves **Q** — the loading — and Q is power. The tank is given the gain it must produce, and the frequency decides how much current it draws while doing so. **In this converter frequency is a power command, not a voltage command.**

### 4.2 Two divergences that cancel

Near the mains zero crossing that equation looks impossible to satisfy: as  $\theta \rightarrow 0$  the required gain rises as  $1/\sin\theta$  without limit. The converter survives because the load disappears at the same time. A unity-power-factor input draws power proportional to  $\sin^2\theta$ , so

$$Q(\theta) = Q_{pk} \sin^2 \theta \quad (12)$$

and an unloaded (lossless) LLC has unlimited gain at its lower resonance  $f_o$ . The two infinities cancel, and the operating point moves down towards  $f_o$  as the mains voltage approaches zero (Section 7.6 draws it for the worked design).  **$f_o$  is therefore the frequency floor of the entire design**, and putting the oscillator clamp below it is one of the few mistakes that destroys hardware.

# Single-Stage PFC LLC Converter Design

## 4.3 Frequency modulation is the power factor correction

Putting these together gives the control law. Over a line half cycle the controller sweeps the switching frequency so the power drawn follows  $\sin^2\theta$ . There is no current loop, no multiplier and no separate PFC stage: **the frequency profile  $f_{sw}(\theta)$  is the power factor correction**. Section 7.6 shows the profile of the worked design over the whole equivalent input range.

Computing that profile seems to need a numerical root search: at each  $\theta$  the gain equation has to be solved for  $f_n$ , and it has two roots: a capacitive one and an inductive one. **It does not.** Substituting  $M_{req} = M_{pk}/\sin\theta$  and  $Q = Q_{pk}\sin^2\theta$  and writing  $x = 1/f_n^2$  turns the gain equation into a cubic in  $x$ :

$$\lambda^2 x^3 + (q - 2\lambda(1 + \lambda))x^2 + ((1 + \lambda)^2 - 2q - \frac{u}{M_{pk}^2})x + q = 0, \quad u = \sin^2 \theta, \quad q = Q_{pk}^2 u^2 \quad (13)$$

One root is always negative: the cubic equals  $+q > 0$  at  $x = 0$  and goes to  $-\infty$  as  $x$  goes to  $-\infty$ , so it crosses zero somewhere below  $x = 0$ . The other two roots are where one curve is crossed twice. For  $x > 0$  the inverse gain  $1/M^2$  of the tank, as a function of  $x$ , starts at infinity, falls to a single minimum and rises again. So a required gain that the tank can produce is met at **exactly two** positive values of  $x$ : the capacitive crossing and the inductive one. All three roots are then real. The physical, inductive root is the **middle** one, because  $x = 1/f_n^2$  decreases as frequency increases. There is no branch to choose: in the trigonometric form of Cardano it is always the  $k = 1$  branch. If the required gain is below that minimum there is no positive root at all, which is the no-solution case of Section 4.8 rather than a numerical failure.

**Why this matters beyond elegance.** A grid search needs an upper bound on  $f_n$ , and a bound taken from the ac maximum instead of the morphing corner quietly leaves out the real operating point. The closed form has no grid, no bracket and no starting guess, so that failure cannot happen. The inverse problem — finding the  $\sin^2\theta$  that gives a required  $f_n$  — is a quadratic and is also exact.

**The zero-crossing dead zone.** Very near  $\theta = 0$  the required gain exceeds anything the tank can produce and the converter simply stops drawing current. This is inherent to the topology and shows up as third-harmonic distortion on the input current. It is one of two mechanisms that set the achievable THD; the other is the voltage loop (Section 6.26).

## 4.4 The feedback pin is a power command

Combining the burst-mode expression in the datasheet with the multiplier in the block diagram gives the physical meaning of the feedback voltage. It is not an error signal in the usual sense:

$$V_{FB} = V_{os} + \frac{2K_{HV}}{K_M K_{FF}} R_{CS} P_{in} = 0.5 \text{ V} + 0.167 \frac{\text{V}}{\Omega \cdot \text{W}} R_{CS} P_{in} \quad (14)$$

Substituting the 2.8 V feedback span gives  $2.8/0.167 = 16.8 \Omega \cdot \text{W}$ , which is exactly the maximum-power rule in the datasheet. **Maximum power limiting, burst entry and overload detection therefore all sit on one scale**, and that scale is  $R_{CS}$ . The outer optocoupler loop commands an *input power*; the internal loop distributes that power over the line cycle as  $\sin^2\theta$ .

This is why the current-sense pin tolerates no filter and no series resistor. The maximum-power law and the over-current threshold read the same pin, so anything that shifts it shifts both.

## 4.5 Which side of resonance this converter runs on

Section 2.5 left this question open. Now it can be answered: a single-stage converter does not pick one side. The equivalent input follows a half sine every 10 ms, the required gain follows it, and the operating point crosses  $f_r$  and

# Single-Stage PFC LLC Converter Design

comes back within one line cycle. How much of the cycle it spends on each side depends on the mains voltage.

Section 7.6 shows the profile of the worked design at four inputs.

**Both sides have to be designed for.** The boosting side sets the gain requirement and the ZVS margin. The bucking side sets the top switching frequency, and it takes the secondary out of zero-current switching — so the rectifier body diode and the SR dead time matter here in a way they do not in a conventional LLC. A design checked only at the line peak, or only at low line, has checked only one of the two.

## 4.6 Why $\lambda$ must be about 0.5

A classic LLC uses  $m = L_p/L_r$  between 5 and 10, that is  $\lambda$  between 0.11 and 0.25. A single-stage converter cannot: it has to produce very high gain near the zero crossing, and peak gain falls as  $\lambda$  falls. The evaluation board ST publishes for this part runs  $\lambda = 0.500$ , and the worked design comes out close to it (Section 7.4): the value comes from the topology, not from one particular design.

The price is circulating current. A small  $L_m$  means a large magnetising current that carries no power but does carry conduction loss, at every load. **A single-stage PFC LLC is less efficient than a two-stage LLC, and that is inherent rather than a design defect.**

## 4.7 $f_{sw,max}$ does not check the tank — it computes it

Two of the four candidates for  $\lambda$  carry the specified maximum switching frequency in a denominator:

$$\lambda_2 = \frac{\lambda_1}{1 - \left(\frac{f_r}{f_{sw,max}}\right)^2} \quad \lambda_{TD} = \frac{\lambda_1}{1 - \frac{\pi^2}{8} \left(\frac{f_r}{f_{sw,max}}\right)^2} \quad (15)$$

$\lambda_1$  is the plain minimum-gain condition. The other two are the same condition with the frequency ceiling folded in: the tank has to reach the lowest required gain *before* it runs out of frequency. With the usual  $f_{sw,max} = 1.5 f_r$  the squared ratio is 0.44, both denominators fall well below one, and **the number typed as  $f_{sw,max}$  roughly doubles the  $\lambda$  the tank is asked for**. Set it close to  $f_r$  and the denominators go to zero and the required  $\lambda$  to infinity.

**So where does  $f_{sw,max}$  come from?** The common rule is  $1.5 \times f_r$ . It is a **choice**, not a limit — and it sits upstream of the tank, so changing it changes  $L_m$ . A maximum in a design sheet usually only checks a result; this one is an input, so a round number typed into it changes the tank without anyone noticing.

It should not be confused with  $f_{Max}$ , the oscillator ceiling. That one *is* a limit:  $R_T$ ,  $C_T$  and the idle time fix it in silicon, the control loop cannot push past it however much gain it wants, and it is checked after the fact by  $k_{ceil}$  (Section 7.3). The two sit at opposite ends of the design: one decides the tank, the other reports whether the oscillator can serve it.

## 4.8 The other bound on $\lambda$ , and where it has no solution

The condition above sets a floor under  $\lambda$ . The ceiling comes from the opposite end: the tank must also be able to reach the *lowest* required gain, which happens at the highest equivalent input and no load. That is where the usual design rule quietly breaks down, because the no-load gain of an LLC does not fall without limit. Raising the frequency only brings it down towards a limit (an asymptote):

$$M_\infty = \lim_{f \rightarrow \infty} M_{no\ load} = \frac{1}{1 + \lambda} \quad (16)$$

**If the required minimum gain lies below  $M_\infty$ , no frequency satisfies the condition at all.** A spreadsheet solving for that frequency returns an error rather than a number, and the error is easy to mistake for a broken formula. It is not: the question simply has no answer.

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Whether a given tank sits above or below the asymptote is a matter of a per cent or so, and no full-load check shows it; Section 7.4 reports it for the worked design.

Losing the solution is not a failure. No load at high line is the region **burst mode** owns, not frequency control. Under load the gain curve falls further and the corner arrives sooner, so every full-load check is unaffected — which is exactly why this can go unnoticed. What matters is to know which of the candidate turns ratios can still regulate by frequency alone at no load, because no full-load number shows it.

## 5 Topology morphing

A resonant tank has a usable gain range of roughly 1.5:1 at full load. A universal mains asks for 2.93:1. No single tank covers that and keeps ZVS, so the L6790A changes the bridge instead.

Below a mains peak of 235 V the part runs a **full bridge**, which drives the tank with twice the rail. Above 245 V<sub>pk</sub> it runs a **half bridge**. The tank therefore sees 173.2 to 332.3 Vac equivalent, a range of 1.92:1, and one tank can cover it.

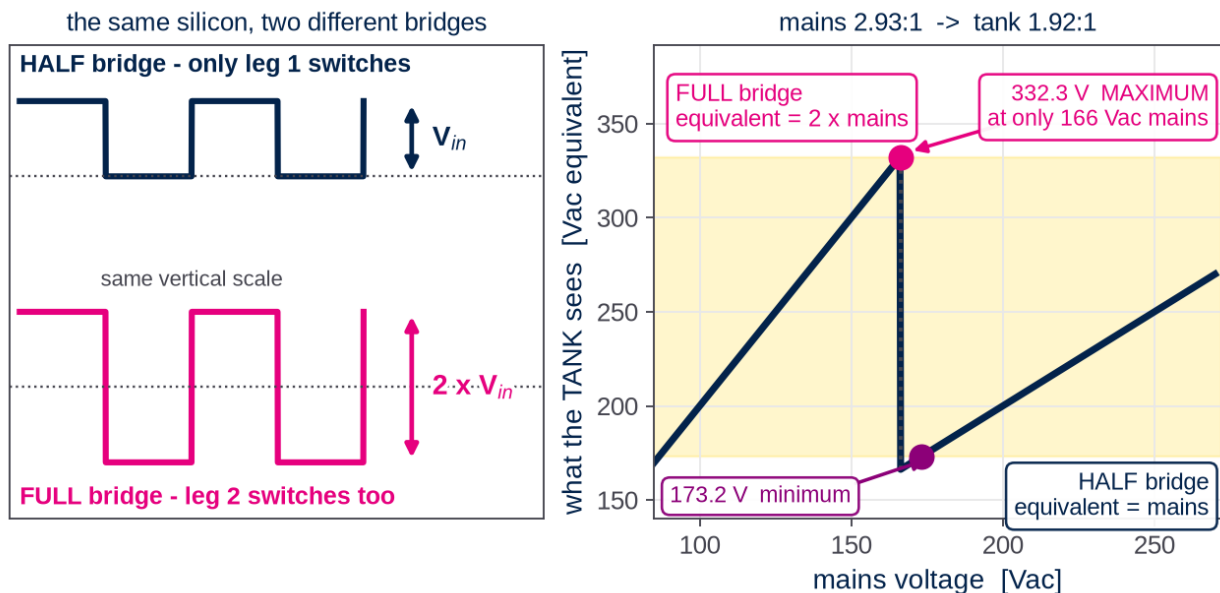


Figure 25: Morphing collapses a 2.93:1 mains range into a 1.92:1 range at the tank. The worst corners are the morphing edges, not the ends of the mains range.

### 5.1 Full bridge

Diagonal pairs conduct together. S1 and S4 put +V<sub>in</sub> across the tank, S2 and S3 put -V<sub>in</sub> across it, so the drive is a square wave of amplitude V<sub>in</sub> and its fundamental is (4/π)V<sub>in</sub>. All four devices switch, and each carries the tank current for half of every period.

# Single-Stage PFC LLC Converter Design

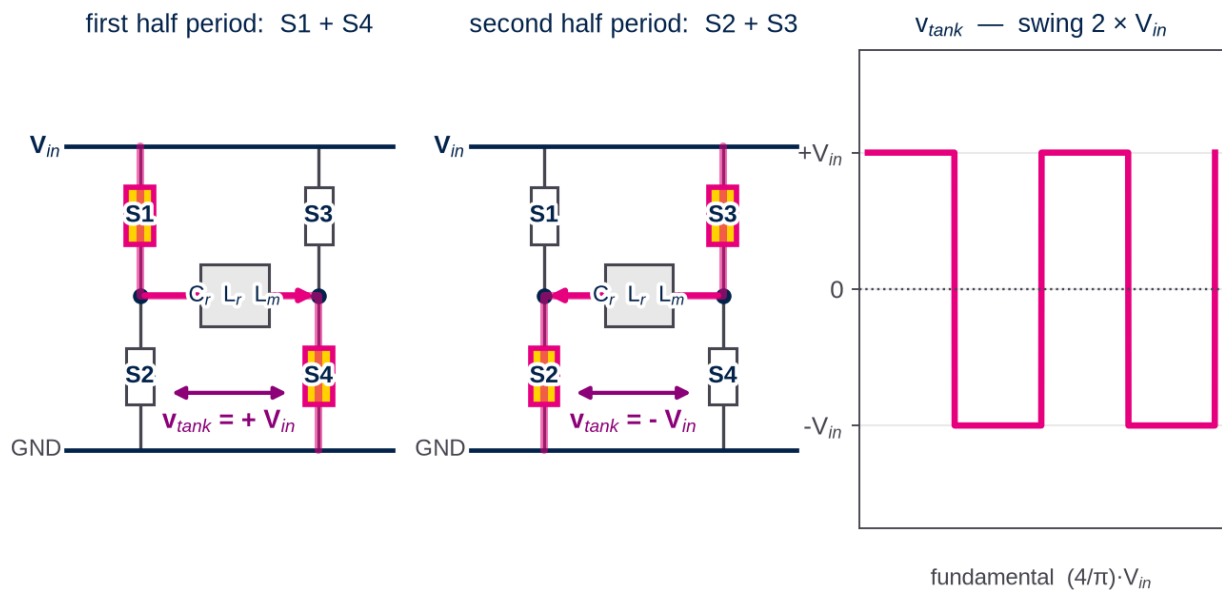


Figure 26: Full bridge: conduction path in each half period, and the resulting tank drive.

## 5.2 Half bridge

LOUT2 is held statically high. S3 never turns on and S4 never turns off, so the leg-2 midpoint becomes ground and the bridge output swings between 0 and  $V_{in}$ .  $C_r$  is already in series with the tank, so it blocks the  $V_{in}/2$  of dc for free and the tank sees  $\pm V_{in}/2$  — a fundamental of  $(2/\pi)V_{in}$ , exactly half. No extra hardware is involved.

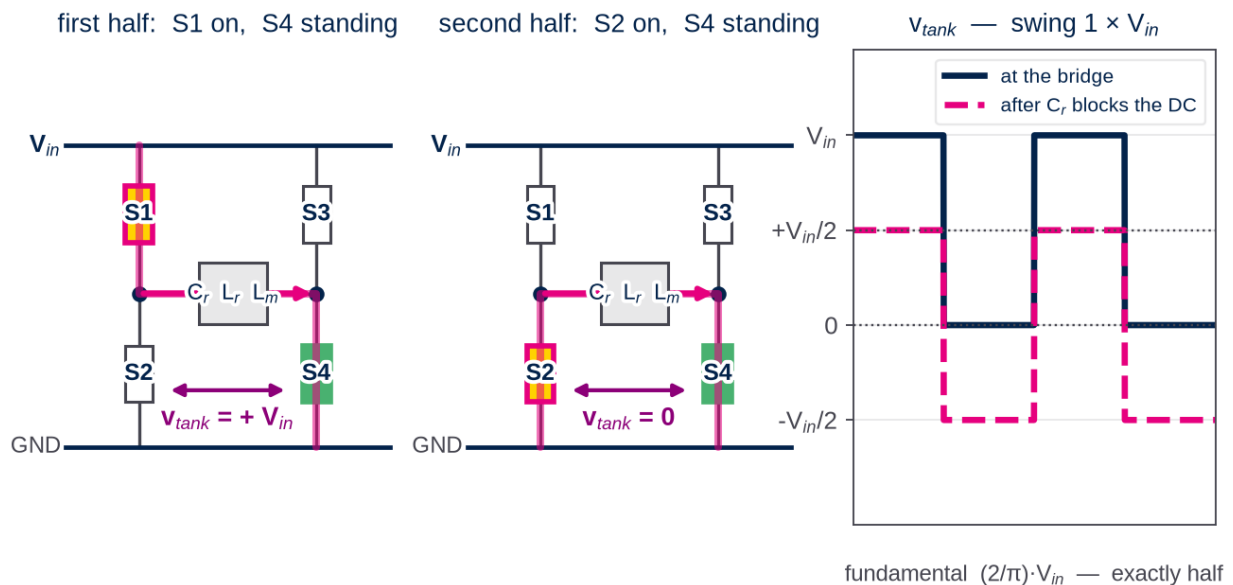


Figure 27: Half bridge: leg 2 stops switching, and  $C_r$  removes the dc that the asymmetric drive creates.

**The standing device is the hottest one.** S4 never switches, but it conducts the full tank current continuously. It is easy to overlook exactly because it is not switching, and in the worked design it is the single largest loss item (Section 7.11).

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## 5.3 How the controller does it

Only one gate signal changes between the two modes, which is why morphing costs nothing in hardware. The thresholds themselves are **fixed constants inside the IC** and cannot be moved by the designer;  $R_{CFG}$  decides only whether morphing is enabled at all and where the brown-out threshold sits.

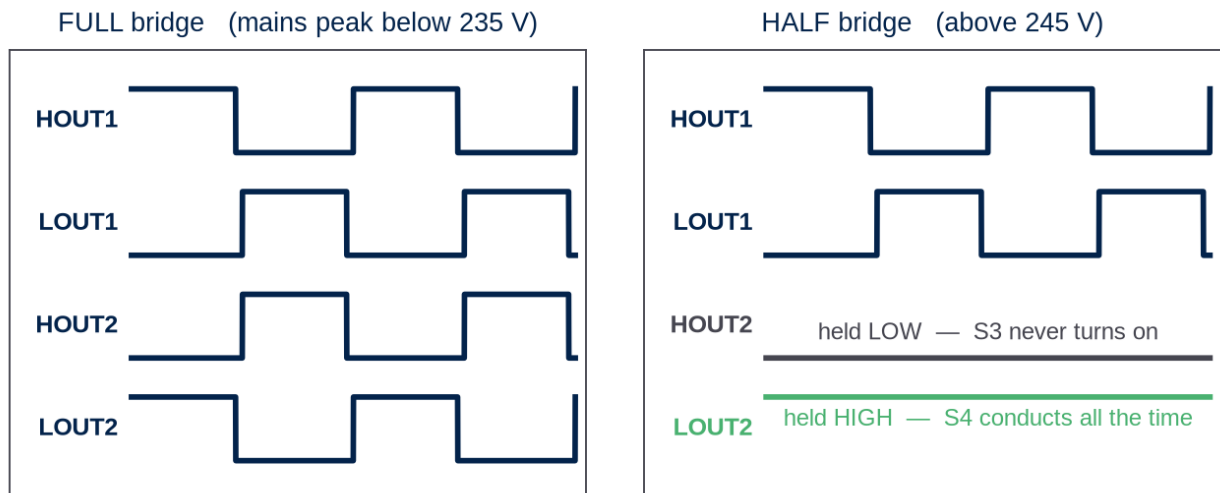


Figure 28: The four bridge drive signals in each mode. LOUT2 held high is the entire mechanism.

## 5.4 The hysteresis band, and the range it really implies

The two thresholds are not the same number: the part drops to half bridge at  $245 V_{pk}$  going up and returns to full bridge at  $235 V_{pk}$  coming down. The 10 V of hysteresis stops the mode from toggling back and forth at the threshold, but it also means that **between 235 and 245  $V_{pk}$  the mains voltage alone does not determine the mode** — the direction of travel does.

Taking each threshold on the side where its mode is guaranteed gives the design range 173.2 to 332.3 Vac (1.92:1). Including the band gives **166.2 to 346.5 Vac (2.08:1)**. This design passes every check at the wider range as well, but a design with less margin would not, and the narrower figure must not be used when tightening a specification.

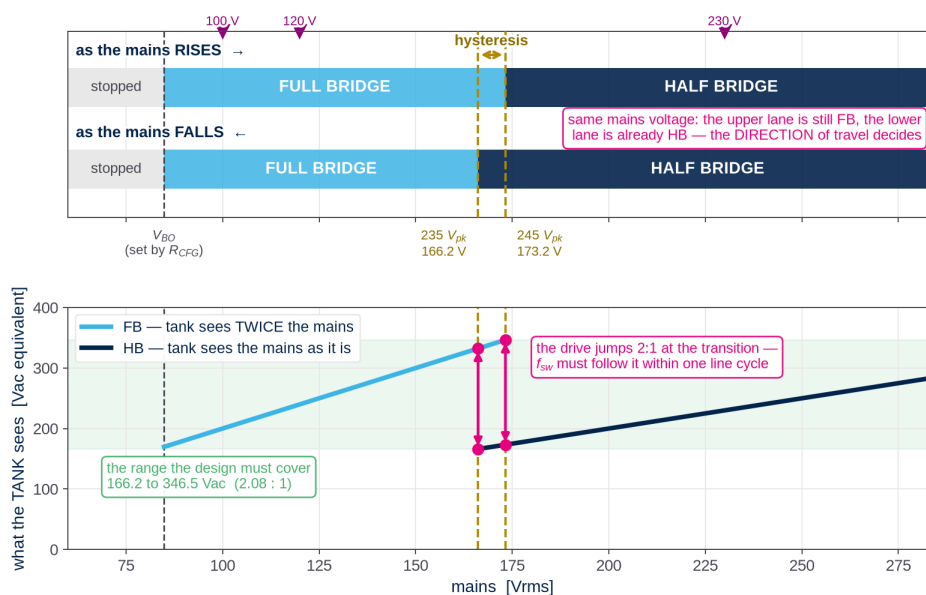


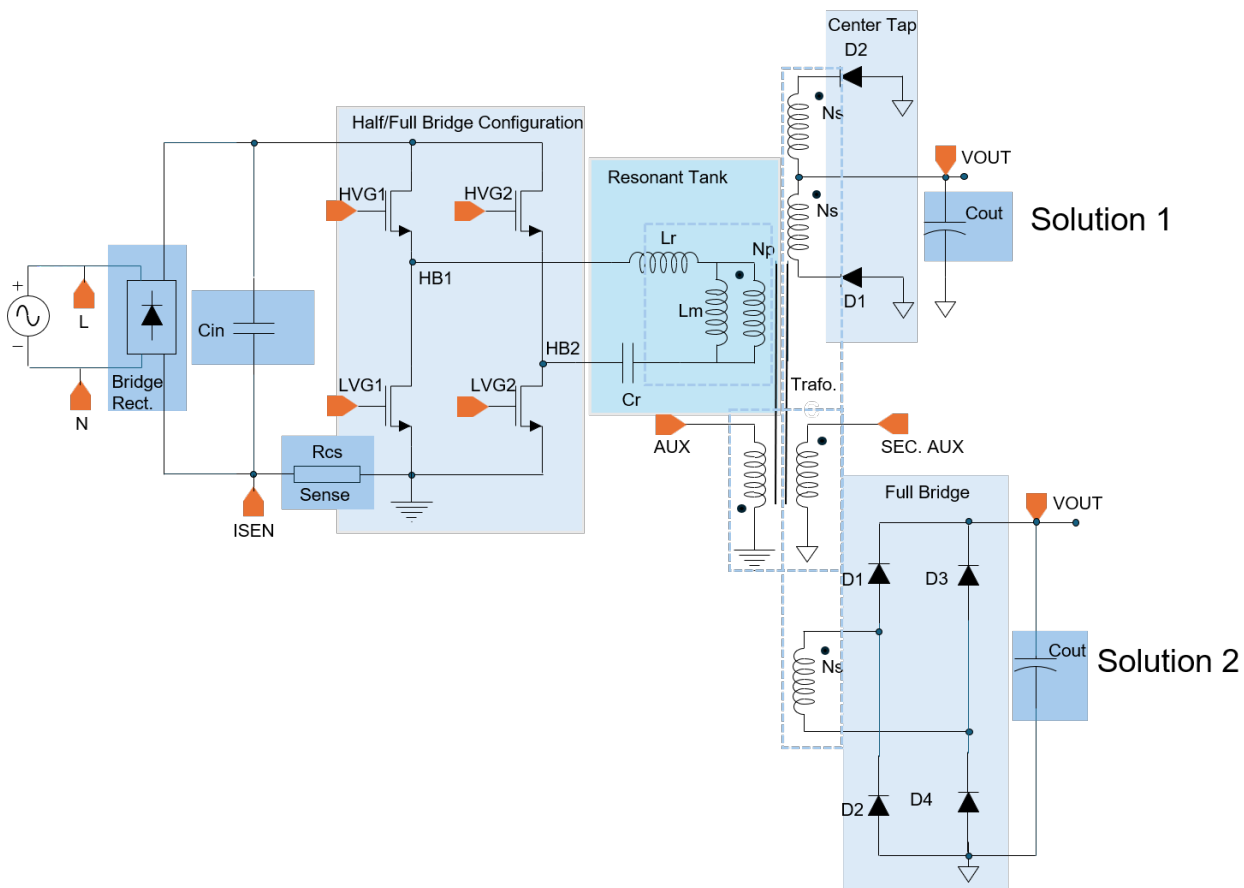
Figure 29: Where each mode applies. Inside the band the mode depends on which direction the mains arrived from.

# Single-Stage PFC LLC Converter Design

**Where you will actually meet the band.** Real mains never sits at 166 to 173 Vrms, so in service the mode is settled at start-up and stays. The band is met on a programmable ac source and in line dip and surge testing. Test it with a **step** across the threshold, not a ramp — a ramp gives the voltage loop time to follow, and a step does not.

**Transition transient.** Crossing a threshold changes the tank drive 2:1 within one line cycle, and  $f_{sw}$  must follow it against a voltage loop that crosses in the tens of hertz. The output droops until the loop recovers. This event is **less severe than the hold-up case**, which removes power entirely for  $T_{hold}$  and is already met, so it is not a reason to enlarge  $C_{out}$  or speed up the loop. That comparison has not yet been checked in the time domain, by simulation or by measurement.

## 6 Design procedure



**Figure 30: The complete power stage. The bridge rectifier feeds the half/full-bridge leg pair directly — there is no bulk capacitor and no boost stage. The secondary is drawn both ways: centre tap (Solution 1) and full bridge (Solution 2).**

Pick the secondary before the tank: it sets  $N_{rect}$ , and  $N_{rect}$  sets the reflected voltage. A centre tap puts one device in the conduction path, a full bridge puts two, so at a low output voltage the centre tap halves the secondary conduction loss. The price is twice the reverse voltage on each device, and a secondary winding that works only half of the time. This design uses the centre tap.

The sequence below is the order in which the quantities actually depend on one another. Working out of order will produce a tank that has to be redone.

# Single-Stage PFC LLC Converter Design

## 6.1 The specification

The worked example in Section 7.1 states the specification in full, sorted by what kind of number each one is; the procedure below refers to it but does not repeat it. Two qualifiers matter to every step here: **the worst line frequency is the lowest**, so everything that depends on  $f_l$  uses  $f_{l,min}$ , and **hold-up is specified at the worst line phase**, the ripple trough.

## 6.2 The equivalent input range

This is the step that is most often done wrong. With morphing the tank does not see 90 Vac at the bottom and 264 Vac at the top. It sees an **equivalent** input given by the bridge mode:

$$V_{ac,eq} = 2 V_{ac} \quad (\text{full bridge}) \quad V_{ac,eq} = V_{ac} \quad (\text{half bridge}) \quad (17)$$

At the half-bridge edge the mains is 245 V<sub>pk</sub> and the tank sees 173.2 Vac equivalent; at the full-bridge edge the mains is 235 V<sub>pk</sub> in full bridge and the tank sees 332.3 Vac equivalent. **These two edges are the design corners, not the ends of the mains range:** the low edge is the worst case for gain and ZVS, the high edge for switching frequency.

## 6.3 Turns ratio and reflected voltage

The turns ratio only ever appears multiplied by the effective output voltage, so what matters is the **reflected voltage**:

$$V_{refl} = n V_{o,eff} = n(V_{out} + N_{rect} V_f) \quad (18)$$

Two ratios have to be kept apart: **n** is the equivalent-model ratio in the gain equation, **n<sub>T</sub>** the physical turns ratio with the leakage folded into L<sub>r</sub>. At  $\lambda \approx 0.5$  they differ by more than 20 %; Section 6.9 has the relation, and the transformer drawing must carry open- and short-circuit inductance as well as turns.

## 6.4 The resonant tank

The equivalent ac load resistance for a centre-tapped secondary is

$$R_{ac} = \frac{4}{\pi^2} \frac{n^2 V_{o,eff}^2}{P_{in,LLC}} \quad (19)$$

with P<sub>in,LLC</sub> the power into the resonant stage. The 4 is the usual 8 with the *line-peak* power 2P in the denominator rather than the average — see the note below.

The gain requirement at the low equivalent corner and the ZVS requirement together cap the quality factor at Q<sub>ZVS</sub>, and R<sub>ac</sub> Q<sub>ZVS</sub> is then the impedance the tank is *sized* from: it fixes C<sub>r</sub> through equation 1, and C<sub>r</sub> with the target f<sub>r</sub> fixes L<sub>r</sub>. Three rules govern what happens next, and they matter more than the arithmetic:

- The three *calculated* figures are not one tank. C<sub>r</sub> comes from the design impedance; L<sub>r</sub> is then whatever pairs with the **selected** C<sub>r</sub>, which is why it is not R<sub>ac</sub> Q<sub>ZVS</sub>/2πf<sub>r</sub>; and the L<sub>m</sub> figure is L<sub>r</sub> divided by the  $\lambda$  that *no load at the high corner* asks for — a condition a design may knowingly not meet (Section 4.8), so it is not a value to round towards.
- **Never round L<sub>m</sub> up.** That lowers  $\lambda$  and takes the ZVS margin with it. C<sub>r</sub> above its calculated value and L<sub>m</sub> below have the opposite effect: Q<sub>pk</sub> falls and  $\lambda$  is held.
- **Stay below the Q<sub>ZVS</sub> cap, not on it.** That gap is where most of a design's ZVS margin comes from, and it is a deliberate choice, not an accident of rounding. L<sub>m</sub> is chosen together with n so that  $n_T = n\sqrt{1+\lambda_{act}}$  lands on a ratio that can actually be wound.

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**The same expression, evaluated at the line peak.** Every LLC text writes  $R_{ac} = (8/\pi^2)n^2V_o^2/P$  with the one output power a two-stage converter has. Here the drawn power is  $p(\theta) = 2P\sin^2\theta$ , so  $R_{ac}$  has to be pinned to one instant. Pinned to the **line peak**, where  $p = 2P$  and the tank is most heavily loaded, the 8 becomes a 4. It is not a different formula and it has nothing to do with half against full bridge — the bridge factor belongs to the equivalent input voltage.  $Q$  then means the quality factor at the line peak, with  $Q(\theta) = Q_{pk}\sin^2\theta$  elsewhere. Using the two-stage value understates the loading by two, and every tank value that follows is wrong.

## 6.5 ZVS verification

The closed-form ZVS estimate in most design guides is a fitted approximation evaluated at the design  $Q_{ZVS}$ , not at the  $Q$  the converter runs at. Against a full sweep it errs in **either** direction by tens of per cent, and the sign depends on the tank; optimistic is the dangerous direction, because it reports ZVS margin that is not there.

**And it has to be told which  $\lambda$  to use.** The shortcut is written with a bare  $\lambda$ , and it means the *design*  $\lambda$  — the largest of the four candidates — together with the design  $Q_{ZVS}$ . Read the same expression with  $\lambda_{act}$ , which is what the symbol means everywhere after the tank is chosen, and the phase can come out **negative** — a capacitive answer for a tank that is well inside the inductive region. The sweep carries no such ambiguity: it uses  $\lambda_{act}$  and the  $Q$  of the moment, which is what the hardware has.

The reliable procedure is to sweep  $\theta$ , input voltage and load with the **selected** tank and take the minimum. Section 7.5 runs it for this design, with numbers on both directions of the closed-form error, and the worst point is not where one would expect.

## 6.6 Which side of resonance the converter runs on

Below  $f_r$  the secondary current is a truncated sine with a dead interval and the conduction ratio  $d = f_{sw}/f_r$  is less than one; above  $f_r$  the current is continuous and  $d$  clamps at one. The rms figures that ratings and losses are built on contain  $d$ , so the difference is real, not a matter of words.

**Which side of resonance the converter runs on is not a separate choice. The turns ratio decides it.** The demand is  $M_{req} = 2nV_{o,eff}/(\sqrt{2}V_{eq})$ , so a larger  $n$  asks for more gain, and more gain means further below resonance. Two candidate transformers on the same tank can end up on opposite sides of  $f_r$  at the same mains voltage.

Worth settling before the transformer is ordered. Comparing candidate turns ratios on gain margin alone hides it, and the two sides do not stress the secondary the same way: below resonance the rectifiers turn off at zero current, above it they do not.

## 6.7 Currents over the line cycle

Ratings come from the worst switching cycle; losses come from the line-cycle rms. They are different numbers and must not be confused. Two further rules apply to a single-stage converter in particular.

- The **composite** tank current is what the primary devices and the sense resistor see. It is the sum of the reflected load current and the magnetising current, and its peak is neither the sum nor the larger of the two peaks, because they occur at different instants.
- Below resonance the secondary conducts for only  $d = f_{sw}/f_r$  of each half period. Omitting that conduction ratio over-estimates the rms, and the loss by the square of that.

Section 7.7 draws the composite current of the worked design.

## 6.8 The transformer

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The tank has fixed  $L_r$ ,  $L_m$  and a ratio; the transformer has turns, an open-circuit inductance and a leakage. Getting from one to the other is where most of the errors in a resonant design are made, because **two different turns ratios and two different inductances carry almost the same names**. Section 2.12 has already set out why this transformer is sized the way it is; this section turns that into the numbers a supplier can measure.

Every number on the drawing is read off one of three waveforms — the primary winding current, the current in each secondary winding, and the secondary winding voltage — or off one bench test. Figure 37, in the design example, marks where on the worked design's waveforms, and Table 2 says what each mark sets and over which interval it is taken. The sections that follow derive the entries.

| Mark | Item                                | Read from — and over what                                                                                                                                                                                                                   | What it sets                                                                    |
|------|-------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|
| 1    | Primary peak current                | Top of $i_p$ : the half sine of load current riding on the magnetising ramp. Worst switching cycle: line peak, minimum equivalent input, full load.                                                                                         | Primary switch and OCP1 headroom. <b>Not</b> the saturation test.               |
| 2    | Primary rms current                 | $i_p^2$ averaged over the whole switching period, then over the line half cycle. Every unit carries the same current: the primaries are in series.                                                                                          | Primary copper: $A_{cu} = I_{rms}/J$ per unit                                   |
| 3    | Magnetising peak                    | The dashed $i_{Lm}$ at the switching instant. Below resonance it depends on neither $f_{sw}$ nor load, only on $V_{out}$ , so it is the same on every cycle.                                                                                | The peak flux; and, scaled to $V_{OVP2}$ , the DC-overlap test current (mark 7) |
| 4    | Secondary peak current              | Top of the pulse in either secondary winding, same worst cycle. Per unit it is $1/N_x$ of the rectifier-leg value: the secondaries are in parallel.                                                                                         | Rectifier peak rating; the foil termination                                     |
| 5    | Secondary rms current, each winding | One pulse per period in each winding, so the rms is taken over the whole period with the idle half counted, then over the line half cycle.                                                                                                  | Secondary copper: $A_{cu} = I_{rms}/J$ , per winding and per unit               |
| 6    | Winding volt-seconds                | The secondary winding voltage is $V_{out}$ for the resonant half period $T_r/2$ while the rectifier conducts. That area is the flux swing $2N_s A_e B_{pk}$ ; $f_r$ is the worst case because the interval never gets longer than $T_r/2$ . | Core area and $N_s$ : $A_e \geq V_{out}/(4 f_r N_s B_{max})$                    |
| 7    | DC-overlap (saturation) test        | Not a waveform. Secondaries open, dc current in the primary, inductance at the primary read against current. The test current is mark 3 scaled to the highest output the controller allows, $V_{OVP2}/V_{out}$ .                            | Core saturation margin; the number the supplier tests to                        |
| —    | $L_{open}$ , $L_{short}$            | LCR meter at the primary with the secondaries open, then shorted. Terminal quantities, so a supplier can test them.                                                                                                                         | $L_m + L_r$ and $L_r$ : the tank itself                                         |

**Table 2: What each mark in the figure sets. Currents are per unit of the assembly: the primaries are in series and the secondaries in parallel.**

## 6.9 Two ratios, two inductances

When all of the leakage is referred to the primary, the tank sees an ideal transformer of ratio  $n$ , a series  $L_r$  and a shunt  $L_m$ . That  $n$  is **not** the wound ratio. With  $k$  the coupling coefficient,

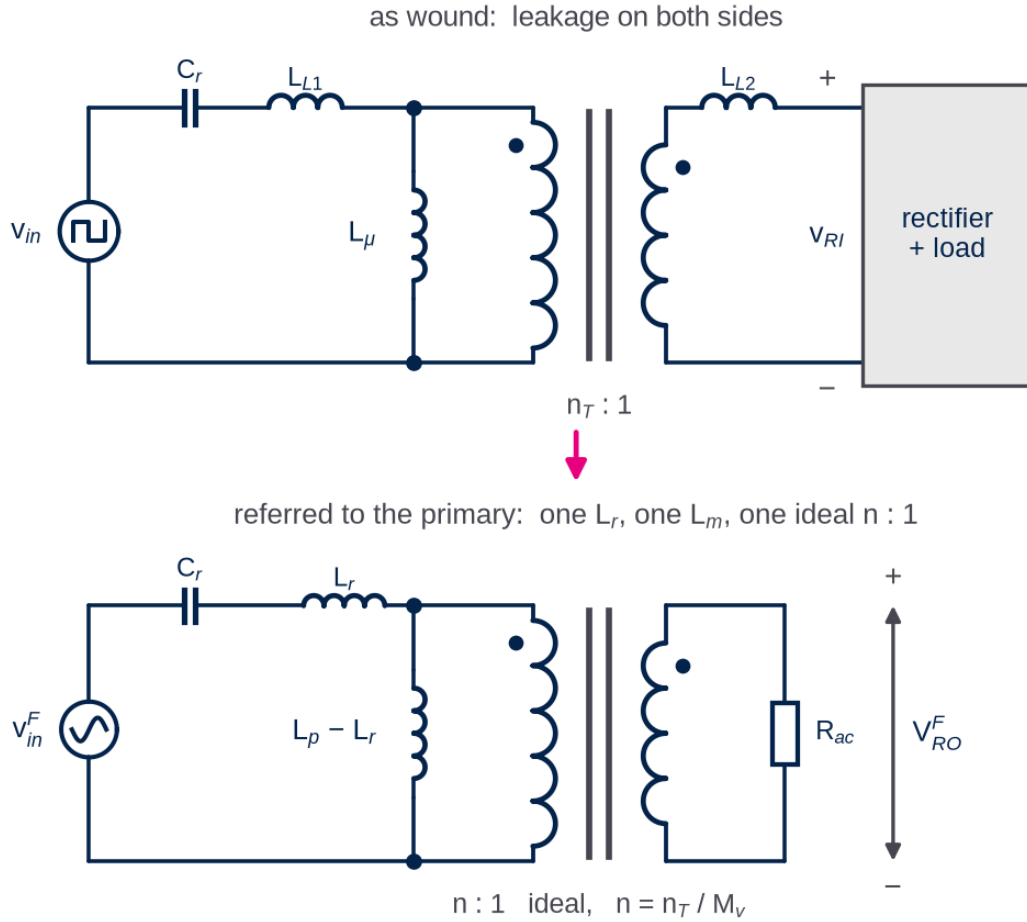
$$n = k n_T, \quad k = \sqrt{\frac{L_m}{L_m + L_r}} = \frac{1}{\sqrt{1 + \lambda}} \quad (20)$$

so  $n_T = n \sqrt{1 + \lambda}$ . At  $\lambda \approx 0.5$  the two differ by more than 20 %, which is far more than any gain margin.

The same split appears in the inductances. The tank model uses  $L_r$  and  $L_m$ ; the physical transformer is described by a magnetising inductance  $L_\mu$  with leakage on both sides:

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$$L_\mu = \sqrt{L_m (L_m + L_r)}, \quad L_{L1} = L_m + L_r - L_\mu, \quad L_{L2} = \frac{L_{L1}}{n_T^2} \quad (21)$$



**Figure 31: The same transformer drawn both ways. Above, as wound: the leakage split between the two sides, the physical magnetising inductance  $L_\mu$  in shunt, and the wound ratio  $n_T : 1$ . Below, referred to the primary: one  $L_r$ , one  $L_m$ , and an ideal ratio  $n : 1$  that is no longer the wound one. The two shunt elements are not the same number —  $L_\mu = \sqrt{L_m(L_m + L_r)}$  — and neither are the two ratios.**

That three-element form assumes the leakage splits evenly between the two sides. The real split is set by the winding arrangement rather than by the design, and it does not matter: the tank sees only the two quantities a meter can read.

### 6.10 What the transformer specification must say

Only two inductances are measurable at terminals, and they are exactly the two the tank needs:

$$L_{open} = L_\mu + L_{L1} = L_m + L_r, \quad L_{short} = L_{L1} + \frac{L_\mu n_T^2 L_{L2}}{L_\mu + n_T^2 L_{L2}} = L_r \quad (22)$$

$L_{open}$  is read at the primary with every other winding open,  $L_{short}$  with the secondary shorted. **Specify those two and the turns, never  $L_\mu$**  — no terminal pair exposes  $L_\mu$ , so a supplier cannot test it, and a part that meets it can

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still miss the tank. Also state the turns as numbers: with one or two secondary turns, the inductance of that winding is dominated by the lead-out loop, so  $\sqrt{L_1/L_2}$  does not give the turns ratio.

## 6.11 Peak flux is set by the secondary, not the primary

Mark **6** of Figure 37: the secondary winding voltage is a rectangle of height  $V_{out}$  and width  $T_r/2$ , and the flux swings from  $-B_{pk}$  to  $+B_{pk}$  across it. The equation is that area divided by  $2N_s A_e$ , and nothing else enters.

Below resonance the secondary conducts for a resonant half period and the output voltage stands across the secondary winding for all of it. Above resonance it conducts for a shorter switching half period, so  $f_r$  is the worst case everywhere on the line cycle:

$$B_{pk} = \frac{V_{out, eff}}{4f_r N_s A_e} \implies A_e \geq \frac{V_{out, eff}}{4f_r N_s B_{max}} \quad (23)$$

**$N_p$  does not appear.** Adding primary turns does not reduce the flux; only adding secondary turns does. That is an uncomfortable result for a low-voltage, high-current output, where the secondary wants to be a single turn: halving  $N_s$  doubles the core area needed. That is why a single-turn secondary is expensive in core area even though it is the obvious choice for the current.

Computing the same flux from an inductance is correct only with the **physical**  $L_\mu$ :  $B_{pk} = L_\mu i_{\mu, pk} / (N_p A_e)$  reduces exactly to the expression above. Using the tank model  $L_m$  with the real  $N_p$  understates the flux by  $\sqrt{1+\lambda}$  — about a quarter at  $\lambda \approx 0.5$  — because the  $L_m$  branch carries  $n V_{out}$  while the winding reflects  $n_T V_{out}$ .

## 6.12 Choosing the core: two areas, and the one that usually decides

A core has to satisfy two independent conditions, and they have nothing to do with each other. The magnetic one is the equation above: enough cross-section  $A_e$  to carry the flux at the turns chosen. The electrical one is the winding window  $A_N$ : enough room for the copper the currents need. Catalogues quote both, and their product is what handbooks call the **area product**. Either can be the binding one.

**Read the flux check against  $A_{min}$ , not  $A_e$ .**  $A_e$  is the effective area for inductance;  $A_{min}$  is the narrowest section the flux actually has to pass. On a PQ core the two differ by ten to twenty per cent, and it is  $A_{min}$  that saturates first: the real peak is  $B_{pk} A_e / A_{min}$ . Both numbers are in the datasheet, and only one of them is usually used.

In a **single-stage PF LLC** there is a **third condition**, and it is **normally the one that decides**: the leakage inductance has to come out at  $L_r$ . Written as a fraction of what a meter reads at the primary,

$$\frac{L_{short}}{L_{open}} = \frac{L_r}{L_m + L_r} = \frac{\lambda}{1 + \lambda} \quad (24)$$

At  $\lambda \approx 0.5$  that is **about a third**. A conventional transformer, wound to couple as well as it can, leaks a few per cent; this one has to leak an order of magnitude more, *on purpose*. Leakage that large is not a winding tolerance, it is a winding **geometry** — and geometry costs window area. A core that passes on  $A_e$  and on copper can still fail here.

## 6.13 The winding arrangement is the leakage

Leakage is the flux that links one winding and not the other, so it is set by how far apart the two windings are and by how much of the window separates them. That gives the designer one knob and a well-known ordering:

- **Interleaved** — primary, secondary, primary, all concentric. Lowest leakage, typically one or two per cent. This is what a conventional forward or flyback transformer wants, and it is the opposite of what is needed here.
- **Concentric, not interleaved** — the whole primary, then the whole secondary, over it. A few per cent. Still far short.

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- **Side by side** — the primary on one half of the winding width, the secondary on the other. Tens of per cent, and adjustable by the gap left between them. **This is the arrangement a single-stage tank asks for**, and it is why the window has to be generous: half of it goes to each winding, and part of what is left is deliberately empty.
- **A magnetic shunt** — a ferrite bar in the window between the two windings, which carries leakage flux on purpose. It buys leakage without spending winding width, at the cost of a part and of a tolerance that is harder to hold.

**This inverts the usual advice.** Every transformer text says to interleave, because in every other topology leakage is loss and overshoot. Here  $L_r$  is a design value that the tank needs, and a supplier who "improves" the coupling has broken the converter —  $f_r$  moves,  $\lambda$  collapses and the gain curve goes with it. Say so on the drawing, in words, next to  $L_{short}$ .

## 6.14 The gap, and why the drawing must not name it

$L_m$  is two orders below what an ungapped core of this size gives, so the core is gapped. The inductance factor that follows from the turns is

$$A_L = \frac{L_{open} N_x}{N_p^2} \quad g \approx \frac{\mu_0 A_e}{A_L} \quad (25)$$

with  $N_x$  the number of units the assembly is built from. The second expression is only an estimate: it assumes the gap carries the whole reluctance and it ignores fringing, which always makes the real gap larger than it predicts. **Specify  $A_L$ , or better  $L_{open}$  itself, and leave the gap to the supplier** — that is the number they grind to, and it is the number a meter can check.

## 6.15 The window: wire, current density and what actually fits

Marks **2** and **5** of Figure 37. The copper is sized on the **line-cycle** rms: the rms of each switching period (the bracket under the trace), averaged in  $I^2$  over the line half cycle (the lower-left panel). The secondary value is per winding and per unit — each winding carries one pulse per period, and its idle half is inside the average.

Each winding needs a conductor cross-section set by its own rms current and the current density the design can cool:

$$A_{cu} = \frac{I_{rms}}{J} \quad \sum_w N_w A_{cu, w} \leq k_u A_N \quad (26)$$

$J$  is an assumption, not a constant: 4 to 5 A/mm<sup>2</sup> is usual for a transformer of this size in free air, less if it is enclosed.  $k_u$  is the window utilisation, and it is brutal — round wire in a round bundle, insulation, the bobbin wall, the margins and the layer-to-layer tape leave **0.3 or less** of the window as copper when the wire is Litz. Sizing on bare copper alone overstates what fits by three times.

**The primary current does not divide between units.** With the primaries in series every unit carries the whole primary current, and only the secondaries share. So the primary copper is the same in each unit as it would be in one big transformer, while the secondary copper is divided — which is exactly why splitting helps a low-voltage, high-current output.

## 6.16 Skin depth, and why the wire is not a wire

At the switching frequency current does not fill a conductor. It crowds into a surface layer of depth

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$$\delta = \sqrt{\frac{\rho}{\pi f \mu_0}} \quad (27)$$

which for a series resonance in the 100 to 200 kHz range is about **0.2 mm** in copper at 100 °C. A conductor thicker than about  $2\delta$  carries no more current than one of  $2\delta$  — it only adds weight. Worse, the field from the *other* turns drives circulating current in each conductor (the proximity effect), and in a side-by-side winding, where the two windings sit in each other's leakage field, that term can dominate the skin term.

- Use **Litz** on the primary: many strands, each well under  $2\delta$ , individually insulated and transposed. Strand diameter around 0.2 mm is a usual choice at this frequency.
- On a two-turn, high-current secondary, **copper foil** is usually better than wire: the thickness can be held near  $\delta$  while the width fills the window, and it terminates well into a centre tap.
- Whatever is chosen, the ac resistance is what heats the winding. The dc resistance  $\rho l_N/A_{cu}$  is a floor, not an answer.

## 6.17 Isolation, margins and what they cost in window

The transformer is the isolation barrier of the whole supply, so the winding that looks like a copper problem is also a safety one. Two arrangements are common, and they spend the window differently. **Margin tape** leaves a creepage margin at each end of the bobbin — typically 3 mm each side for reinforced isolation from a universal mains — so it takes 6 mm off the usable winding width and costs nothing in material. **Triple-insulated wire** needs no margin, so the winding uses the full width, but the wire is thicker for the same copper and it costs more.

On a side-by-side winding the margin is taken twice over: once for the isolation, and once again for the gap that sets the leakage. Work the window budget with both in it, or the part that is ordered will not be the part that was calculated.

## 6.18 Loss, and the temperature the drawing has to survive

- **Core loss** comes off the material curve at the operating flux and frequency, not off the headline figure. A datasheet quotes  $P_v$  at one point — 100 or 200 mT at 100 kHz — and a design at a different flux and a different frequency has to be read from the curve.
- **Copper loss** is  $I^2 R_{ac}$ , and  $R_{ac}$  is the number the previous section is about. Both windings count, and in a side-by-side arrangement the proximity term is not small.
- **The temperature rise** is what actually limits the design, and it depends on the surface area and the airflow, neither of which is in any of the equations above. Treat the computed loss as an input to a thermal measurement, not as an answer.

**Flux and loss do not move together here.**  $B_{pk}$  is set at  $f_r$  by the secondary volt-seconds, so it does not change when the converter runs faster — but core loss does, and in a single-stage converter the frequency sweeps over the line cycle. The core sees its worst flux at the bottom of the sweep and its worst loss per cycle count at the top.

## 6.19 The saturation test is not the peak winding current

Mark **3** and the bench panel of Figure 37. The current to ask for is the peak of the dashed magnetising trace at the switching instant, scaled to the over-voltage ceiling — not the peak of the winding current at mark **1**.

A DC-overlap test leaves every other winding open, so none of the primary ampere-turns is cancelled by the secondary and the same current makes far more flux than it does in operation. It is the flyback condition of Section 2.12, reached deliberately. Ask for the current that reproduces the operating flux in that test:

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$$I_{eq} = \frac{B_{pk} N_p A_e}{L_\mu} \quad (28)$$

**That denominator is  $L_\mu$ , not  $L_{open}$ .** In the open-circuit test the primary current is entirely magnetising current, so what links the core is the  $L_\mu$  part of the flux linkage and the rest is primary leakage, which goes round no core at all. Dividing by  $L_{open}$  understates the current by  $1/\sqrt{1+\lambda}$  — about a fifth at  $\lambda \approx 0.5$ . There is a check that costs nothing: the answer **must** come out equal to  $i_{\mu,pk}$ , because in that test they are the same current.

Giving the supplier the peak tank current instead asks for a flux density no ferrite reaches, and the supplier will either oversize the core or ask why. Nor is an arbitrary margin wanted on top of  $I_{eq}$ : the flux ceiling the core has to survive is already defined, by the highest output voltage the controller allows before it shuts down. Scale  $I_{eq}$  by that ratio and there is no number left to invent.

$$I_{sat} = I_{eq} \frac{V_{OVP2}}{V_{out,eff}} \quad (29)$$

## 6.20 The specification, and what it is not allowed to leave out

**The open-circuit inductance tolerance cannot simply be the usual round number.**  $f_o$  goes as  $1/\sqrt{L_{open}}$ , so a low part raises the frequency floor towards the oscillator clamp. The fall a tank can tolerate follows from  $k_{floor}$  and is typically a few per cent, so the usual  $\pm 10\%$  can allow a part that meets its specification to hard switch. Section 6.25 derives the figure and gives the two ways out. Ask the supplier for a tighter low-side limit, or first make room in the oscillator setting and then ask for  $\pm 10\%$ .

Core, bobbin, wire and winding order are the supplier choice, and insulation and creepage follow the applicable safety standard. Nothing above constrains them; everything above is measurable at the terminals.

## 6.21 If one transformer is not practical

At a low output voltage and high current the secondary is a single heavy turn and the core comes out large, which may not fit a height or footprint limit. The transformer can then be built as  $N_x$  identical units with **primaries in series and secondaries in parallel**. Series primaries carry the same current, so the paralleled secondaries share automatically without balancing resistors.

- Open-circuit and leakage inductance divide by  $N_x$ , and so do the secondary currents. **The primary current does not divide.**
- The DC-overlap test current does not change: flux and primary ampere-turns scale together.
- The wound ratio of the assembly becomes  $N_x N_p / N_s$ , so **the achievable ratios are quantised in steps of  $N_x / N_s$** . That has to be respected when the tank is chosen, not discovered afterwards.

## 6.22 The input capacitor

With no bulk capacitor,  $C_{in}$  is a film capacitor that absorbs switching ripple and nothing else. A large value is not the safe choice: it holds charge across the zero crossing and distorts the line current, so it is a THD term, not a filter term.

$$C_{in} = 3 \frac{nF}{W} \times P_{in} \quad (30)$$

That is a **lower** bound, so round up to the next standard value and rate it for the full line voltage.

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## 6.23 The output capacitor bank

Two conditions apply, and the larger wins. The ripple condition is

$$C_{out} \geq \frac{P_{out}}{2\pi f_{l,min} \Delta v V_{out}^2} \quad (31)$$

and the hold-up condition is

$$C_{out} \geq \frac{2P_{out}T_{hold}}{(V_{out} - \frac{1}{2}\Delta v_{pp})^2 - V_{o,min}^2} \quad (32)$$

**Hold-up starts at the worst line phase, and that belongs in the sizing.** If the mains disappears at the trough of the  $2f_l$  ripple the bank is already half a ripple below nominal when it starts, which is why  $V_{out}$  above carries the  $-\frac{1}{2}\Delta v$  term. The difference is not small: it costs about a tenth of the ride-out time. Note also that  $\Delta v_{pp}$  there is the ripple the bank actually produces, not the allowance. The condition therefore depends on its own answer, which is why it is checked once the bank is chosen.

Which one wins is a question about the specification and not about the output voltage: both conditions scale as  $1/V_{out}^2$ , so  $V_{out}$  cancels when they are divided. Writing  $k = V_{o,min}/V_{out}$ , ripple dominates when

$$\left(1 - \frac{\Delta v}{2}\right)^2 - k^2 > 4\pi f_l \Delta v T_{hold} \quad (33)$$

This screening form uses the *allowed*  $\Delta v$  rather than the achieved ripple, because it is asked before the bank exists. Section 7.10 evaluates it for the worked design.

- Include the  **$2f_l$  component** in the ripple current. The switching-frequency part alone is far too small: the  $2f_l$  envelope carries a comparable rms of its own, and the two are orthogonal so they add in quadrature.
- With a centre-tapped secondary, judge the capacitor ripple current at the **output node**, not per winding.
- Use the ESR at the **switching frequency**. An electrolytic  $\tan\delta$  is quoted at 120 Hz and is five to ten times larger there.

**A risk this architecture creates.** The controller start-up window is finite, and a bank of this size is a far heavier start-up load than the few millifarads a conventional design presents. Cold start into the full bank should be one of the first things measured on hardware.

## 6.24 Semiconductor requirements

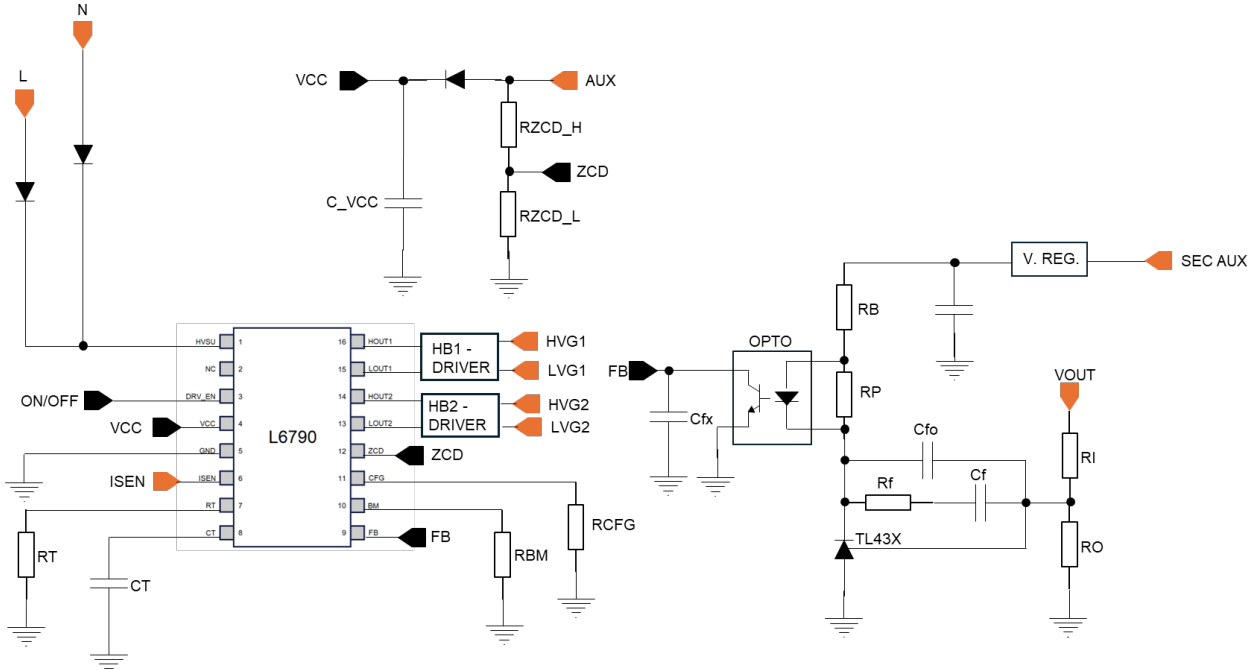
State requirements rather than pick parts: what to pick depends on package, thermal design and cost. Four rules decide whether the requirement is even stated correctly.

- Rate the primary switches on the **composite tank peak**, not on the reflected load current. The load component alone under-rates the device by more than a tenth here, because the magnetising current is missing from it.
- Divide currents by the number of devices actually in parallel before judging stress. Tabulated currents are **per switch position** on the primary and **per leg** on the secondary; a centre-tapped leg carries the whole secondary current, not half of it.
- Compute loss from  $R_{DS(on)}$  at  $T_{j,max}$ , not at 25 °C. The datasheet maximum is process spread at 25 °C; temperature is a separate multiplier, close to two for a superjunction device, read from its normalised  $R_{DS(on)}$  curve at  $T_{j,max}$ . Using the 25 °C value to size a heatsink is optimistic by about a factor of two.
- A curve plotted against  $T_a$  may still be read as  $T_j$  **if it is a pulse test** — the die does not self-heat during the pulse. For any non-pulsed curve this is not valid.

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**The standing device in half-bridge morphing is the one to watch** (Section 5.2). It never switches and carries the whole tank current, and it is where the conduction-loss budget is met or knowingly missed — Section 7.11 shows which happened here.

## 6.25 Controller network



**Figure 32: The controller and its passive network.** HVBU is fed from the AC side of the bridge; the auxiliary winding drives ZCD through a divider;  $R_T$  and  $C_T$  set the oscillator limits;  $R_{CFG}$  and  $R_{BM}$  are read at power-up.

Decide first what the auxiliary winding is for. When it also supplies  $V_{CC}$ , the rectified auxiliary voltage at OVP2 must stay under the  $V_{CC}$  rating, and that caps its turns ratio. Here  $V_{CC}$  comes from elsewhere, so the winding only senses, the only constraint is that the turns come out whole, and the datasheet ratio check is not applicable (Section 7.19).

Each of these parts follows from something already fixed, and Section 7.14 sizes them one at a time in the order they have to be done. The one that deserves the most attention throughout is  $R_T$ , because it sets the oscillator floor and that floor must stay above  $f_o$ .

**That floor is usually the thinnest margin in the design.**  $f_{Min}$  must sit above  $f_o$ , or nothing but the anti-capacitive protection stands between the converter and the capacitive region. It also depends on the oscillator idle time, which the draft datasheet states inconsistently, so whatever it computes to has to be confirmed by measuring  $f_{sw}(\theta)$  on hardware.

That ratio is also what the transformer tolerance has to fit inside, and this is easy to miss because the two numbers are in different documents. Since  $f_o = 1/(2\pi\sqrt{L_{open}C_r})$ , a **low** open-circuit inductance raises  $f_o$ . The fall that can be tolerated is

$$\frac{\Delta L}{L} = 1 - \frac{1}{k_{floor}^2} \quad (34)$$

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A floor margin of a few per cent allows a fall of a few per cent in inductance. A transformer drawing asking for the usual  $\pm 10\%$  does **not** fit inside that: at the bottom of the tolerance the converter hard switches. There are two ways out — tighten the tolerance on the low side, or lower  $R_T$  to lift  $f_{Min}$  and make the room. **Check this before the transformer is ordered**, not after.

## 6.26 Voltage loop and compensation

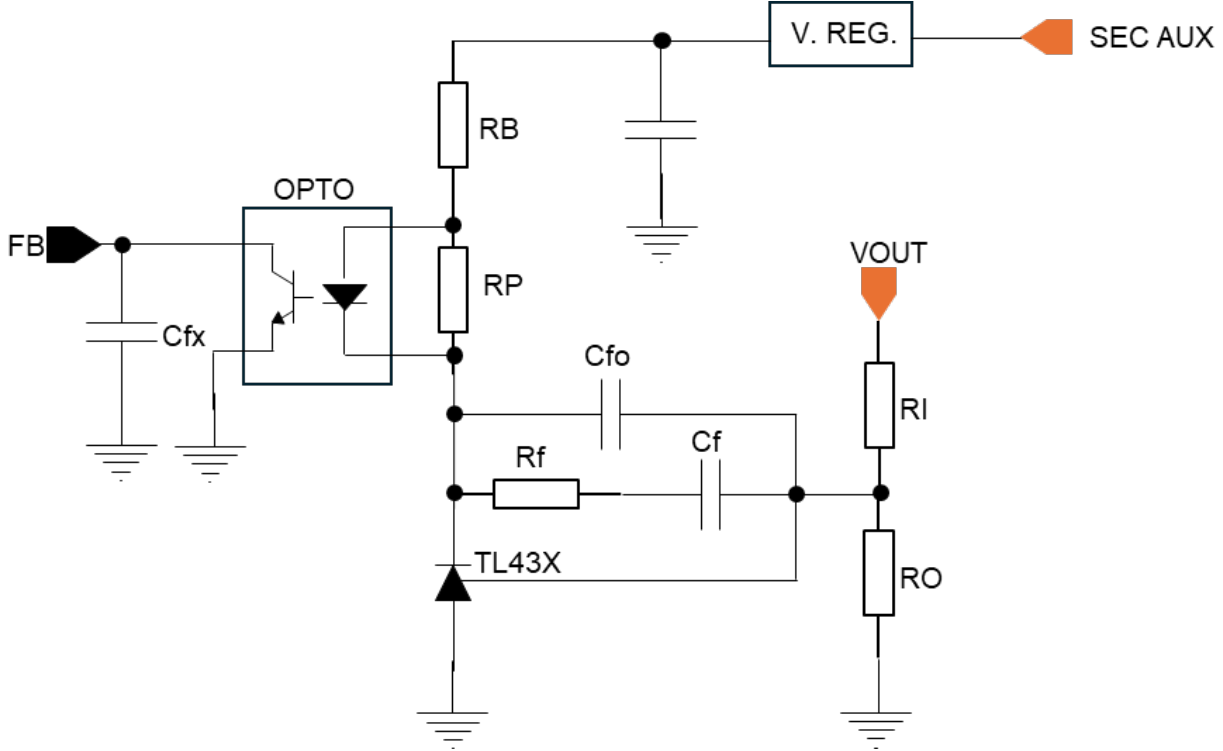


Figure 33: The secondary-side error amplifier and the optocoupler.  $R_I$  and  $R_O$  set the regulated output,  $R_f C_f$  and  $C_{fo}$  shape the compensator, and  $R_p$  and  $R_B$  bias the shunt regulator and the optocoupler diode.

The output is a capacitor fed by a constant-power source, so the control-to-output transfer is a **pure integrator**:  $-90^\circ$  at every frequency. That makes the loop easy to stabilise but hard to place, because the crossover frequency is set by a constraint that has nothing to do with stability.

Any  $2f_r$  ripple that survives the loop and reaches the feedback pin becomes a power command that varies over the line cycle, and therefore input current distortion — mostly third harmonic. Raising the crossover reduces output ripple and **increases** distortion. The two constraints pull in opposite directions, and the practical answer is a crossover between 15 and 20 Hz.

The compensator is a Type II network placed by the Venable K-factor method. With  $\alpha_v$  the ratio between the error-amplifier gain at  $2f_r$  and the gain the loop needs there,

$$K_v = \frac{1}{2\alpha_v} \left[ (1 + \alpha_v^2) \tan \Phi_M + \sqrt{(1 + \alpha_v^2 \tan \Phi_M)^2 + 4\alpha_v^2} \right] \quad (35)$$

$K_v$  puts the zero and the pole the same factor below and above the crossover. That places the phase boost where it is needed:

$$f_{MB} = \frac{1}{2\pi} \sqrt{\frac{G_o K_v E A_o}{\Gamma_v}}, \quad f_p = K_v f_{MB}, \quad f_z = \frac{f_{MB}}{K_v} \quad (36)$$

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$f_{MB}$  is the point they sit around: the zero goes below it and the pole above it, by the same factor. A third pole  $f_{px}$  is added near a kilohertz to roll off switching noise.

## 6.27 Gain margin, and why it needs checking

Phase margin alone does not prove stability. Above  $f_{px}$  this network has one zero against two poles, so the phase heads for  $-270^\circ$  and  **$-180^\circ$  is crossed at a finite frequency**. That crossing needs no root search either: setting the zero and pole phase contributions equal and taking the tangent of both sides leaves one square root,

$$f_{180} = \sqrt{f_p f_{px} - f_z(f_p + f_{px})}, \quad GM = -20 \log_{10} |T(f_{180})| \quad (37)$$

The value under the root is positive whenever the zero sits well below both poles, which is what the K-factor placement gives. If it ever came out negative the loop would not be extra safe: the two poles would pull the phase below  $-180^\circ$  before the zero could lift it, at every frequency, so the phase margin at any crossover would be negative. Judge the gain margin against 6 dB as a floor and 10 dB as a comfortable target. With a crossover in the tens of hertz the margin is usually large in this topology — which is a reason to compute it, not a reason to assume it.

## 6.28 Feedback ripple against the burst threshold

A crossover this low leaves  $2f_l$  ripple on the feedback pin. Because that pin is a power command, the ripple moves the commanded power up and down every half line cycle; if it crosses the burst-entry threshold the converter chatters in and out of burst. The ripple amplitude at the burst point is

$$\Delta V_{FB} = \frac{P_{in, BM}}{V_{out}} \frac{1}{2\pi f_l C_{out}} G_{EA}(2f_l) \quad (38)$$

Lowering  $R_{BM}$  by  $\Delta R_{BM} = \Delta V_{FB} / (2 \times 0.01 \text{ V/k}\Omega)$  moves the burst entry point below that ripple. **Raising the crossover would shrink the ripple but worsen the distortion**, so trading it against  $R_{BM}$  is the practical answer rather than speeding up the loop.

**A loop this slow cannot recover from a large load step.** The controller's anti-saturation circuit does that instead, which is why the optocoupler bias resistor must sit inside its permitted window. Checking the window is not optional.

## 7 Design example

One design, from the specification to the component values. Every number below is either given, chosen, assumed, or computed from those, and each computed value shows the arithmetic it came from. The specification is **90 to 264 Vac in, 25 V / 26.3 A = 657.5 W out**, centre-tapped synchronous rectification, no bulk capacitor and no boost stage.

### 7.1 The specification, sorted by what kind of number it is

The only useful question about a number is whether it can be moved, so the kind of each one is stated: **given** by the load and the mains, **chosen** by the designer, or **assumed** in place of a measurement that has not been made yet.

# Single-Stage PFC LLC Converter Design

| Item                            | Symbol                | Value                             | Kind    | Note                                                                                                  |
|---------------------------------|-----------------------|-----------------------------------|---------|-------------------------------------------------------------------------------------------------------|
| Mains input                     | $V_{ac}, f_l$         | 90 to 264 Vac, 47 to 63 Hz        | given   | the worst line frequency is the lowest, so every $f_l$ term uses 47 Hz                                |
| Output                          | $V_{out}, I_{out}$    | 25 V, 26.3 A (657.5 W)            | given   | $V_{o,eff} = V_{out} + N_{rect} V_f = 25.0 \text{ V}$ with $V_f = 0$                                  |
| Output ripple, $2f_l$ pk-pk     | $\Delta v$            | $\leq 5 \%$ of $V_{out}$ (1.25 V) | given   | a property of the load, not of the converter                                                          |
| Hold-up                         | $T_{hold}, V_{o,min}$ | 12 ms down to 19 V, at 100 Vac    | given   | starts at the ripple trough, the worst line phase                                                     |
| Secondary rectifier             | $N_{rect}$            | 1 (centre tap, SR)                | chosen  | halves the secondary conduction loss; doubles the device reverse voltage                              |
| Target series resonance         | $f_r$                 | 150 kHz                           | chosen  | sets the magnetics size                                                                               |
| Specified maximum $f_{sw}$      | $f_{sw,max}$          | 225 kHz (1.5 $f_r$ )              | chosen  | an input to $\lambda$ , Section 4.7                                                                   |
| Bridge dead time                | $t_D$                 | 220 ns                            | chosen  | what the ZVS sweep has to beat                                                                        |
| $V_{CC}$ source                 | —                     | external 12 V rail                | chosen  | no turns-ratio ceiling on the auxiliary winding                                                       |
| LLC stage efficiency            | $\eta_{HB}$           | 98 %                              | assumed | a 95 % outcome moves $R_{ac}$ and $R_{CS}$ by about 3 %; nothing downstream is sensitive              |
| Oscillator idle time            | $T_{idle}$            | 250 ns                            | assumed | the draft datasheet also says 700 ns, which would close the floor margin — the first thing to measure |
| $R_{DS(on)}$ temperature factor | $k_T$                 | 1.8 / 1.9                         | assumed | primary / secondary, read off the datasheet curves at $T_{j,max}$                                     |
| Burst entry point               | $r_{BM}$              | 75 W (11 % of rated)              | assumed | sets $R_{BM}$ , then checked against the feedback ripple                                              |

**Table 3: The specification. Everything else in this chapter is computed from these.**

**Harmonic class.** IEC 61000-3-2 Class D reaches only 600 W. This design draws 685 W at the input, so the absolute Class A limits apply.

## 7.2 What the design came out as

| Block       | Values                                                                                                                                                |
|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| Tank        | $C_r$ 100 nF, $L_r$ 11 $\mu\text{H}$ , $L_m$ 20 $\mu\text{H}$ , $\lambda$ 0.55, $f_r$ 151.7 kHz, $f_o$ 90.4 kHz                                       |
| Transformer | $N_p:N_s$ 15:2, $n$ 6.02, $n_T$ 7.50, $L_{open}$ 31.0 $\mu\text{H}$ , $L_{short}$ 11.0 $\mu\text{H}$ , $A_e \geq 103 \text{ mm}^2$ at $B_{pk}$ 0.20 T |
| Frequency   | $f_{sw}$ 127.1 kHz at the low corner, 249.0 kHz at the high corner                                                                                    |
| Output      | $470 \mu\text{F} \times 160 = 75.2 \text{ mF}$ , ripple 1.18 V (4.74 %), hold-up 13.42 ms                                                             |
| Controller  | $R_T$ 11 k $\Omega$ , $C_T$ 470 pF, $R_{CS}$ 24.0 m $\Omega$ , $R_{CFG}$ 30 k $\Omega$ , $R_{BM}$ 30 k $\Omega$                                       |
| Loop        | $f_{cross}$ 17.57 Hz, $\Phi_M$ 49.64°, third harmonic 4.23 %                                                                                          |

**Table 4: Principal values. The rest of the chapter shows where each one comes from.**

## 7.3 What a verification margin is

Every check has the same shape: what the design has, divided by what it needs.

# Single-Stage PFC LLC Converter Design

$$k = \frac{X_{\text{act}}}{X_{\text{req}}} \implies \text{pass when } k > 1 \quad (39)$$

$k = 1.05$  is five per cent of room,  $k = 0.9$  is ten per cent short. The ratios have no units and all point the same way, so a frequency check and a power check can sit in one column, and the smallest  $k$  is the thinnest part of the design.

| Check                                           | $k = \text{has} / \text{needs}$         | Substituted             | $k$          |
|-------------------------------------------------|-----------------------------------------|-------------------------|--------------|
| ZVS at the worst point of the sweep             | $T_{\text{ZC,min}} / t_{\text{D}}$      | 401 ns / 220 ns         | 1.824        |
| Oscillator floor clears the lower resonance     | $f_{\text{Min}} / f_{\text{o}}$         | 92.25 kHz / 90.39 kHz   | <b>1.021</b> |
| Oscillator ceiling clears the operating maximum | $f_{\text{Max}} / f_{\text{sw,max}}$    | 319.61 kHz / 249.03 kHz | 1.283        |
| Over-current threshold clears the tank peak     | $I_{\text{OCP1}} / I_{\text{Lr,pk}}$    | 22.92 A / 18.32 A       | 1.251        |
| Hold-up achieved against hold-up required       | $t_{\text{hold}} / T_{\text{hold}}$     | 13.42 ms / 12 ms        | 1.119        |
| Secondary loss against its budget               | $P_{\text{budget}} / P_{\text{SR,leg}}$ | 3.00 W / 2.83 W         | 1.063        |
| Primary loss against its budget                 | $P_{\text{budget}} / P_{\text{mos,dc}}$ | 3.00 W / 9.65 W         | <b>0.311</b> |

**Table 5: The verification margins.**

**Two rows need reading with care.**  $k_{\text{Ploss}} = 0.311$  is a result, not a failed calculation: the standing primary device spends 9.65 W against a 3 W budget, and Section 6.24 says why the design goes ahead anyway.  $k_{\text{floor}} = 1.021$  looks like a safe 2 %, but the transformer tolerance and the oscillator idle time both move it, and neither is known to 2 % yet.

## 7.4 Following the numbers through

Every quantity in the order it was computed, with the equation it came from and the numbers put into it. Nothing appears before the things it needs, which is why the design procedure is in this order.

# Single-Stage PFC LLC Converter Design

| #  | Quantity                         | Eq.  | Computed as                                                                                                                                        | Result                             |
|----|----------------------------------|------|----------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|
| 1  | Input power $P_{in}$             | —    | $P_{out} + \text{LLC loss at } \eta_{HB} + \text{EMI} + \text{bridge budgets} = 657.5 + 13.42 + 8.34 + 5.52$                                       | <b>684.8 W</b>                     |
| 2  | Power into the tank $P_{in,LLC}$ | —    | $P_{in} - \text{bridge} - \text{EMI} = 684.8 - 5.52 - 8.34$                                                                                        | <b>670.9 W</b>                     |
| 3  | Equivalent input, low corner     | 17   | $245 V_{pk} / \sqrt{2}$ in half bridge                                                                                                             | <b>173.2 Vac</b>                   |
| 4  | Equivalent input, high corner    | 17   | $2 \times 235 V_{pk} / \sqrt{2}$ in full bridge                                                                                                    | <b>332.3 Vac</b>                   |
| 5  | Turns ratio $n$                  | 20   | $n_T / \sqrt{(1+\lambda_{act})} = 7.500/1.2450$ — the wound ratio 15:2 is the input; the plain calculation would give 6.046                        | <b>6.024</b>                       |
| 6  | Reflected voltage                | 18   | $n V_{o,eff} = 6.024 \times 25.0$                                                                                                                  | <b>150.6 V</b>                     |
| 7  | Gain demanded, low corner        | 11   | $2 n V_{o,eff} / (\sqrt{2} V_{ac,eq}) = 301.2/245.0$                                                                                               | <b>1.2294</b>                      |
| 8  | Gain demanded, high corner       | 11   | the same at 332.3 Vac                                                                                                                              | <b>0.6409</b>                      |
| 9  | AC load resistance $R_{ac}$      | 19   | $(4/\pi^2) n^2 V_{o,eff}^2 / P_{in,LLC} = 0.4053 \times 36.29 \times 625 / 670.9$ — the 4, not 8, is the line-peak power                           | <b>13.70 <math>\Omega</math></b>   |
| 10 | Quality-factor cap $Q_{ZVS}$     | 4    | the smaller of the gain limit and the ZVS limit at the low corner                                                                                  | <b>1.7019</b>                      |
| 11 | Design impedance                 | —    | $R_{ac} Q_{ZVS} = 13.70 \times 1.7019$                                                                                                             | <b>23.32 <math>\Omega</math></b>   |
| 12 | $C_r$                            | 1    | $1/(2\pi f_r \times 23.32 \Omega) = 45.50 \text{ nF}$ , taken <b>up</b> on purpose to gain Q margin                                                | <b>100 nF</b>                      |
| 13 | $L_r$                            | 1    | $1/((2\pi f_r)^2 C_r)$ with the selected $C_r = 11.26 \mu\text{H}$                                                                                 | <b>11 <math>\mu\text{H}</math></b> |
| 14 | $\lambda$ required               | 15   | the largest of $\lambda_1$ 0.560, $\lambda_2$ 1.009, $\lambda_{TD}$ 1.241                                                                          | <b>1.241</b>                       |
| 15 | $L_m$                            | —    | $L_r/\lambda_{req} = 8.87 \mu\text{H}$ is the <i>no-load</i> condition, knowingly not met; chosen with $n$ so that $n_T$ lands on a windable ratio | <b>20 <math>\mu\text{H}</math></b> |
| 16 | Realised $\lambda_{act}$         | —    | $L_r/L_m = 11/20$                                                                                                                                  | <b>0.550</b>                       |
| 17 | Realised $f_r, f_o$              | 1, 2 | $1/(2\pi\sqrt{L_r C_r})$ and the same with $L_r + L_m = 31 \mu\text{H}$                                                                            | <b>151.7 / 90.4 kHz</b>            |
| 18 | Realised $Q_{pk}$                | 3    | $Z_0 = \sqrt{L_r/C_r} = 10.49 \Omega$ , divided by $R_{ac}$ — well under the cap of row 10                                                         | <b>0.766</b>                       |
| 19 | Physical turns ratio $n_T$       | 20   | $n \sqrt{(1+\lambda_{act})} = 6.024 \times 1.2450 = 15 : 2$ across the assembly, 5 : 2 in each of the 3 units                                      | <b>7.500</b>                       |

**Table 6: The design, step by step.**

**Rows 12, 15 and 18 are the judgement calls.**  $C_r$  was taken to 100 nF against a calculated 45.5 nF and  $L_m$  to 20  $\mu\text{H}$  against 8.9  $\mu\text{H}$ ; both lower  $Q_{pk}$  and hold  $\lambda$ , which is where the ZVS margin comes from. Everything else in the table is arithmetic any two engineers would reproduce identically.

**No-load solution: none.** The no-load gain of this tank bottoms out at  $M_\infty = 0.6452$ , and the high corner asks for 0.6409, below it. No frequency reaches that gain without load, and the margin is under one per cent, so no full-load check would show it. Burst mode owns that region (Section 4.8).

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## 7.5 ZVS over the whole operating space

The closed form of Section 6.5 reads 292 ns on this tank against a swept 460 ns, so it understates the margin by 36 %; on another tank in the same family it overstates by about 23 %. Read with  $\lambda_{act} = 0.550$  instead of the design  $\lambda = 1.241$  it returns -471 ns, a capacitive answer for a tank well inside the inductive region. So this design is judged by the sweep.

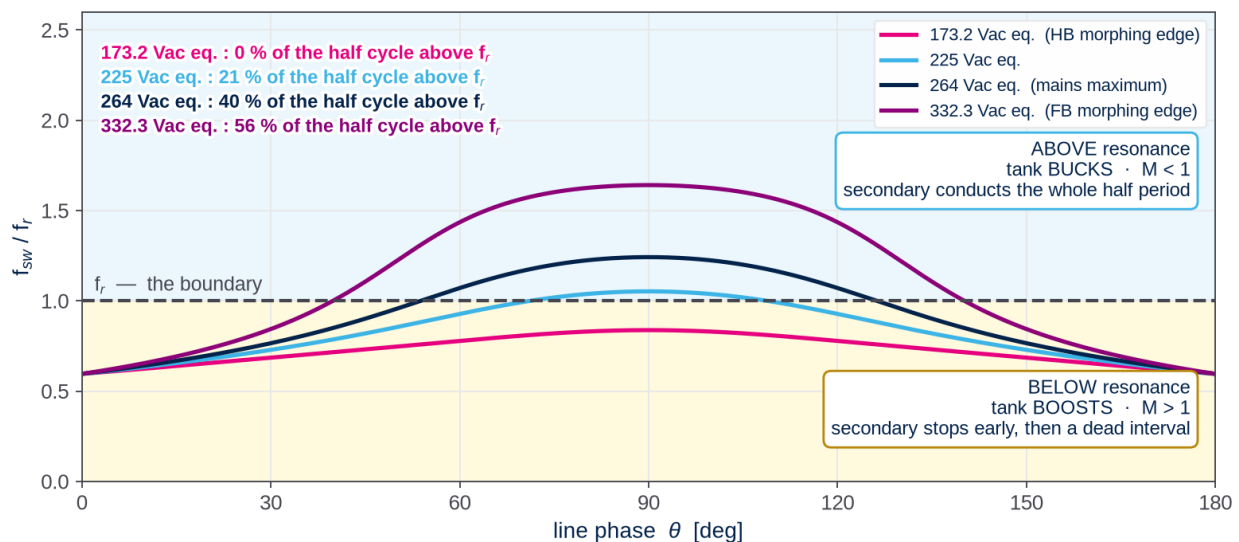
The capacitive edge of the full-load curve is where  $\arg Z_{in} = 0$ , at 110.1 kHz; the gain peak sits lower, at 105.4 kHz, and there the bridge still sees 8.4° of capacitive phase (Section 2.9). Neither is used for the verdict: the sweep measures  $T_{ZC}$  directly.

**The worst point is not the corner the tank was designed at.** Full load at the low equivalent input is where the gain requirement is set, and it gives  $T_{ZC} = 460$  ns, a margin of 2.09. The smallest number in the table is **401 ns at 25 % load and the 332.3 Vac equivalent corner**, a margin of 1.82 against  $t_D = 220$  ns. That is the number this design is held to.

**Why light load at high line can be the ZVS corner.** The charge that swings the bridge node is carried by the magnetising current, and its peak goes as  $V_{in}/(4f_{sw}L_m)$ . Shed load and the controller raises  $f_{sw}$  to cut the gain, so that current falls while the node still has the same capacitance to move. Sweeping only the design corner therefore reports a margin that is not there.

## 7.6 Which side of resonance this design runs on

Section 6.6 says the turns ratio decides this. With  $n_T = 7.50$ , at all but the lowest line conditions the tank spends part of every line cycle above  $f_r$ , where the secondary loses zero-current turn-off, so the rectifier body diode and the SR dead time both need checking.

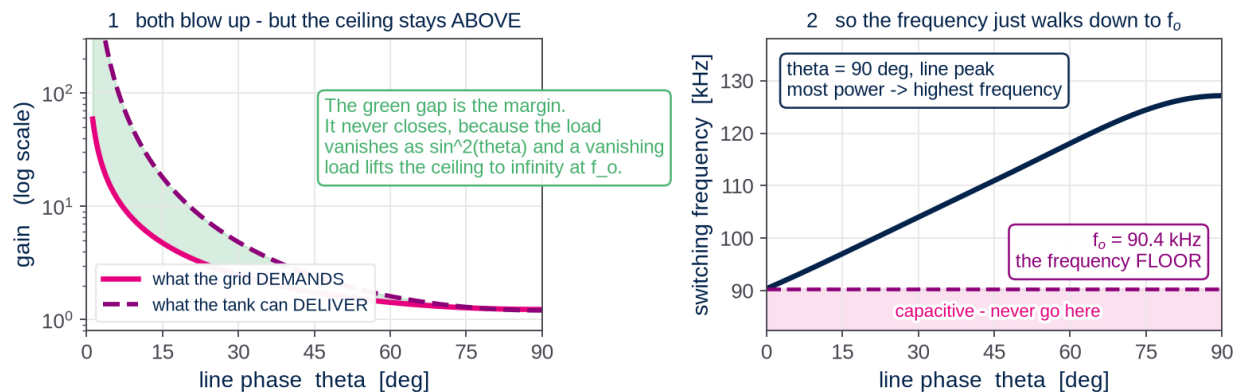


**Figure 34: Which side of  $f_r$  the converter is on over a line half cycle, at four equivalent inputs. At the low corner it never leaves the boosting region; at the high corner it spends most of the cycle bucking. Every curve converges on  $f_o$  at the zero crossing.**

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| Line condition | $V_{eq}$ | peak $f_{sw}$ | side of $f_r$ |
|----------------|----------|---------------|---------------|
| minimum line   | 180 V    | 131.3 kHz     | below         |
| tank minimum   | 173 V    | 127.1 kHz     | below         |
| 110 Vac        | 220 V    | 156.4 kHz     | <b>above</b>  |
| nominal        | 225 V    | 159.7 kHz     | <b>above</b>  |
| 230 Vac        | 230 V    | 163.2 kHz     | <b>above</b>  |
| maximum line   | 264 V    | 188.5 kHz     | <b>above</b>  |
| FB threshold   | 332 V    | 249.0 kHz     | <b>above</b>  |

**Table 7: Peak  $f_{sw}$  over the half cycle against  $f_r = 151.7$  kHz. 5 of the seven line conditions cross into above-resonance operation for part of the cycle.**



**Figure 35: Near the zero crossing the required gain diverges and the load vanishes together, so the operating point converges on  $f_o = 90.4$  kHz (Section 4.2).**

## 7.7 The currents this design has to carry

Ratings come from the worst switching cycle and losses from the line-cycle rms, so both are listed. The row to watch is the composite tank peak: it is what the primary devices and  $R_{CS}$  see, and it is neither the sum of the two component peaks nor the larger of them.

## Single-Stage PFC LLC Converter Design

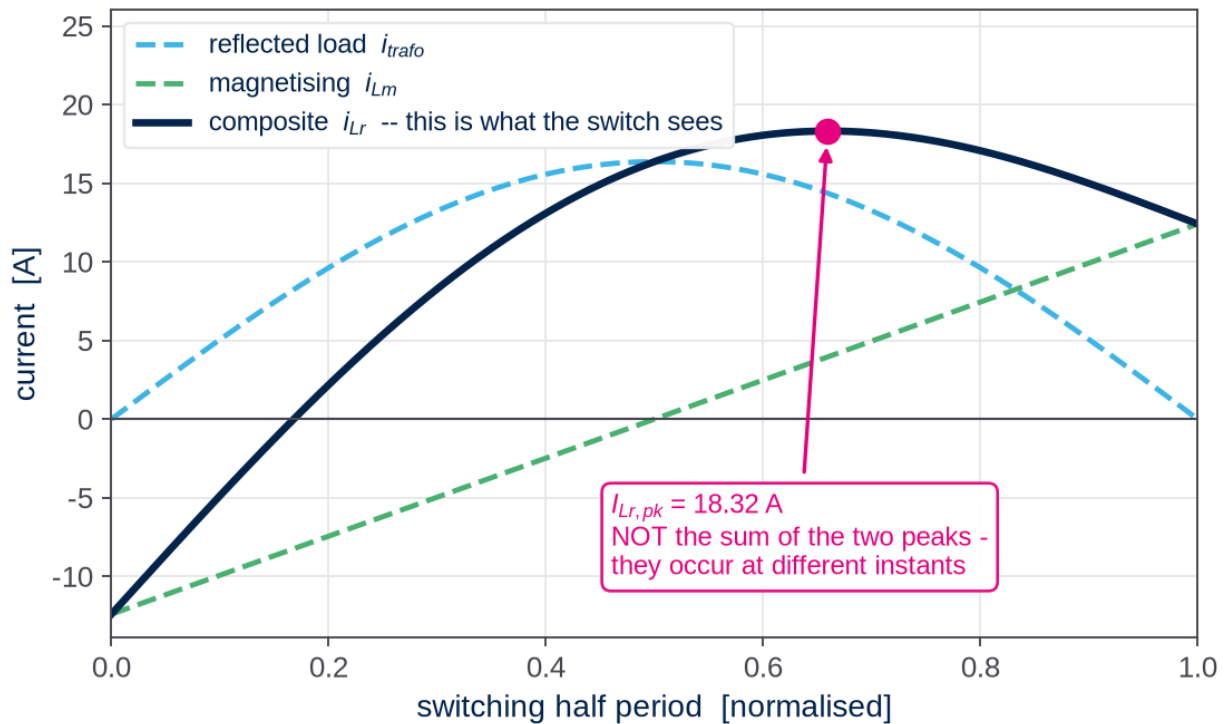


Figure 36: The composite tank current at the low equivalent corner, full load, and why its peak is not the sum of the two component peaks: they occur at different instants.

| Quantity                            | Value          | Where it is used                         |
|-------------------------------------|----------------|------------------------------------------|
| $f_{sw}$                            | 127.1 kHz      | oscillator floor check                   |
| $I_{sec,pk}$                        | 98.6 A         | secondary rectifier peak                 |
| $I_{trafo,pk}$                      | 16.37 A        | reflected load component                 |
| $I_{Lm,pk}$                         | 12.41 A        | magnetising component                    |
| <b>composite tank peak</b>          | <b>18.32 A</b> | <b>primary device rating, OCP margin</b> |
| $I_{pri,rms}$ (worst cycle)         | 13.76 A        | primary rms rating                       |
| $I_{pri}$ (line-cycle rms)          | 10.93 A        | primary conduction loss                  |
| $I_{rect}$ per leg (line-cycle rms) | 28.34 A        | secondary conduction loss                |
| $I_{Cout}$ (line-cycle rms)         | 30.24 A        | output bank ripple current               |

Table 8: Currents at the low equivalent corner, full load,  $\theta = 90^\circ$ .

### 7.8 The transformer, as built

The tank asks for  $L_r = 11 \mu\text{H}$ ,  $L_m = 20 \mu\text{H}$  and  $n_T = 7.50$ . At 26.3 A out the secondary wants to be a few heavy turns, and a single core that carries the flux at that few turns comes out too large, so the transformer is 3 identical units, primaries in series and secondaries in parallel (Section 6.21). The secondary of each unit is two windings of  $N_s = 2$  turns, NS2 and NS3, joined at the centre tap; one conducts in each half period.

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| Quantity                  | Per unit                    | Assembly               | Computed as                                                                                                                                       |
|---------------------------|-----------------------------|------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| Turns                     | $N_p$ 5 T; NS2 2 T, NS3 2 T | $N_p$ 15 T : $N_s$ 2 T | primaries in series, secondaries in parallel                                                                                                      |
| Open-circuit inductance   | 10.3 $\mu$ H                | 31.0 $\mu$ H           | $L_r + L_m$ , every other winding open                                                                                                            |
| Short-circuit inductance  | 3.67 $\mu$ H                | 11.0 $\mu$ H           | this is $L_r$ : no separate resonant inductor                                                                                                     |
| $A_L$                     | 413 nH                      | —                      | open-circuit inductance per unit / $N_p^2$                                                                                                        |
| Peak flux density         | 182 mT                      | —                      | eq. 23: $V_{o,eff} / (4 f_r N_s A_e) = 25.0 / (4 \times 151.75 \text{ kHz} \times 2 \times 113.1 \text{ mm}^2)$ ; $N_p$ does not enter            |
| Core area needed          | $\geq 103 \text{ mm}^2$     | —                      | the same at $B_{max} = 0.20 \text{ T}$ ; halving $N_s$ would double it                                                                            |
| Magnetising peak $I_{eq}$ | 12.41 A                     | —                      | eq. 28: $B_{pk} N_p A_e / L_{\mu} = 0.1821 \text{ T} \times 5 \times 113.1 \text{ mm}^2 / 8.30 \mu\text{H}$ , equal to $i_{\mu,pk}$ as it must be |
| Saturation test current   | 15.2 A                      | —                      | eq. 29: $I_{eq} \times V_{OVP2} / V_{out} = 12.41 \times 1.2244$ — <b>not</b> the 18.3 A tank peak                                                |

Table 9: The transformer as wound and specified, with the arithmetic behind the computed rows.

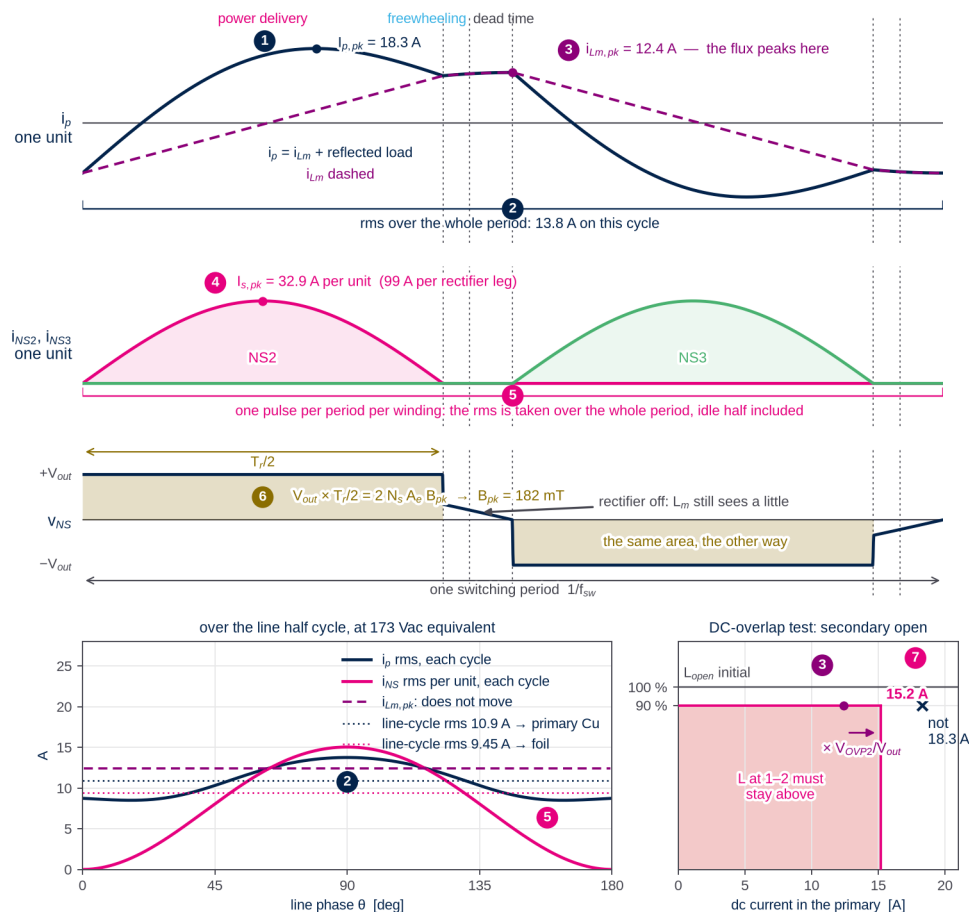
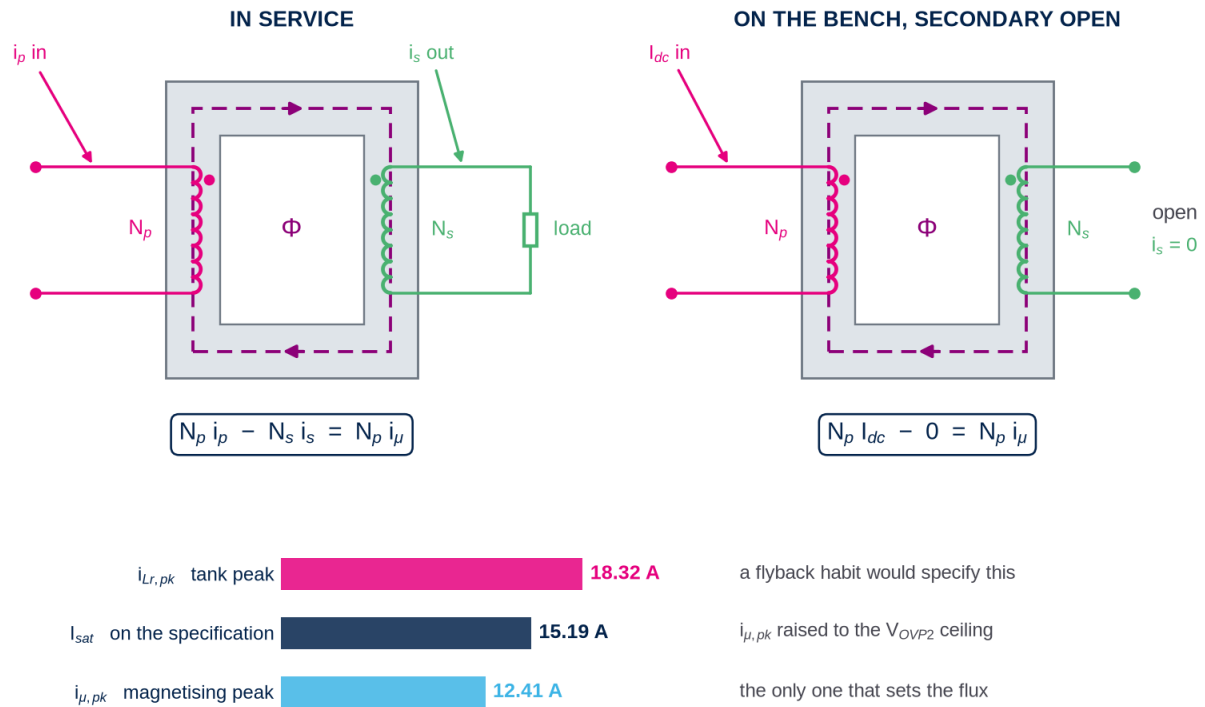


Figure 37: Where each of those numbers is read. Top: the primary winding current of one unit, the current in its two secondary windings and the secondary winding voltage over one switching period, at the worst cycle (line peak, 173 Vac equivalent, full load). Lower left: the rms of the two winding currents over the line half cycle, and the line-cycle values the copper is sized on. Lower right: the DC-overlap test, a bench condition. The circled marks are the rows of Table 2.

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**Figure 38: Why the test current is the magnetising peak and not the tank peak. In service the secondary cancels most of the primary ampere-turns and only the difference magnetises the core; on the bench, with the secondary open, the whole test current is magnetising current, so the design flux is reached at 12.41 A, not at the 18.32 A the primary carries.**

## 7.9 Choosing the core, and building the winding on it

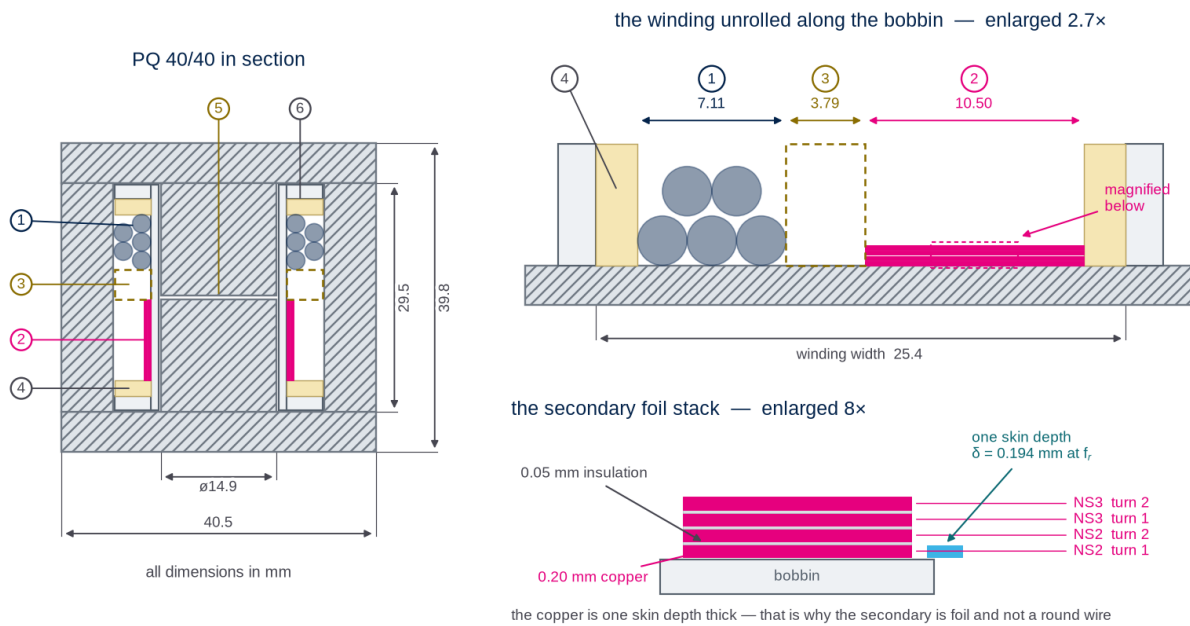
The tank has fixed what the transformer must be:  $L_{open} = 10.3 \mu H$  and  $L_{short} = 3.67 \mu H$  per unit, 5 : 2 turns, and a leakage that is 35 % of the open-circuit inductance. Two standard TDK PQ sets are tried against those three conditions.

|                                | PQ 32/30              | PQ 40/40              | what it has to beat                         |
|--------------------------------|-----------------------|-----------------------|---------------------------------------------|
| $A_e$                          | 153.8 mm <sup>2</sup> | 189.0 mm <sup>2</sup> | $\geq 103 \text{ mm}^2$ at 0.20 T           |
| $A_{min}$                      | 127.5 mm <sup>2</sup> | 174.0 mm <sup>2</sup> | the section that really saturates           |
| $B_{pk}$                       | 134 mT                | 109 mT                | at $A_{min}$ : 162 and 118 mT               |
| $N_s$ needed                   | 1.34 → 2              | 1.09 → 2              | a whole number, so 2 on either              |
| window $A_N$                   | 104 mm <sup>2</sup>   | 309 mm <sup>2</sup>   | $\geq 68 \text{ mm}^2$ for the copper alone |
| window used                    | 66 %                  | 22 %                  | before margins and the separation           |
| mean turn $l_N$                | 62 mm                 | 87 mm                 | copper length, so copper loss               |
| gap for $A_L = 413 \text{ nH}$ | 0.47 mm               | 0.57 mm               | first estimate, before fringing             |

**Table 10: Two candidate cores against the three conditions. Datasheet figures on the left, this design on the right.**

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**The window decides, not the flux.** Both cores pass the flux condition ( $A_e$  clears the requirement by 1.49 and 1.83) and either takes the same two secondary turns. The copper alone takes 66 % of the smaller window and 22 % of the larger, before the margins and the separation that the leakage needs, so the smaller core has nothing left to give. **PQ 40/40 is taken**, N95 or an equivalent low-loss grade, gapped to  $A_L$ ; the price is a mean turn 40 % longer and a set 57 % heavier.



**Figure 39: The chosen core in section, the winding unrolled along the bobbin, and the secondary foil stack, to scale from the TDK drawings. Primary and secondary sit side by side, and the 3.79 mm left between them is the deliberate separation that makes  $L_{short}$  come out at 3.67  $\mu$ H per unit. The circled numbers are the rows of Table 11.**

| Mark | Item                     | Value               | Note                                                                                                                |
|------|--------------------------|---------------------|---------------------------------------------------------------------------------------------------------------------|
| 1    | Primary NP1, pins 1–2    | 5 T                 | Litz 310 $\times$ $\varnothing$ 0.10 mm, bundle $\varnothing$ 2.37 mm, 2 layers (3 + 2); conducts every half period |
| 2    | Secondary NS2, pins 3–5  | 2 T                 | foil 0.20 $\times$ 10.5 mm; conducts in the first half period                                                       |
| 2    | Secondary NS3, pins 4–6  | 2 T                 | the same foil, wound on top of NS2; conducts in the second half period                                              |
| 3    | Separation               | 3.79 mm             | the width left over; it is the leakage, $L_{short}/L_{open} = 35 \%$                                                |
| 4    | Margin tape              | 2.0 mm each flange  | creepage to the flange; assumed                                                                                     |
| 5    | Centre-leg gap           | $\approx$ 0.57 mm   | ground to $A_L = 413$ nH; the estimate ignores fringing                                                             |
| 6    | Window $A_N$             | 309 mm <sup>2</sup> | 20.5 mm <sup>2</sup> of bare copper in it                                                                           |
| —    | Auxiliary NAUX, pins a–b | 1 T                 | any wire; ZCD sense only, no load                                                                                   |

**Table 11: Legend of the figure: the windings of one unit.**

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| Step                                               | Primary NP1                                                                                                                                                                  | Secondary NS2                                                                                          |
|----------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|
| Line-cycle rms current per unit (marks 2 and 5)    | 10.93 A — the whole primary current, the primaries are in series                                                                                                             | 28.34 A / 3 units = 9.45 A                                                                             |
| Copper at $J = 4.5 \text{ A/mm}^2$                 | $10.93 / 4.5 = \mathbf{2.43 \text{ mm}^2}$                                                                                                                                   | $9.45 / 4.5 = \mathbf{2.10 \text{ mm}^2}$                                                              |
| Conductor ( $2\delta = 0.39 \text{ mm}$ at $f_r$ ) | Litz: $2.43 \text{ mm}^2 / (\pi \times 0.10^2/4) = \mathbf{310 \text{ strands}}$ of $\varnothing 0.10 \text{ mm}$ ; served bundle $\varnothing 2.37 \text{ mm}$ at 55 % fill | Foil 0.20 mm thick (one skin depth, all of it conducts): $2.10 / 0.20 = \mathbf{10.5 \text{ mm}}$ wide |
| Turns                                              | 5, in 2 layers (3 + 2)                                                                                                                                                       | 2 (NS3 another 2 on top)                                                                               |
| Width taken on the bobbin                          | $3 \times 2.37 = \mathbf{7.11 \text{ mm}}$                                                                                                                                   | the foil width, $\mathbf{10.5 \text{ mm}}$                                                             |
| Radial build                                       | $2 \times 2.37 = 4.74 \text{ mm}$                                                                                                                                            | NS2 + NS3: $4 \times (0.20 + 0.05 \text{ insulation}) = 1.00 \text{ mm}$                               |
| Left for the separation                            | $25.4 - 2 \times 2.0 - 7.11 - 10.5 = \mathbf{3.79 \text{ mm}}$ (mark 3)                                                                                                      |                                                                                                        |
| Bare copper in the window                          | $5 \times 2.43 + 4 \times 2.10 = \mathbf{20.5 \text{ mm}^2}$ ; at $k_u = 0.30$ that is $68 \text{ mm}^2$ , 22 % of $A_N$                                                     |                                                                                                        |

**Table 12: From current to copper: how the numbers in the figure come about, for one unit. NS3 is identical to NS2.**

Reduced to what a supplier can measure at the terminals, this gives the specification sheet.  $L_\mu$  is deliberately absent (Section 6.10).

| Item                            | Value                                                    | Condition                                           |
|---------------------------------|----------------------------------------------------------|-----------------------------------------------------|
| Turns                           | $N_p = 15 \text{ T}$ (5 T per unit); NS2 2 T and NS3 2 T | two secondary windings, centre tap outside the part |
| Open-circuit inductance         | $31.0 \mu\text{H}$ , -4.1 % at worst                     | primary, all other windings open                    |
| Leakage inductance              | $11.0 \mu\text{H} \pm 10 \%$                             | primary, secondary shorted                          |
| DC overlap                      | $\geq 90 \%$ of initial inductance                       | test current 15 A (mark 7)                          |
| Primary current                 | 10.9 A rms / 18.3 A pk                                   | line-cycle rms (mark 2), composite peak (mark 1)    |
| Secondary current, each winding | 28.3 A rms / 99 A pk                                     | line-cycle rms (mark 5), peak (mark 4)              |
| Core area                       | $A_e \geq 103 \text{ mm}^2$                              | holds $B_{pk}$ at or below 0.20 T (mark 6)          |
| Switching frequency             | 127 to 249 kHz                                           | at full load                                        |

**Table 13: Transformer specification for the worked design.**

## 7.10 The output bank, as sized

Both conditions of Section 6.23, on this specification. The larger wins.

| Condition                       | Eq. | Computed as                                                                                                                                                                             | Asks for               |
|---------------------------------|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|
| $2f_r \text{ ripple} \leq 5 \%$ | 31  | $P_{out} / (2\pi f_{l,min} \Delta v V_{out}^2) = 657.5 / (2\pi \times 47 \times 0.05 \times 625)$                                                                                       | <b>71.25 mF</b> — wins |
| Hold-up 12 ms to 19 V           | 32  | $2P_{out} T_{hold} / ((V_{out} - \frac{1}{2}\Delta v_{pp})^2 - V_{o,min}^2) = 2 \times 657.5 \times 0.012 / (24.41^2 - 19^2)$ ; started at $V_{out}$ instead it would ask only 59.77 mF | 67.22 mF               |
| Selected                        | —   | 470 $\mu\text{F} \times 160$ , the next assembly up from 71.2 mF                                                                                                                        | <b>75.2 mF</b>         |

**Table 14: The two sizing conditions, and what was fitted.**

The two conditions are only 1.060 apart. The screening inequality of Section 6.23 reads 0.373 against 0.354 at  $k = 0.760$ , so ripple wins; relaxed to 10 % ripple, the two sides would swap.

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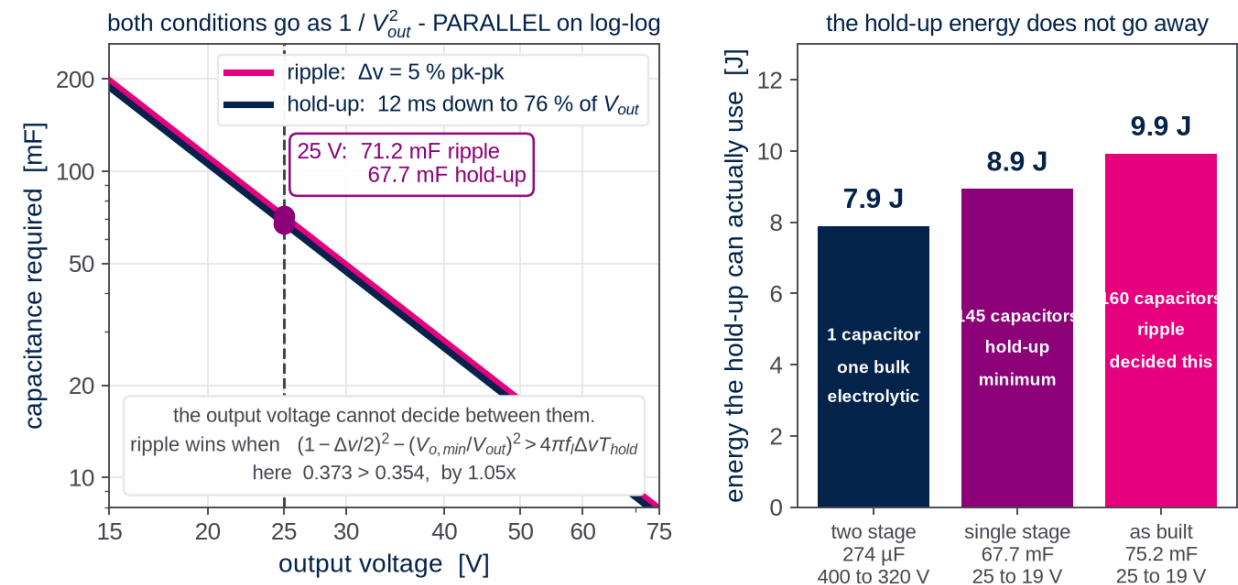


Figure 40: Left: both sizing conditions fall as  $1/V_{out}^2$ , so only the specification separates them; at 25 V ripple asks for 71.2 mF and hold-up for 67.2 mF. Right: the hold-up energy is the same 7.9 J either side of the boost stage; on a 400 V bus falling to 320 V it is a 274  $\mu$ F part, on this output it is 67.2 mF, 245 times more, because the usable voltage window is so much smaller. That ratio is the price of the architecture.

| Quantity                            | Value              | Against                                                                    |
|-------------------------------------|--------------------|----------------------------------------------------------------------------|
| Achieved ripple                     | 1.18 V (4.74 %)    | 5 % allowed                                                                |
| Achieved hold-up                    | 13.42 ms           | 12 ms required — 15.10 ms if started at $V_{out}$ instead of at the trough |
| Ripple current, switching component | 23.8 A rms         | —                                                                          |
| Ripple current, $2f_l$ component    | 18.6 A rms         | orthogonal to the above                                                    |
| Ripple current, total               | <b>30.24 A rms</b> | 0.19 A in each of 160 parts                                                |
| Bank ESR                            | 0.1875 m $\Omega$  | 30 m $\Omega$ each, 160 in parallel                                        |
| Ripple and noise                    | 17.4 mV            | with the 10 $\mu$ F ceramic bypass                                         |

Table 15: What the selected bank then delivers, and what it has to survive.

## 7.11 Where the power goes

The specification allowed 27.3 W of loss between  $P_{in} = 684.8$  W and  $P_{out} = 657.5$  W, split three ways before any component existed. The device losses computed afterwards do not have to agree with that split, and here they do not.

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| Item                                | Value          | Note                                                              |
|-------------------------------------|----------------|-------------------------------------------------------------------|
| <b>Budget</b> — input bridge        | 5.52 W         | synchronous bridge assumed                                        |
| <b>Budget</b> — EMI filter          | 8.34 W         | —                                                                 |
| <b>Budget</b> — LLC stage           | 13.42 W        | $\eta_{HB} = 98\%$ of $P_{in,LLC}$                                |
| Primary conduction, standing device | 9.65 W         | the half-bridge leg that never switches — the single largest item |
| Primary switching, three positions  | 4.83 W         | not a problem                                                     |
| Secondary rectifiers                | 5.65 W         | 1.411 W per device, 2 in parallel per leg                         |
| Current-sense resistors             | 2.86 W         | 4.54 W at the worst switching cycle                               |
| Output capacitor ESR                | 0.162 W        | —                                                                 |
| <b>Itemised total</b>               | <b>23.15 W</b> | and the magnetics are not in it                                   |

**Table 16: The budget as assumed, and the device losses as computed. The second block is not a breakdown of the first.**

**The efficiency assumption is optimistic.** The itemised device losses already come to 23.2 W against the 13.4 W that  $\eta_{HB} = 98\%$  allows the LLC stage, before the transformer. Nothing electrical depends on it (95 % moves  $R_{ac}$  and  $R_{CS}$  by about 3 %); the thermal design does, and that is settled by measurement.

## 7.12 What the semiconductors have to be

Requirements, not part numbers; the four rules of Section 6.24 decide whether they are stated correctly.

| Item                            | Requirement                                                          |
|---------------------------------|----------------------------------------------------------------------|
| Primary drain-source voltage    | $\geq 485$ V, so a 600 V class part                                  |
| Primary body diode              | fast recovery: in full bridge the body diode conducts before turn-on |
| Primary $R_{DS(on)}$ per device | $\leq 25.2$ m $\Omega$ <b>hot</b> to meet the loss budget            |
| Primary $C_{o(tr)}$             | small enough that the tank can swing the node inside 220 ns          |
| Secondary drain-source voltage  | $\geq 80$ V including the centre-tap doubling                        |
| Secondary rectifier current     | 28.34 A rms per leg, 98.6 A peak                                     |
| Secondary package               | check the lead and clip rating, not only the die                     |

**Table 17: Semiconductor requirements for the worked design.**

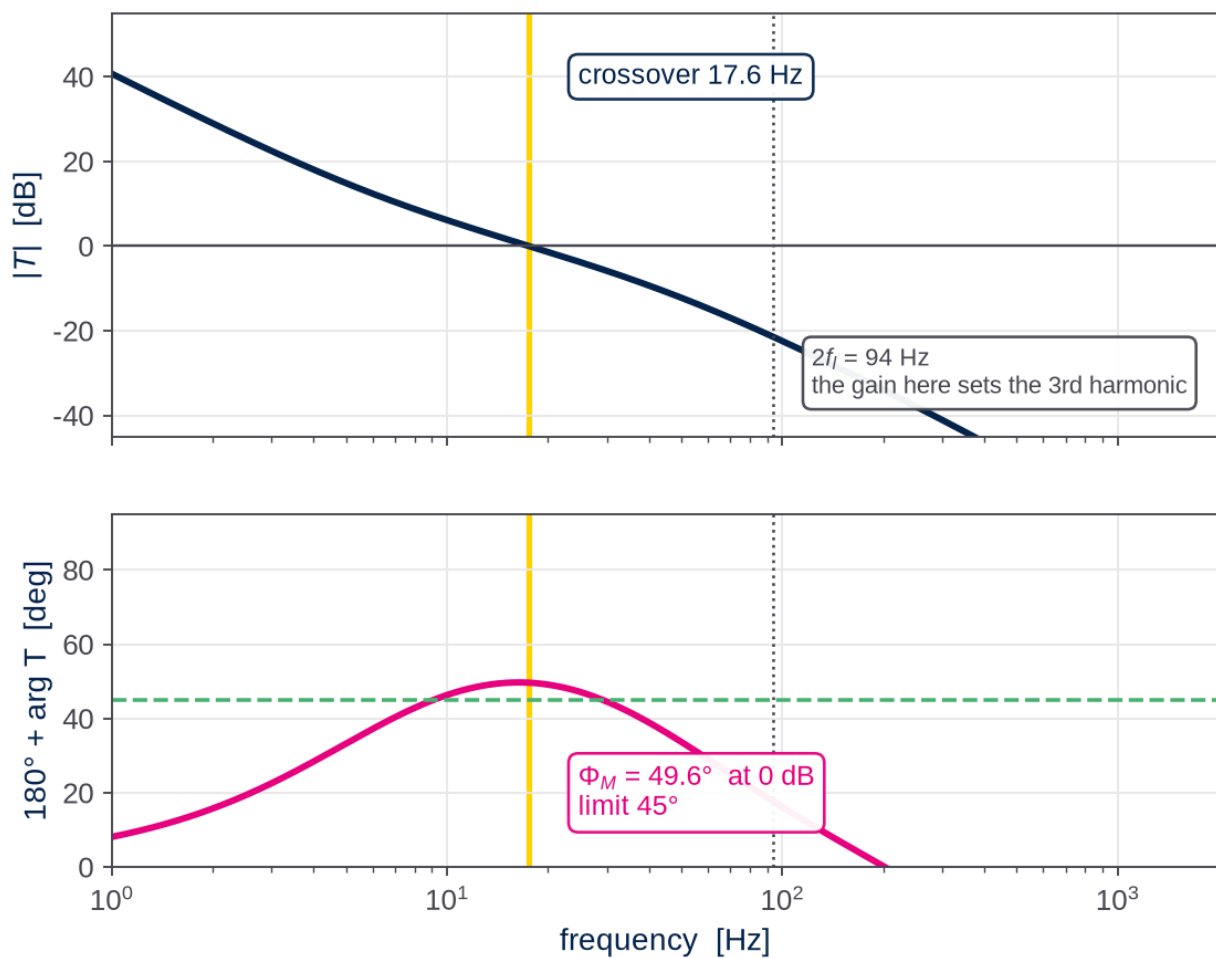
## 7.13 The voltage loop, as built

Section 6.26 gives the method; these are its numbers on this converter, in the order they are found.

# Single-Stage PFC LLC Converter Design

| # | Quantity                              | Value                                                                | From                                                                                     |
|---|---------------------------------------|----------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| 1 | Output divider                        | $R_i$ 100 k $\Omega$ / $R_o$ 11 k $\Omega$                           | sets the regulated output against the 2.495 V reference                                  |
| 2 | Plant gain $G_o$                      | 130.06 rad/s                                                         | a pure integrator — the output is a capacitor fed by a power source                      |
| 3 | Error-amplifier gain needed at $2f_l$ | 0.4541                                                               | what the loop needs there, against what the amplifier has                                |
| 4 | K factor $K_v$                        | 2.7218                                                               | places the zero and pole symmetrically                                                   |
| 5 | Geometric mean $f_{MB}$               | 16.921 Hz                                                            | the point they are placed about                                                          |
| 6 | Zero / pole                           | 6.217 / 46.057 Hz                                                    | $f_{MB}/K_v$ and $K_v f_{MB}$                                                            |
| 7 | High-frequency pole $f_{px}$          | 999.4 Hz                                                             | added to roll off switching noise                                                        |
| 8 | Components                            | $C_{F0}$ 82 nF, $C_F$ 560 nF, $R_F$ 47 k $\Omega$ , $C_{fx}$ 1.55 nF | the nearest standard values to the pole and zero above                                   |
| 9 | Optocoupler bias                      | $R_p$ 1.3 k $\Omega$ , $R_B$ 6.2 k $\Omega$                          | $R_B$ must land inside 5.76 to 6.48 k $\Omega$ or the anti-saturation circuit cannot act |

**Table 18: The compensator, from the plant to the parts.**



**Figure 41: Open-loop gain of the selected compensation network, and  $180^\circ + \arg T$ . That distance is the phase margin only where  $|T|$  crosses 0 dB.**

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| Quantity                            | Value              | Against                                |
|-------------------------------------|--------------------|----------------------------------------|
| Crossover $f_{cross}$               | 17.57 Hz           | the 15 to 20 Hz this topology lands in |
| Phase margin                        | 49.64°             | 45° as a floor                         |
| Gain margin                         | 34.18 dB at 202 Hz | 6 dB floor, 10 dB comfortable          |
| Third harmonic on the input current | 4.23 %             | the price of crossing this low         |
| Feedback ripple at the burst point  | 52 mV              | move $R_{BM}$ down 2.60 kΩ to clear it |

Table 19: And what that loop then measures as.

## 7.14 The parts around the controller

These are the results. The sections after this one derive each value in the order it has to be done, because every one of them is sized against something already fixed.

| Part         | Value                            | Result                                      |
|--------------|----------------------------------|---------------------------------------------|
| $C_T / R_T$  | 470 pF / 11 kΩ                   | $f_{Min}$ 92.2 kHz, $f_{Max}$ 319.6 kHz     |
| $R_{CS}$     | 24.0 mΩ (120 mΩ × 5 in parallel) | OCP1 at 1.251 × the composite peak          |
| $R_{CFG}$    | 30 kΩ                            | $V_{BO}$ 84.8 V, morphing enabled           |
| $R_{BM}$     | 30 kΩ                            | burst-mode entry point                      |
| ZCD divider  | 330 kΩ / 19 kΩ                   | OVP1 28.16 V, OVP2 30.61 V                  |
| $C_{in}$     | 2200 nF film                     | about 3.2 nF/W — there is no bulk capacitor |
| Compensation | 82 nF / 560 nF / 47 kΩ / 1.55 nF | $f_{cross}$ 17.57 Hz, $\phi_M$ 49.64°       |

Table 20: Controller network for the worked design.

## 7.15 The oscillator: $C_T$ first, then $R_T$

The VCO charges  $C_T$  from the sum of a fixed current  $V_{ref}/R_T$  and the error-amplifier current  $I_{EA}$ , up to  $V_{ref} = 1.5$  V, then adds a fixed idle time:

$$\frac{T_{sw}}{2} = \frac{C_T V_{ref}}{\frac{V_{ref}}{R_T} + I_{EA}} + T_{idle} \quad (40)$$

$I_{EA}$  is the only control input in the converter. More current means a higher frequency, which means less power. Everything the control loop does, it does through this one term.

$C_T$  is chosen first because it sets the *span*, and  $R_T$  after it because it sets the *floor*:

$$C_{T,max} = \frac{I_{EA,max}}{2V_{ref}} \cdot \frac{(1 - 2T_{idle}f_{sw,max})(1 - 2T_{idle}f_{sw,min})}{f_{sw,max} - f_{sw,min}}, \quad C_{T,min} = \frac{1}{R_{T,max}} \left( \frac{1}{2f_{sw,min}} - T_{idle} \right) \quad (41)$$

$$R_{T,ceil} = \frac{1}{C_T} \left( \frac{1}{2f_{sw,min}} - T_{idle} \right) \quad (42)$$

with  $f_{sw,min}$  set to  $f_o$  — the frequency floor of the whole design, for the reason in Section 4.2. Here that window is 176 to 703 pF, the datasheet allows 270 to 1000 pF, and the part is taken from the middle of the overlap rather than from either edge:  $C_T = 470$  pF, COG. Then  $R_{T,ceil} = 11.24$  kΩ gives  $R_T = 11$  kΩ.

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**$R_{T,ceil}$  is a maximum, not a minimum**, whatever the name suggests.  $f_{Min} = 1/[2(C_T R_T + T_{idle})]$  falls as  $R_T$  grows. Round this number up and the VCO clamp drops below  $f_o$ , leaving only the anti-capacitive protection between the converter and the capacitive region. Round it **down**.

The three frequencies the selected pair actually produce are

$$f_{Min} = \frac{1}{2(C_T R_T + T_{idle})}, \quad f_{Max} = \frac{1}{2\left(\frac{C_T}{\frac{I_{EA,max}}{V_{ref}} + \frac{1}{R_T}} + T_{idle}\right)} \quad (43)$$

Putting the selected pair into the first of those, with  $T_{idle} = 250$  ns:  $f_{Min} = 1/[2(470 \text{ pF} \times 11 \text{ k}\Omega + 0.25 \text{ }\mu\text{s})] = 1/(2 \times 5.420 \text{ }\mu\text{s}) = \mathbf{92.25 \text{ kHz}}$ , which is what has to clear  $f_o$ .

| Quantity  | Value              | Limit                                  | Margin        |
|-----------|--------------------|----------------------------------------|---------------|
| $f_{Min}$ | 92.2 kHz           | above $f_o = 90.4 \text{ kHz}$         | 1.021         |
| $f_{Max}$ | 319.6 kHz          | above $f_{sw,max} = 249.0 \text{ kHz}$ | 1.283         |
| $f_{SU}$  | 503.5 kHz          | below the 675 kHz silicon ceiling      | 1.341         |
| $R_T C_T$ | 5.17 $\mu\text{s}$ | 2.5 to 12 $\mu\text{s}$                | 2.068 / 2.321 |
| $R_T$     | 11 k $\Omega$      | 5 to 30 k $\Omega$                     | 2.200 / 2.727 |
| $C_T$     | 470 pF             | 270 to 1000 pF                         | 1.741 / 2.128 |

**Table 21: Oscillator: what the two parts produce, and the limits each result has to clear.**

**$T_{idle}$  is the weakest number in this chapter.** The draft datasheet gives 700 ns in the text and a table that back-solves to 250 ns; this design is worked at 250 ns. At 700 ns the same  $R_T$  and  $C_T$  put  $f_{Min}$  essentially on top of  $f_o$  and the floor margin disappears. **It is not a precision question, it is whether the two clamps still bracket the operating range**, and it is the first thing to measure on hardware.

## 7.16 The current-sense resistor, which sets three things at once

$R_{CS}$  is asked for by two conditions and the smaller one wins:

$$R_{CS,max1} = \frac{16.8 \Omega \cdot W}{P_{in}} \quad R_{CS,max2} = \frac{0.55 \text{ V}}{I_{Lr,pk}} \quad (44)$$

The first is the maximum-power law of Section 4.4; the second is the OCP1 trip point. Opened out, equation 44 is two divisions —  $16.8/P_{in}$  and  $0.55/I_{Lr,pk}$  — and they give 24.53 and 30.03 m $\Omega$ , so the power law is the limiting one. Dissipation then fixes how many resistors that has to be split over. At the worst switching cycle that is 4.54 W, so **5 in parallel**; 120 m $\Omega$  each gives  **$R_{CS} = 24.0 \text{ m}\Omega$** .

**The denominator of  $R_{CS,max2}$  is the composite tank peak, not the reflected load current.**  $R_{CS}$  sits in the bridge return, so it carries the magnetising component too. Using  $I_{trafo,pk} = 16.37 \text{ A}$  instead of  $I_{Lr,pk} = 18.32 \text{ A}$  loosens the limit by about 12 % and the over-current protection with it.

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| Result              | Value                      | Against                                                |
|---------------------|----------------------------|--------------------------------------------------------|
| Maximum input power | 700.0 W                    | $P_{in} = 684.8 \text{ W}$                             |
| OCP1 trip           | 22.92 A                    | composite peak 18.32 A, margin 1.251                   |
| OCP2 trip           | 31.25 A                    | immediate stop, 50 $\mu\text{s}$ , restart at $f_{SU}$ |
| Sense dissipation   | 4.54 W total, 0.909 W each | 1 W parts, margin 1.100                                |

Table 22: What the selected  $R_{CS}$  then fixes.

## 7.17 Burst mode: one resistor, read once at power-up

$$R_{BM} = 16.7 \frac{\text{k}\Omega}{\sqrt{2}} R_{CS} P_{in, BM}, \quad V_{BM, eq} = \frac{R_{BM}}{100 \text{ k}\Omega} + 0.5 \text{ V} \quad (45)$$

Entering burst at 75 W of input power — about 11 % of rated — asks for 30.06 k $\Omega$ , so  $R_{BM} = 30 \text{ k}\Omega$  and the equivalent threshold on the FB pin is 0.800 V. The valid range is 15 to 140 k $\Omega$ . Tie the pin to ground and burst mode is off altogether; deep burst stays active either way.

This threshold and the feedback ripple have to be checked against each other, or the converter chatters in and out of burst once per half line cycle — Section 6.28.

## 7.18 Brown-out and bridge configuration: the CFG pin

One resistor does two unrelated jobs, and both are read once at power-up:

$$V_{BO, pk} = R_{CFG} \times 4 \frac{\text{V}}{\text{k}\Omega}, \quad V_{BO, rms} = \frac{V_{BO, pk}}{\sqrt{2}} \quad (46)$$

The brown-out threshold is compared against the *peak* of the mains, so it has to be converted before it is compared with an rms specification — a  $\sqrt{2}$  that is easy to drop. Asking for brown-out just below the minimum line gives  $R_{CFG, max} = 31.82 \text{ k}\Omega$ ; **this is a maximum, so it rounds down, to 30 k $\Omega$** , and  $V_{BO} = 84.85 \text{ V}_{rms}$  against a 90 Vac minimum — margin 1.061. Rounding up instead would stop the converter starting at low line.

| $R_{CFG}$            | LOUT2  | Configuration                                         |
|----------------------|--------|-------------------------------------------------------|
| 15 k $\Omega$        | open   | morphing, fixed brown-out (60 V / 70 V peak)          |
| 15 to 47 k $\Omega$  | open   | <b>morphing, adjustable brown-out — this design</b>   |
| 47 to 100 k $\Omega$ | open   | fixed full bridge                                     |
| 15 k $\Omega$        | to GND | fixed half bridge, split $C_r$ , fixed brown-out      |
| 15 to 100 k $\Omega$ | to GND | fixed half bridge, split $C_r$ , adjustable brown-out |

Table 23: What  $R_{CFG}$  and LOUT2 select together. The 235 and 245 V thresholds are fixed inside the IC and cannot be moved.

**For a universal-input design there is only one window.** The datasheet floor is 15 k $\Omega$  and brown-out at the minimum line is the ceiling at 31.8 k $\Omega$ , so the 47 k $\Omega$  morphing limit never comes into play. And hang nothing on LOUT2. The pin-strap drives 300  $\mu\text{A}$  into it and needs 2.5 V, so any pull-down under about 8 k $\Omega$  reads as a request for a fixed half bridge.

## 7.19 Output sensing and over-voltage: the ZCD divider

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The auxiliary winding is divided down to the ZCD pin, and the divider ratio alone sets both over-voltage thresholds — the absolute resistor values only set the bias current:

$$R_{ZCD,L} = \frac{2.3 \text{ V}}{I_{bias}}, \quad R_{ZCD,H} = R_{ZCD,L} \left( \frac{n_{aux}}{n_{sec}} \frac{V_{OVP1,out}}{2.3 \text{ V}} - 1 \right) \quad (47)$$

$$V_{OVP1} = \frac{2.3 \text{ V}}{n_{aux}/n_{sec}} \left( \frac{R_{ZCD,H}}{R_{ZCD,L}} + 1 \right), \quad V_{OVP2} = \frac{2.5}{2.3} V_{OVP1} \quad (48)$$

Aiming OVP1 10 % above the output gives 27.50 V as a target. With  $n_{aux}/n_{sec} = 1.5$  the calculated pair is 19.17 / 321.8 kΩ. The parts fitted are **19 kΩ and 330 kΩ**. The upper one is deliberately the *next E24 value above* the calculation, so OVP1 lands clear of the ripple crest rather than on it. The result is OVP1 28.16 V and OVP2 30.61 V, and start-up hands over to the loop at 16.53 V of output.

**Two ways to get this wrong, and both stop the converter completely.** Swapping the two resistors inverts the ratio and OVP1 trips at well under a volt of output, so the converter never starts. And the winding polarity must make ZCD *positive* when the low side of leg 1 is on; reversed, the bridge hard switches from the first pulse.

**The turns-ratio ceiling in the datasheet does not apply here.** It asks that the rectified auxiliary voltage stay under the  $V_{CC}$  rating at OVP2, which is a question only when that winding supplies  $V_{CC}$ . In this design  $V_{CC}$  comes from a housekeeping rail and the winding senses only, so its voltage is free — 37.5 V nominal, 45.9 V at OVP2 — and the only remaining constraint is that the turns come out whole. Carrying that check over from a self-supplied design reports a failure that is not one.

## 7.20 HVSU and $V_{CC}$

- **HVSU connects ahead of the input bridge**, on the ac side, through one 1000 V diode per line. Behind the bridge both the X-capacitor discharge and the brown-out detection stop working, because neither can see the mains disappear.
- The  $V_{CC}$  capacitor has to hold the IC from  $V_{CCCon} = 17 \text{ V}$  down to  $V_{CCOff} = 8 \text{ V}$  for as long as it takes the supply to take over. The start-up unit stops 120 ms after  $V_{CCCon}$  is reached, whether or not the supply has taken over by then.
- Where a linear regulator feeds  $V_{CC}$ , put a diode between its output and the pin so the start-up charging current cannot be pushed back into it.
- A 100 nF bypass sits at the pin itself.

## 7.21 Controller pin rules

These follow from the datasheet and each one causes immediate misbehaviour if broken.

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| Pin          | Rule                                                                                                                                   |
|--------------|----------------------------------------------------------------------------------------------------------------------------------------|
| ISEN         | No filter and no series resistor. The maximum-power law and the over-current threshold share this pin.                                 |
| CFG, BM      | No capacitor. Both are strapped at power-up. An out-of-range CFG resistor latches the controller off.                                  |
| DRV_EN, RT   | No capacitor. Both are sampled briefly every few milliseconds and a capacitor is read as a fault.                                      |
| LOUT2        | No pull-down if morphing is used. A few kilohms to ground reads as a fixed half bridge.                                                |
| HVSU         | Connect ahead of the input bridge, on the ac side. Behind the bridge, both X-capacitor discharge and brown-out detection stop working. |
| ZCD          | Auxiliary winding polarity must make ZCD positive when the low side of leg 1 is on. Reversed, the bridge hard switches.                |
| HOUTx, LOUTx | Logic-level outputs, not gate drivers. External half-bridge drivers are required.                                                      |

**Table 24: Pin rules that must not be broken.**

## 7.22 Protections, and where each threshold is set

Every one of these is a design constant fixed by a resistor this note has already chosen, so a change made for one reason moves a threshold set for another. The table is here because the thresholds are otherwise spread over five sections.

| Protection                        | Set by                                                | This design                                                                |
|-----------------------------------|-------------------------------------------------------|----------------------------------------------------------------------------|
| Brown-out, on the rectified mains | $R_{CFG}$ , read at power-up                          | $V_{BO}$ 84.8 V, clear of the 90 Vac minimum by 1.061                      |
| Cycle-by-cycle over-current, OCP1 | $R_{CS}$ — the same resistor as the maximum-power law | 22.92 A against a composite peak of 18.32 A, a margin of 1.251             |
| Second-level over-current, OCP2   | $R_{CS}$ , fixed ratio to OCP1                        | 31.25 A                                                                    |
| Output over-voltage, OVP1         | the ZCD divider on the auxiliary winding              | 28.16 V, above the ripple crest and below OVP2                             |
| Second-level over-voltage, OVP2   | the same divider                                      | 30.61 V                                                                    |
| Capacitive-mode protection        | internal; it watches the ZCD edge against the gate    | the last line of defence only — the frequency floor $f_{Min}$ is the first |
| Burst mode at light load          | $R_{BM}$ , read at power-up                           | 30 k $\Omega$ ; the $2f_l$ feedback ripple is 52 mV, so see Section 6.28   |
| Loss of ZVS at the worst corner   | not a protection — a design margin                    | swept, not estimated; Section 6.5                                          |

**Table 25: Protections and the component that sets each one.**

**Two of these share one resistor.**  $R_{CS}$  sets the over-current thresholds *and* the maximum-power law, because the controller reads both from the same pin. Changing it to fix a power limit moves OCP1, and the other way round. That is also why the pin tolerates no filter.

## 7.23 System design rules

The same material as a checklist, in the order a design meets it. Every line is argued for somewhere above; collected here so a reviewer can run down the list without re-reading the argument.

# Single-Stage PFC LLC Converter Design

- **Design the tank at the real ends of the equivalent range.** With morphing that is 173.2 and 332.3 Vac, not 2×90 and 264. Looking only at the mains range misses both worst corners.
- **Expect  $\lambda$  near 0.5.** The 5-to-10 inductance ratio of a two-stage LLC will not start at low line.
- **Keep the oscillator floor above  $f_o$ ,** and treat the anti-capacitive protection as the last line, not the first.
- **Size the output bank from both conditions** — ripple and hold-up — take the larger, and then check the rms ripple current separately. The capacitance and the current are different questions.
- **Put the voltage-loop crossover in the tens of hertz,** then deal with the feedback ripple that survives by moving  $R_{BM}$ , not by speeding the loop up.
- **Measure both morphing edges,** at full load and at light load. The half-bridge edge is the worst case for gain and ZVS, the full-bridge edge for switching frequency.
- **Check the turns ratio as a reflected voltage,**  $nV_{o,eff}$ , never as a bare ratio — that is what keeps  $n$  and  $n_T$  from being interchanged.
- **Judge the over-current margin on the composite tank peak,** not on the reflected load current.
- **Include the  $2f_l$  component** in the output capacitor ripple current, and judge a centre tap at the output node rather than per winding.
- **Carry the conduction ratio  $\sqrt{d}$  into the secondary rms.** Dropping it overstates the rms by about 12 % here, and the loss — which goes as the square — by about 26 %.
- **Decide how  $L_r$  is built before specifying the transformer,** and put open-circuit and short-circuit inductance on the drawing alongside the turns.
- **Use capacitor ESR at the switching frequency,** not the 120 Hz  $\tan\delta$ .
- **Never round  $L_m$  up.** It lowers  $\lambda$  and takes the ZVS margin with it.
- **Verify ZVS by sweeping the selected tank,** not with the closed form — and if the closed form is used anyway, feed it  $\lambda_{act}$  and the  $Q$  of the moment, and know that its error has no fixed sign.
- **Record calculated and selected values separately,** and make every downstream check read the selected one. A sheet that quietly reads the calculated value reports margins the hardware does not have.

## 7.24 What to measure first on hardware

In order. The first four decide whether the design is sound at all; the rest confirm margins that are computed but never yet seen.

- **$f_{sw}(\theta)$  over a line half cycle.** This settles  $T_{idle}$ , and with it both oscillator clamps — the thinnest margin in the design. Everything else in the controller network hangs off it.
- **Cold start into the full output bank.** A 75.2 mF bank is a far heavier start-up load than a conventional design presents, and the start-up window is finite. This risk is new to this architecture, so there is no earlier design to compare with.
- **ZVS at the half-bridge morphing edge** ( $245 V_{pk}$ , full load), on the gate and mid-point waveforms. The calculation says 460 ns against 220 ns; confirm it is not hard switching, and that the adaptive dead time actually settles where it is assumed to. **Then measure the worst point of the sweep as well** — 25 % load at the 332.3 Vac equivalent corner, 401 ns — because that, not the design corner, is the figure this design is held to.
- **Switching frequency at the full-bridge morphing edge** ( $235 V_{pk}$ , full load). Calculated 249.0 kHz; check it does not run into the VCO ceiling and get power-limited.
- **Input current and THD at 90 Vac, full load,** to put a number on the zero-crossing dead zone.
- **Output ripple against what the load actually tolerates.** The bank is sized from a specification; this is where the specification is tested.
- **Ripple current and temperature rise in the output bank** — 30.24 A rms calculated, 0.19 A per capacitor.
- **Device temperatures, primary and secondary.** This is where the accepted loss-budget shortfall is decided: measure it in **half-bridge morphing**, above  $245 V_{pk}$ , because in full bridge the standing device is switching and

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dissipates far less.

- **Feedback ripple against the burst threshold** at light load.
- **An ac dropout at the worst line phase** — cut the mains at the trough of the ripple and confirm the output is still above 19 V after 12 ms.
- **A step across the morphing band**, both directions, on a programmable source. A ramp gives the loop time to follow and hides the transition.

## 8 References

- STMicroelectronics, *L6790A LLC-PFC controller*, preliminary datasheet, 30 April 2026. **Draft.**
- STMicroelectronics, *EVL6790\_670W* evaluation board schematics.
- Infineon Technologies, *LLC Converter Design Note*, AN 2013-03, V1.0, March 2013.
- Infineon Technologies, *Resonant LLC converter: operation and design*, AN 2012-09, V1.0, September 2012.
- ON Semiconductor / Fairchild, *Half-bridge LLC resonant converter design using FSFR-series Fairchild power switch*, AN-4151.
- Monolithic Power Systems, *Understanding LLC operation*, part 2.
- Toshiba Electronic Devices & Storage, *Power factor correction (PFC) circuits*, application note.
- ROHM Semiconductor, TechWeb, *LLC operating regions* (region framing of Figure 12).
- W. Wenbo et al., *A single-stage 1.65 kW ac-dc LLC converter*, IEEE ECCE.

## 9 Errata and open items

Everything in this chapter is something the note works around rather than something it solves. It is last because none of it is needed to follow the design, and first to be re-read before a production release — most of it is a question that only the released datasheet or a prototype can close.

### 9.1 Datasheet: the draft is not self-consistent

| Item                            | What the draft says                                                                                                                                          | What is used here, and why                                                                                                                |
|---------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| Oscillator idle time $T_{idle}$ | Section 5.3.2 says 700 ns; Table 5 back-solves to about 250 ns from the frequency expressions                                                                | <b>250 ns.</b> The two clamps bracket the operating range at 250 ns and stop doing so at 700 ns, so this is a measurement, not a rounding |
| Brown-out expression            | $V_{BO} = \min(60 \text{ V}, R_{CFG} \cdot 4 \text{ V/k}\Omega)$ — read literally this is 60 V for every resistor, which contradicts the configuration table | Read as $R_{CFG} \cdot 4 \text{ V/k}\Omega$ with 15 k $\Omega$ as the floor. That is the only reading consistent with the table           |
| Brown-out units                 | the threshold is a <i>peak</i> voltage; board notes and design tools quote rms                                                                               | Converted explicitly wherever it is compared ( $\sqrt{2} = 1.414$ )                                                                       |
| Thermal resistance              | TBD                                                                                                                                                          | No junction temperature can be predicted from the datasheet alone; the loss budget is checked against measured rise instead               |
| ZCD absolute maximum            | lower limit given as TBD                                                                                                                                     | The divider is sized from the OVP thresholds only                                                                                         |
| Driver naming in Section 5.3.1  | LOUT1 and LOUT2 are transposed in one sentence against the block diagram and the configuration table                                                         | The block diagram and the table are taken as correct: LOUT2 is the pin that is strapped and held high in half bridge                      |

**Table 26: Points where the draft datasheet contradicts itself or leaves a value open. Every one of them must be re-checked against the released document.**

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## 9.2 Traps in the obvious way of building a design sheet

These are not datasheet problems. They are places where the obvious way to set a design sheet up gives an answer that is wrong in a way nothing flags, so they are worth stating as traps rather than as corrections to any particular tool.

| Where                                     | The trap                                                                                           | Consequence if missed                                                                                                                            |
|-------------------------------------------|----------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| Equivalent input range                    | taking the maximum from the ac maximum instead of the full-bridge morphing edge                    | The frequency corner is never evaluated. Here it is the difference between 264 and 332.3 Vac equivalent, and it also sets $\lambda$              |
| $\lambda$ from the minimum-gain condition | the same corner again, one step earlier                                                            | The required $\lambda$ comes out far too small — a factor of over a hundred in one case — and $L_m$ with it                                      |
| Line-cycle rms currents                   | evaluating one phase, usually $\theta = \pi/4$ , and calling it the line-cycle value               | The output bank ripple current and the rectifier loss are both understated. A Simpson rule over the half cycle costs nothing and is exact enough |
| Secondary rms above resonance             | squaring a full half sine when the current is truncated                                            | Up to 8 % at the high corner. Below resonance the factor is exactly 1, which is why it can sit unnoticed for a long time                         |
| Primary device rating                     | rating on the reflected load current $I_{\text{trafo,pk}}$                                         | About 12 % under-rated — the switch carries the <i>composite</i> tank current                                                                    |
| Loss and thermal resistance               | computing loss from the 25 °C $R_{\text{DS(on)}}$ and then asking for a heatsink that holds 125 °C | The required thermal resistance comes out only about half as demanding as it really is                                                           |
| ZVS check                                 | the closed-form shortcut, at the design Q and with the design $\lambda$                            | Error of either sign; on this tank 36 % conservative, on another about 23 % optimistic. Optimistic is the dangerous direction                    |
| Selected versus calculated                | a check that reads the calculated value while the board carries the selected one                   | Every downstream margin is reported for a design that was not built                                                                              |

Table 27: Traps that produce a plausible wrong number.

## 9.3 Constants used here without a derivation

- The  $\pi^2/8$  in the dead-time form of the  $\lambda$  condition. It behaves like a correction for the fundamental content of a square wave, but no source states it. It is indicative; the binding check is the ZVS sweep.
- The **exponent 5** in the closed-form ZVS shortcut. This is a fitted number, and it is the main reason the shortcut errs unpredictably near the boundary.
- The **0.744** in the input-voltage margin factor of the compensator design. It scales the crossover target and nothing downstream is sensitive to it at the percent level.
- The **16.8  $\Omega \cdot W$**  maximum-power constant is consistent with the feedback span and the multiplier gains quoted in the datasheet —  $2.8 \text{ V} / 0.167 = 16.8$  — so it is derived rather than assumed, and is listed here only because the gains themselves are draft values.

## 9.4 Open items in this design

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| Item                             | State                                                                                                      | What closes it                                                                                                                                            |
|----------------------------------|------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| $T_{idle}$                       | two candidate values, 250 and 700 ns                                                                       | Measure $f_{sw}(\theta)$ on the first board                                                                                                               |
| Primary conduction loss          | 0.311 against the budget — <b>not met, and accepted</b>                                                    | No single 600 V device meets a 3 W budget in the standing position. Whether it needs a heatsink is a thermal measurement, not a calculation               |
| Secondary loss budget            | 1.063 — essentially exhausted                                                                              | One more device in parallel per leg recovers it; the decision waits on the measured temperature                                                           |
| Transformer inductance tolerance | the tank tolerates a 4.1 % fall in open-circuit inductance and the usual specification asks for $\pm 10$ % | Either tighten the low side with the supplier, or lower $R_T$ to lift $f_{Min}$ first. <b>Before the transformer is ordered</b>                           |
| Output bank volume and height    | 75.2 mF, 160 parts                                                                                         | A mechanical question, not an electrical one, and it can send the whole architecture back to the output voltage                                           |
| Standby power                    | burst entry assumed at 75 W                                                                                | Measure, then trade $R_{BM}$ against the feedback ripple                                                                                                  |
| Efficiency assumption            | $\eta_{HB} = 98$ % assumed                                                                                 | Optimistic. Taking 95 % instead moves $R_{CS}$ and $R_{ac}$ by about 3 %, so nothing downstream is sensitive — but it should be replaced by a measurement |

**Table 28: What is not settled, and what would settle it.**

**Every cross-check here is a consistency check, not a correctness check.** The numbers agree across three independent implementations, which means the three implement the same equations — not that the equations describe the hardware. That is what the measurement list in Section 7.24 is for.